



# AVK INDUSTRIAL PRODUCTS



## *Engineering Manual*

**MARKET LEADERS IN GLOBAL ENGINEERED PRODUCTS**

ISO 9001-1994

QS-9001-1998

ISO 14001-1996



## WELCOME TO THE WORLD OF AVK

**AVK INDUSTRIAL PRODUCTS**, a QS9000-98/ISO9001-94 and ISO14001-96 registered company located in Valencia, CA, is a member of PCC's "SPS Fastener Division Group". AVK manufactures blind installed threaded fasteners for the transportation and general industrial markets worldwide. We feature product lines of both unified (INCH) and metric fasteners along with numerous special designs that meet specific customer application requirements.



**AVK is dedicated to...**

**"Improving The Way We Assemble The World."™**

## BLIND INSTALLED THREADED INSERTS AND STUDS

A blind installed threaded fastener is defined as a fastener with internal or external threads that can be installed into a panel, tube or other structure from the front side without need to see or access the backside, or "blind" side to complete the installation. Once installed the fastener remains captive to which a mating component can be attached using standard hardware.

This engineering manual contains technical information on all AVK standardized product lines including sales drawings and information on installation tooling.



## WARRANTY

### LIMITED WARRANTY AND EXCLUSIVE REMEDY

AVK Industrial Products division of Avibank Mfg., Inc. – which is a subsidiary of SPS Technologies, Inc. ("Seller"). Seller warrants that products sold hereunder conform to industry standards specified herein and will be free from defects in materials and workmanship. THIS WARRANTY IS EXPRESSLY GIVEN IN LIEU OF ANY AND ALL OTHER EXPRESS OR IMPLIED WARRANTIES, INCLUDING ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, AND IN LIEU OF ANY OTHER OBLIGATION ON THE PART OR THE SELLER. Seller will, at its option, repair or replace free of charge (excluding all shipping and handling costs) any products which have not been subject to misuse, abuse or modification and which in its sole determination were not manufactured in compliance with the warranty given above.

It is expressly understood that any technical advice furnished by or on behalf of Seller with respect to the use of its goods or services is given without charge, and Seller assumes no obligations or liability for the advice given or results obtained. All such results being given and accepted is at Buyer's Risk.

THE REMEDY PROVIDED FOR HEREIN SHALL BE THE EXCLUSIVE REMEDY FOR ANY BREACH OF WARRANTY OR ANY CLAIM ARISING IN ANY WAY OUT OF THE MANUFACTURE, SALE, OR USE OF THESE PRODUCTS. In no event shall Seller be liable for consequential, incidental or any other damages of any nature whatsoever except those specifically provided herein for any breach of warranty or any claim arising in any way out of the manufacture, sale, or use of these products. No other person is authorized by Seller to give any other warranty, written or oral, pertaining to the products.

# ENGINEERING MANUAL

This engineering manual contains the full AVK product line, application ideas, features and benefits, sales drawings with dimensions and tolerances, material and finish specifications, and technical information on the selection and use of all installation tooling.

## PRODUCT INDEX

| ITEM                                                                                                                                                                                                                     | PAGE  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Welcome to the World of AVK .....                                                                                                                                                                                        | 2     |
| AVK Product Index .....                                                                                                                                                                                                  | 3     |
| Capabilities .....                                                                                                                                                                                                       | 4     |
| Product Introduction .....                                                                                                                                                                                               | 5     |
| Spinwall Technology .....                                                                                                                                                                                                | 6     |
| Advantages .....                                                                                                                                                                                                         | 7     |
| Typical AVK Applications .....                                                                                                                                                                                           | 8-9   |
|   A-L Series™ Knurled Threaded Insert.....             | 10-11 |
|   A-K Series™ Knurled Threaded Insert.....             | 12-13 |
|   A-H Series™ Threaded Hex Insert.....                 | 14-15 |
|   A-R Series™ Threaded Insert .....                  | 16-17 |
|   A-L Series™ Sealed Head Threaded Insert .....    | 18    |
|   A-H Series™ Sealed Head Threaded Hex Insert..... | 19    |
|   A-S Series™ Captive Threaded Stud.....           | 20-21 |
|   A-T Series™ Knurled Threaded Insert.....         | 22-23 |
|   A-W Series® Knurled Threaded Insert.....         | 24    |
|   A-O Series™ Threaded Insert .....                | 25    |
|   R-N Series® Rivet Nut Threaded Insert .....      | 26-27 |
|   ARO Spin-Spin Pneumatic Tools .....              | 28-33 |
|   SPP™ Spin Pull to Pressure Tools.....            | 34-35 |
|   Dyna-Set™ Automation Tools.....                  | 36    |
|  Miscellaneous Pneumatic/Hydraulic Tools .....                                                                                        | 37    |
|  Hand Tools and Kits .....                                                                                                            | 38-40 |
|  Guidelines for Successful Applications.....                                                                                          | 41    |
|  Basic Test Data .....                                                                                                                | 42    |
|  Dimensional Conversions .....                                                                                                        | 43    |



## CAPABILITIES



### AVK SALES PROFESSIONALS:

AVK's sales representatives and customer service staff are dedicated to assist our customers in providing demonstrations, samples, and technical support. Our global network of Authorized Distributors assist in these efforts and provide for our customer's logistical product needs.

### ENGINEERING SUPPORT:

AVK's engineering and installation tool support staff transforms our customers needs and ideas into robust designs that have become industry standards for innovation and reliability.



### MANUFACTURING:

Housed in a 90,000 sq. ft. facility, AVK utilizes state of the art cold forming equipment to produce net shape products at speeds of up to 240 pieces per minute. Secondary customized internal thread rolling, assembly equipment and a "Lean" manufacturing philosophy help to produce product that exceeds our customers expectations from quality and delivery.

### QUALITY:

Our staff of quality experts are dedicated to the principles of QS9000 and ISO9001. They are continuously working to improve an already world class quality product to even higher levels.

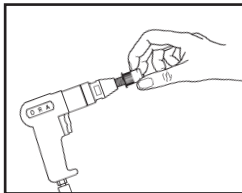


## PRODUCT INTRODUCTION

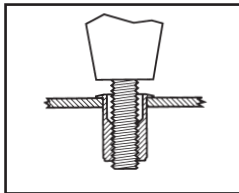
AVK has developed specific manufacturing technology that sets our products apart from other blind installed fastener manufacturers. We call what we do "Spinwall Technology." AVK's Spinwall Technology™ blind installed fasteners consist of two types. Internally threaded inserts and externally threaded studs. Both products can be installed into a flat, tubular or other shaped materials using hand operated or pneumatic hand held tools right on your assembly line without adjustment even if the parent material varies in thickness. Our products can be installed after paint or other finishes is applied to your product which eliminates the need for thread masking.

## INSTALLATION SEQUENCE

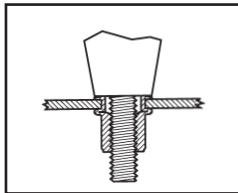
Here is how the AVK product is installed "blind."



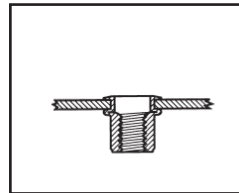
1/4 turn the AVK fastener onto the tool's threaded mandrel.



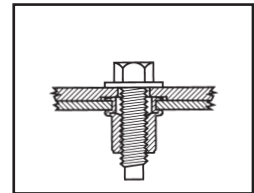
The fastener is placed into the hole.



The forward trigger is depressed and the fastener is installed properly.



The reverse trigger is depressed and the tool unthreads from the installed fastener.



The mating component is then attached using conventional hardware.

## SPINWALL TECHNOLOGY™ PRODUCT DESIGN



Round serrated body threaded insert for excellent spin out resistance in drilled or punched holes. Available in steel, aluminum, brass, and monel.



Hex Body threaded insert for punched holes provides exceptional spin out resistance. Available in steel, brass and 302 stainless steel.

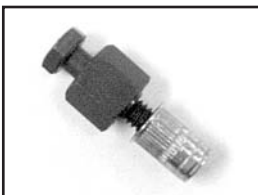


Pre-Bulbed slotted body design for exceptional pull out resistance in drilled or punched holes in plastics, composites and thin sheet metal applications. Available in steel.



Round serrated body threaded stud is ideal as a location device to support heavy components before final installation with a mating nut. Available in steel.

## INSTALLATION TOOL TYPES



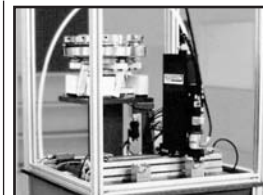
The expendable tool is used for lower consumer cost or field installations.



The lever or plier style tools are used for experimental or field installations.



The pneumatic tool is used for production line work.



The Dyna-Set™ automation system is used for automated installations.



## SPINWALL TECHNOLOGY™

### Spinwall Technology™...Sets AVK apart

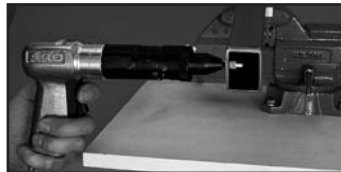
You will notice throughout this catalog references to the phrase Spinwall Technology™. This phrase describes the philosophy of our product's design, manufacturing and installation systems.

Our Spinwall Technology™ products are manufactured on high-speed, state-of-the-art cold forming equipment on which very precise tolerances are achieved. This capability allows AVK to produce products with unique mechanical and installation properties that result in the AVK product filling the hole prior to backside flange formation. Read more about hole fill and the other advantages of Spinwall Technology™ on the separate product profile pages.

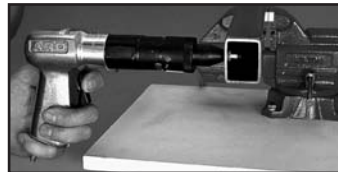
**Our Spinwall Technology™ products can be installed using three different types of installation tooling...**

### ARO PNEUMATIC STALL TORQUE TOOL

The ARO pneumatic stall torque type tool installs AVK as follows...



1/4 turn the fastener onto the ARO tool mandrel and place the fastener into the hole.



Depress the top trigger until the tool stalls and the AVK fastener is collapsed.



Depress the lower trigger and the tool unthreads from the installed part.

### AVK SPP™ SPIN PULL TO PRESSURE TOOL

The SPP tool utilizes an ARO pneumatic stall torque tool and incorporates an integrated hydraulic cylinder powered by a remote hydraulic power pack system.



1/4 turn the fastener onto the SPP tool mandrel and place the fastener into the hole.



Depress the tool trigger and the tool spins into the fastener and automatically exerts a pressure controlled pull installing the fastener.



Depress the tool reverse trigger and the tool spins out of the installed fastener.

### Dyna-Set™ Technology



The patent-pending design of the Dyna-Set™ automated insert system and Material Handling Module utilizes spin pull technology. The Dyna-Set™ will replace antiquated canister pneumatic hydraulic spin pull to stroke tools and provide greater reliability and maximize assembly capability.

Some of the Dyna-Set™ benefits are as follows:

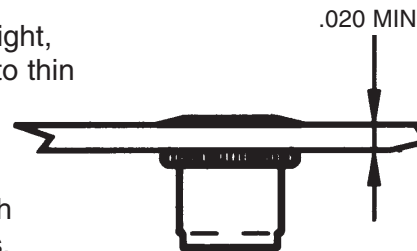
- ◆ Labor savings
- ◆ Single or multi-simultaneous insert installation
- ◆ Maximum up-time providing optimal production output
- ◆ Multiple work station configurations are available
- ◆ Robotic arm with hole locating vision system is available
- ◆ Insert collapse load can be verified
- ◆ Dyna-Set™ systems are self diagnostic

**See pages 28 through 39 for more info on AVK's installation tooling.**

## ADVANTAGES OF ASSEMBLY WITH AVK

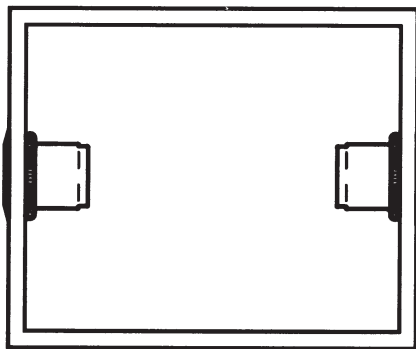
Here are a few reasons why the use of AVK's blind installed fasteners are "Improving The Way We Assemble The World"™

Material thickness is being reduced to save weight, fuel-pollution and raw material cost. Fastening to thin materials is simple and reliable with AVK. AVK fasteners can be installed into thinner materials with greater ease than can be accomplished with weld nuts, pierce nuts, clinch nuts, thread tapping and thread forming screws.



\*Minimum grips vary per thread size

### INSTALLS BLIND



New material structures such as hydro formed tubing, aluminum extrusions and composite panels are being specified due to their strength to weight ratios. These materials form blind applications are ideal for AVK fasteners versus other types of fasteners.

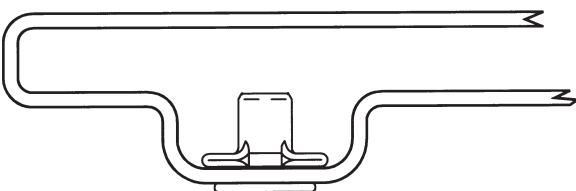
Welding and the attachment of weld fasteners are being replaced with AVK due to health and clean air requirements for workers. The alternatives are expensive capital equipment

vacuuming and air scrubber equipment. Weld fasteners also burn away pre-applied galvanized finishes requiring re-work to prevent corrosion. Pre-painted materials cannot be used with weld fasteners as the paint prevents weld nut attachment. Use of pre-painted materials can eliminate painting facility costs and environmental issues. Weld fasteners must be applied before a product is painted. Thread masking procedures are eliminated by the use of AVK as our products can be installed after paint.

### NEAT AND CLEAN



Plastics and composites are being used for products to take advantage of their molding, corrosion resistance, coloring and strength to weight ratios.

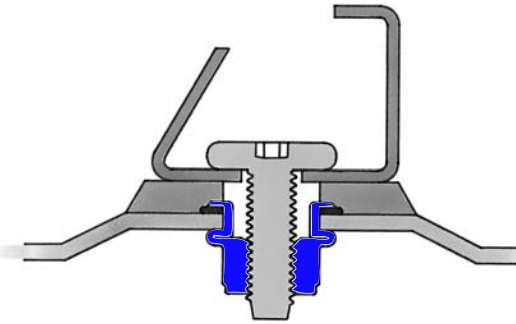


AVK has specific products for blow, rotational, compression, vacuum, scrimp, honeycomb, processed plastic and composites.

Products designed for consumer assembly to reduce in plant assembly costs use AVK to provide strong threads for the attachment of component parts using conventional hardware.

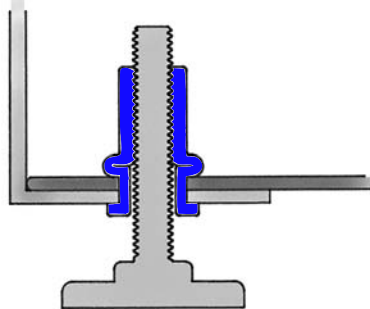
# TYPICAL AVK APPLICATIONS

E-COAT/PAINTED SHEET METAL



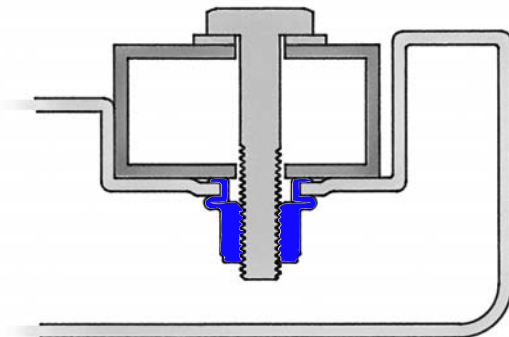
VEHICLE ROOF — LUGGAGE RACK RAIL

FORMED SHEET METAL



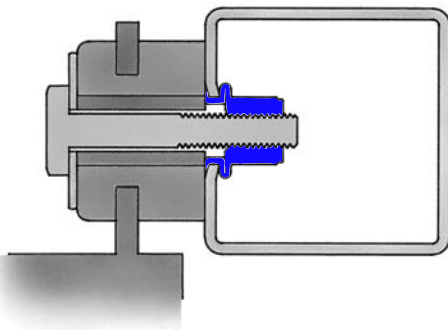
OFFICE FURNITURE — LEG LEVELING

BLOW OR ROTATIONAL MOLDED PLASTICS



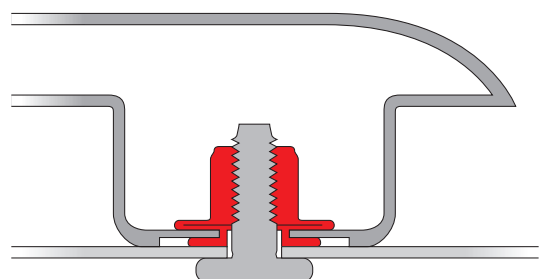
FARM EQUIPMENT — GUARD TO TUBULAR FRAME

HYDROFORMED STEEL TUBING



VEHICLE — RADIATOR ATTACHMENT

BLOW MOLDED PLASTIC



AUTOMOBILES — SPOILER TO TRUNK

## ◆ LAWN/GARDEN FARM IMPLEMENT

Mirrors/Lights to Cab • Components to Decking •  
Guards to Framing • Seats to Framing • Access  
Panels to Frame

## ◆ MEDICAL EQUIPMENT

Wheelchair Hand Rims • Wheelchair  
Seats/Backs • Handles/Casters to Carts •  
Components to Hospital Beds • Aluminum Walker  
Assembly

## ◆ MILITARY GROUND SUPPORT

Ducts/Wireways to Shelters • Electronics to  
Shelters • Missile Container Assembly  
• Antenna Assembly • Hardware to Shelters

## ◆ MILITARY VEHICLES

Side Mirrors to Body • Armor to Body  
• Bulletproof Glass to Body • Instruments  
to Dash • Cargo Hold Down to Floor Pan

## ◆ OFFICE FURNITURE

Modular Unit Stacking • Hardware/Hinges  
to Unit • Leg Leveling • Knock-Down Assembly •  
Partition Electrical Components

## ◆ RECREATION

Playground Equipment Assembly • Bicycle Frame  
Water Bottle Attachment • Basketball Pole  
Assembly • Golf Cart Roof Supports to Body •  
Consumer Assembly of Toys

## ◆ REFRIGERATION

Hinges to Doors/Cabinet • Handles to Doors  
• Shelf Brackets to Wall • Compressor to  
Base Pan • Leg Leveling

## ◆ RV INDUSTRY

Instruments to Boat Dash • Bow Rails to Deck •  
Components to Snowmobile • Components  
to Motorcycles • RV Awning Assembly

## ◆ TRUCKS/TRAILERS

Mirrors/Lights to Cab • Grab Handles to Cab •  
Firewall Attachments • Instruments to Dash •  
Exhaust/Wind Deflectors

# TYPICAL AVK APPLICATIONS

## ◆ AEROSPACE

Galley Equipment Casters to Frame • Aircraft Seating Footrests to Frame • Aircraft Seat Trays to Frame • Bulkhead Partition Mounting Brackets • Shipping Container Hinges and Latches to Frame

## ◆ APPLIANCES

Refrigerator Hinge to Cabinet • Refrigerator Handle to Door • Leg Leveler • Components to Cabinet • Under the Counter Attachments

## ◆ ARCHITECTURAL

Vinyl Window Hardware to Frame • Aluminum Door Hardware to Frame • Threshold Sweeps to Frame • Aluminum Railing "T" Joints • Patio Enclosure Construction

## ◆ AUTO/SPORT-UTILITY VEHICLES

Luggage Racks to Roof • Spoilers to Trunk Lids • Option Controls to Dash Panel • Under Hood Option Items • Grab Handles • Air Bag Attachments

## ◆ ELECTRONICS CABINETRY

Hardware to Cabinet • Hinges to Cabinet • Leg Levelers • Components to Frame • Lifting Anchors

## ◆ EXERCISE EQUIPMENT

Stationary Bike Floor Supports to Frame • Treadmill Controls to Frame • Sheet Metal Covers over Motors • Weightlifting Frame Assembly • Electronics to Unit

## ◆ FOOD SERVICE

Leg Leveling • Fixed Leg Attachment • Coin Box to Unit • Hardware to Cabinet • Casters to Frame

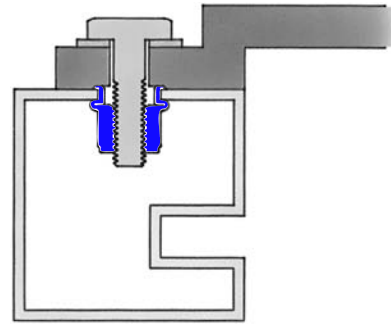
## ◆ FURNITURE

Aluminum Furniture Assembly • Leg Leveling • Brass Headboards to Frame • Patio Table Assembly • Tubular Bed Frame Assembly

## ◆ HEATING/AIR CONDITIONING

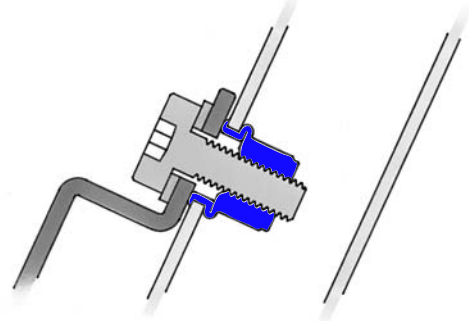
Compressors to Base Pans • Access Doors to Cabinet • Motors to Blower Housing • Blower Housing to Unit • Burner Assembly to Unit

### ALUMINUM EXTRUSIONS



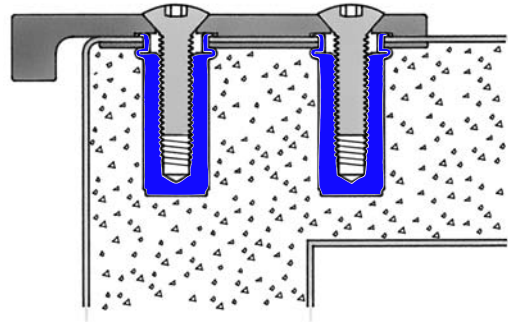
ENTRY DOORS — PUSH BARS TO DOOR FRAME

### POWDER COATED STEEL TUBE



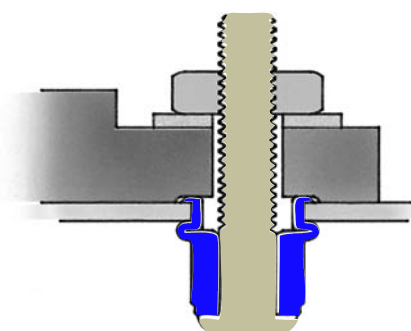
BICYCLE FRAME — WATER BOTTLE BRACKET

### PREPAINTED FOAM CORE SHEET METAL



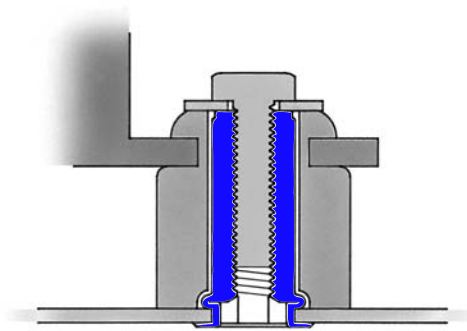
REFRIGERATION EQUIPMENT — HINGES TO CABINET

### E-COAT/PAINTED SHEET METAL



VEHICLE DASH PANEL — ABS MODULE MOUNTING

### GALVANIZED SHEET METAL



AIR CONDITIONER BASE PAN — COMPRESSOR MOUNTING

## A-L SERIES INSERT PROFILE

The **A-L Series Insert** features a knurled body and large diameter—low profile head making it ideal for use in punched or drilled holes. It offers the highest all around strength characteristics and has been designed to be used with Grade 5 or Metric 8.8/9.8 mating screws. The A-L Series is AVK's most versatile performer.

The A-L Series Insert can be installed using AVK's ARO brand pneumatic tools or AVK's SPP™ pneumatic/hydraulic tooling. These tools can be located at any position on your assembly line. The A-L Series can be installed either before or after finish.

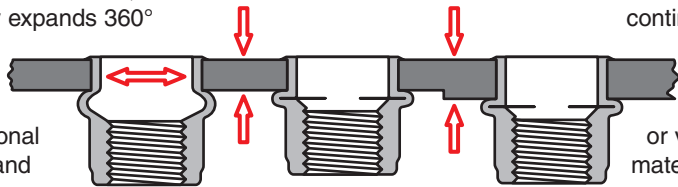
## SPINWALL TECHNOLOGY™

### HOW HOLE FILL WORKS FOR YOU

As the A-L Series is installed, the knurled body expands 360°

**FILLING THE HOLE.**

This feature provides exceptional torque strength and vibration resistance.



The installation tool then continues to install the insert forming a backside flange even in multiple or variable thickness materials—**WITHOUT ADJUSTMENT.**

## DESIGN BENEFITS

- ◆ **EXCEPTIONAL TORQUE STRENGTH** is achieved as the insert's knurled body expands **FILLING THE HOLE.**
- ◆ **QUALITY INSTALLATIONS** even in variable thickness materials are assured by AVK's spin/spin ARO pneumatic tools and our SPP pneumatic hydraulic tools.
- ◆ **SUPERIOR THREAD STRENGTH** is provided due to our internal rolled thread manufacturing process.
- ◆ **THREADS GAUGE** before and after installation due to the increased cross-sectional thickness of the thread area. Thread dilation is prevented.
- ◆ **INVENTORY REDUCTION** is possible because of the A-L Series' wide grip range capacity. It is 2.5 times greater than conventional rivet nuts.
- ◆ **SUPERIOR CORROSION RESISTANCE** is provided by our standard zinc/yellow dichromate finish (96 hours. salt spray to white corrosion). For exceptional corrosion protection we offer a tin/zinc alloy finish.
- ◆ **AVAILABLE** in steel from stock. Additional materials such as aluminum, brass and monel are available by special order. Contact AVK for details.

## ADDITIONAL DESIGN TYPES

### CLOSED END

Thread area is enclosed eliminating leakage past the threads from either side of the application. See page 11.



### SEALED HEAD

A PVC foam seal is bonded to the underside of the head and when installed provides a weathertight seal. (Also available in the closed end version.) See page 18 for important grip information.

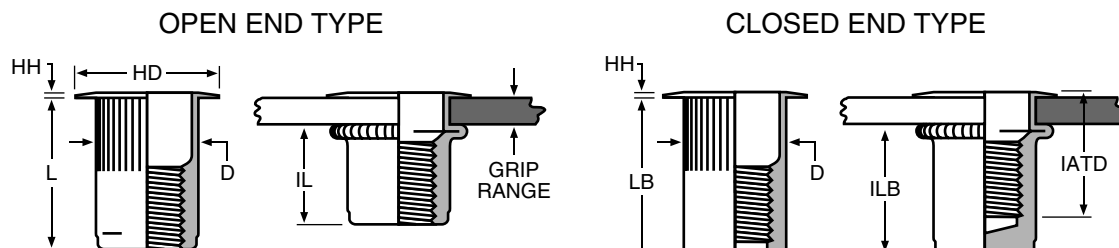


### WEDGE HEAD

The addition of wedges under the head provides even greater torque capability, especially in soft or thin materials, and is excellent for electrical grounding applications. Contact AVK for a sales drawing.



# UNIFIED (INCH) AND METRIC THREAD SIZES



Thread Specifications: Unified 2B/21 per ASME B1.1  
Metric 6H/21 per ASME B1.13M

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+.006<br>-.000 | HD<br>±.010<br>±.025* | HH<br>±.003 | L<br>±.015 | D<br>MAX. | IL<br>MAX. | LB<br>±.015 | ILB<br>MAX. | IATD*<br>MAX. |
|-------------|-----------------|------------|---------------|-----------------------------|-----------------------|-------------|------------|-----------|------------|-------------|-------------|---------------|
| 6-32 UNC    | 632             | .020-.080  | 80            | 17/64 (.2656)               | .390                  | .030        | .420       | .265      | .305       | .740        | .640        | .610          |
| 6-32 UNC    | 632             | .080-.130  | 130           | 17/64 (.2656)               | .390                  | .030        | .470       | .265      | .305       | .740        | .580        | .670          |
| 8-32 UNC    | 832             | .020-.080  | 80            | 17/64 (.2656)               | .390                  | .030        | .420       | .265      | .305       | .740        | .640        | .610          |
| 8-32 UNC    | 832             | .080-.130  | 130           | 17/64 (.2656)               | .390                  | .030        | .470       | .265      | .305       | .740        | .580        | .670          |
| 10-24 UNC   | 1024            | .020-.130  | 130           | 19/64 (.2969)               | .415                  | .030        | .475       | .296      | .315       | .990        | .845        | .730          |
| 10-24 UNC   | 1024            | .130-.225  | 225           | 19/64 (.2969)               | .415                  | .030        | .585       | .296      | .315       | .990        | .735        | .840          |
| 10-32 UNF   | 1032            | .020-.130  | 130           | 19/64 (.2969)               | .415                  | .030        | .475       | .296      | .315       | .990        | .845        | .730          |
| 10-32 UNF   | 1032            | .130-.225  | 225           | 19/64 (.2969)               | .415                  | .030        | .585       | .296      | .315       | .990        | .735        | .840          |
| 1/4-20 UNC  | 420             | .027-.165  | 165           | 25/64 (.3906)               | .500                  | .030        | .580       | .390      | .380       | 1.190       | 1.005       | .895          |
| 1/4-20 UNC  | 420             | .165-.260  | 260           | 25/64 (.3906)               | .500                  | .030        | .680       | .390      | .380       | 1.190       | .905        | 1.035         |
| 5/16-18 UNC | 518             | .027-.150  | 150           | 17/32 (.5312)               | .685*                 | .035        | .690       | .530      | .470       | 1.390       | 1.175       | .995          |
| 5/16-18 UNC | 518             | .150-.312  | 312           | 17/32 (.5312)               | .685*                 | .035        | .805       | .530      | .425       | 1.390       | 1.025       | 1.120         |
| 3/8-16 UNC  | 616             | .027-.150  | 150           | 17/32 (.5312)               | .685*                 | .035        | .690       | .530      | .470       | 1.390       | 1.175       | .995          |
| 3/8-16 UNC  | 616             | .150-.312  | 312           | 17/32 (.5312)               | .685*                 | .035        | .805       | .530      | .425       | 1.390       | 1.025       | 1.120         |
| 1/2-13 UNC  | 813             | .063-.200  | 200           | 11/16 (.6875)               | .865*                 | .047        | 1.150      | .685      | .850       | 2.365       | 2.070       | 1.505         |
| 1/2-13 UNC  | 813             | .200-.350  | 350           | 11/16 (.6875)               | .865*                 | .047        | 1.300      | .685      | .850       | 2.365       | 1.920       | 1.505         |
| 1/2-13 UNC  | 813             | .350-.500  | 500           | 11/16 (.6875)               | .865*                 | .047        | 1.450      | .685      | .860       | 2.365       | 1.770       | 1.505         |

| THREAD SIZE    | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+0.15<br>-0.00 | HD<br>±0.25<br>±0.64* | HH<br>±0.08 | L<br>±0.38 | D<br>MAX. | IL<br>MAX. | LB<br>±0.38 | ILB<br>MAX. | IATD*<br>MAX. |
|----------------|-----------------|------------|---------------|-----------------------------|-----------------------|-------------|------------|-----------|------------|-------------|-------------|---------------|
| M4 x 0.7 ISO   | 470             | 0.50-2.00  | 2.0           | 6.75                        | 9.91                  | 0.76        | 10.67      | 6.73      | 7.75       | 18.80       | 16.26       | 15.49         |
| M4 x 0.7 ISO   | 470             | 2.00-3.30  | 3.3           | 6.75                        | 9.91                  | 0.76        | 11.94      | 6.73      | 7.75       | 18.80       | 14.73       | 17.02         |
| M5 x 0.8 ISO   | 580             | 0.50-3.30  | 3.3           | 7.60                        | 10.54                 | 0.76        | 12.07      | 7.52      | 8.00       | 25.15       | 21.46       | 18.54         |
| M5 x 0.8 ISO   | 580             | 3.30-5.70  | 5.7           | 7.60                        | 10.54                 | 0.76        | 14.86      | 7.52      | 8.00       | 25.15       | 18.67       | 21.34         |
| M6 x 1.0 ISO   | 610             | 0.70-4.20  | 4.2           | 10.00                       | 12.70                 | 0.76        | 14.73      | 9.91      | 9.65       | 30.23       | 25.53       | 22.73         |
| M6 x 1.0 ISO   | 610             | 4.20-6.60  | 6.6           | 10.00                       | 12.70                 | 0.76        | 17.27      | 9.91      | 9.65       | 30.23       | 22.99       | 26.29         |
| M8 x 1.25 ISO  | 8125            | 0.70-3.80  | 3.8           | 13.50                       | 17.40*                | 0.89        | 17.53      | 13.46     | 11.94      | 35.31       | 29.85       | 25.27         |
| M8 x 1.25 ISO  | 8125            | 3.80-7.90  | 7.9           | 13.50                       | 17.40*                | 0.89        | 20.45      | 13.46     | 10.80      | 35.31       | 26.04       | 28.45         |
| M10 x 1.5 ISO  | 1015            | 0.70-3.80  | 3.8           | 13.50                       | 17.40*                | 0.89        | 17.53      | 13.46     | 11.94      | 35.31       | 29.85       | 25.27         |
| M10 x 1.5 ISO  | 1015            | 3.80-7.90  | 7.9           | 13.50                       | 17.40*                | 0.89        | 20.45      | 13.46     | 10.80      | 35.31       | 26.04       | 28.45         |
| M12 x 1.75 ISO | 12175           | 1.60-5.10  | 5.1           | 17.45                       | 21.97*                | 1.19        | 29.21      | 17.4      | 21.59      | 60.07       | 52.58       | 38.23         |
| M12 x 1.75 ISO | 12175           | 5.10-8.90  | 8.9           | 17.45                       | 21.97*                | 1.19        | 33.02      | 17.4      | 21.59      | 60.07       | 48.77       | 38.23         |
| M12 x 1.75 ISO | 12175           | 8.90-12.7  | 12.7          | 17.45                       | 21.97*                | 1.19        | 36.83      | 17.4      | 21.84      | 60.07       | 44.96       | 38.23         |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Additional UNF fine threads are available. Contact AVK for details. **NOTE 3:** Additional grip lengths are available. Contact AVK for details. \*Dimensions in minimum grip condition.

## PART NUMBERING SYSTEM

| AL                 | ( )             | ( )                | ( )                                                                                                  | ( )        |
|--------------------|-----------------|--------------------|------------------------------------------------------------------------------------------------------|------------|
| Product Series     | MATERIAL        | FINISH             | Thread Call Out                                                                                      | TYPE       |
| CALL OUT           | Grade           | CALL OUT           | Specification                                                                                        | CALL OUT   |
| S                  | Steel 1010/1008 | 4                  | STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING | BLANK      |
| * BY SPECIAL ORDER |                 | * BY SPECIAL ORDER |                                                                                                      | B          |
| A                  | Aluminum 5056   | 9                  | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING                      | Closed End |
| B                  | Brass 270/260   | 1                  | ALUMINUM OR MONEL - CLEAR PROTECTIVE COATING                                                         |            |
| M                  | Monel 400       | 2                  | BRASS - CAD PER QQP 416 TYPE I, CLASS 3 WITH CLEAR PROTECTIVE COATING                                |            |

SAMPLE NUMBER:  
ALS4-420-165

\* BY SPECIAL ORDER  
S Sealed  
(See Pg. 18)

\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 28 and 30



## A-K SERIES INSERT PROFILE

The **A-K Series Insert** features a knurled body and a reduced profile head to allow for virtually flush installation. Countersink drilling or dimpling of the parent material can be eliminated. The A-K Series is designed to be used with Grade 5 or Metric Class 8.8/9.8 mating screws.

The A-K Series Insert can be installed using AVK's ARO brand pneumatic tools or AVK's SPP<sup>TM</sup> pneumatic/hydraulic tooling. These tools can be located at any position on your assembly line. The A-K Series Insert can be installed either before or after finish.

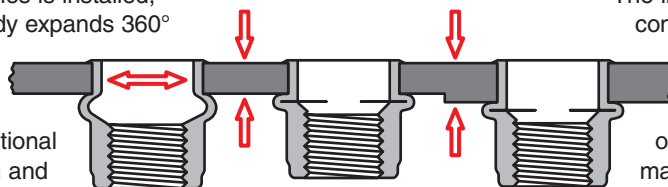
## SPINWALL TECHNOLOGY<sup>TM</sup>

### HOW HOLE FILL WORKS FOR YOU

As the A-K Series is installed, the knurled body expands 360°

**FILLING THE HOLE.**

This feature provides exceptional torque strength and vibration resistance.



The installation tool then continues to install the insert forming a backside flange even in multiple or variable thickness materials—**WITHOUT ADJUSTMENT.**

## DESIGN BENEFITS

- ◆ **VIRTUALLY FLUSH INSTALLATIONS** are achieved without special hole preparation due to the A-K Series minimal head profile.
- ◆ **EXCEPTIONAL TORQUE STRENGTH** is achieved as the insert's knurled body expands **FILLING THE HOLE.**
- ◆ **QUALITY INSTALLATIONS** even in variable thickness materials are assured by AVK's spin/spin ARO pneumatic tools and our SPP<sup>TM</sup> pneumatic/hydraulic tools.
- ◆ **SUPERIOR THREAD STRENGTH** is provided due to our internal rolled thread manufacturing process.
- ◆ **THREADS GAUGE** before and after installation due to the increased cross-sectional thickness of the thread area. Thread dilation is prevented.
- ◆ **INVENTORY REDUCTION** is possible because of the A-K Series' wide grip range capacity. It is 2.5 times greater than conventional rivet nuts.
- ◆ **SUPERIOR CORROSION RESISTANCE** is provided by our standard zinc/yellow dichromate finish (96 hours. salt spray to white corrosion). For exceptional corrosion protection we offer a tin/zinc alloy finish.
- ◆ **AVAILABLE** in steel from stock. Additional materials such as aluminum, brass and monel are available

## ADDITIONAL DESIGN TYPES

### CLOSED END

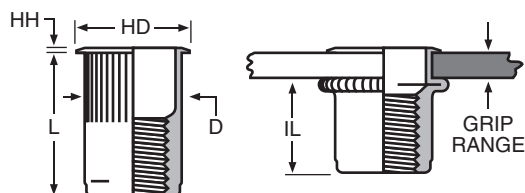
Thread area is enclosed eliminating leakage past the threads from either side of the application.

See page 13.

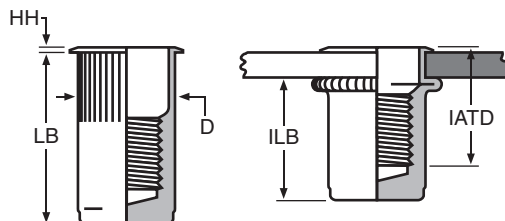


# UNIFIED (INCH) AND METRIC THREAD SIZES

OPEN END TYPE



CLOSED END TYPE



Thread Specifications: Unified 2B/21 per ASME B1.1  
Metric 6H/21 per ASME B1.13M

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+.006<br>-.000 | HD<br>±.010 | HH<br>±.002 | L<br>±.015 | D<br>MAX. | IL<br>MAX. | LB<br>±.015 | ILB<br>MAX. | IATD*<br>MAX. |
|-------------|-----------------|------------|---------------|-----------------------------|-------------|-------------|------------|-----------|------------|-------------|-------------|---------------|
| 6-32 UNC    | 632             | .020-.080  | 80            | 17/64 (.2656)               | .310        | .019        | .420       | .265      | .305       | .740        | .640        | .610          |
| 6-32 UNC    | 632             | .080-.130  | 130           | 17/64 (.2656)               | .310        | .019        | .470       | .265      | .305       | .740        | .580        | .670          |
| 8-32 UNC    | 832             | .020-.080  | 80            | 17/64 (.2656)               | .310        | .019        | .420       | .265      | .305       | .740        | .640        | .610          |
| 8-32 UNC    | 832             | .080-.130  | 130           | 17/64 (.2656)               | .310        | .019        | .470       | .265      | .305       | .740        | .580        | .670          |
| 10-24 UNC   | 1024            | .020-.130  | 130           | 19/64 (.2969)               | .340        | .019        | .475       | .296      | .315       | .990        | .845        | .730          |
| 10-24 UNC   | 1024            | .130-.225  | 225           | 19/64 (.2969)               | .340        | .019        | .585       | .296      | .315       | .990        | .735        | .840          |
| 10-32 UNF   | 1032            | .020-.130  | 130           | 19/64 (.2969)               | .340        | .019        | .475       | .296      | .315       | .990        | .845        | .730          |
| 10-32 UNF   | 1032            | .130-.225  | 225           | 19/64 (.2969)               | .340        | .019        | .585       | .296      | .315       | .990        | .735        | .840          |
| 1/4-20 UNC  | 420             | .027-.165  | 165           | 25/64 (.3906)               | .455        | .022        | .580       | .390      | .380       | 1.190       | 1.005       | .895          |
| 1/4-20 UNC  | 420             | .165-.260  | 260           | 25/64 (.3906)               | .455        | .022        | .680       | .390      | .380       | 1.190       | .905        | 1.035         |
| 5/16-18 UNC | 518             | .027-.150  | 150           | 17/32 (.5312)               | .595        | .022        | .690       | .530      | .470       | 1.390       | 1.175       | .995          |
| 5/16-18 UNC | 518             | .150-.312  | 312           | 17/32 (.5312)               | .595        | .022        | .805       | .530      | .425       | 1.390       | 1.025       | 1.120         |
| 3/8-16 UNC  | 616             | .027-.150  | 150           | 17/32 (.5312)               | .595        | .022        | .690       | .530      | .470       | 1.390       | 1.175       | .995          |
| 3/8-16 UNC  | 616             | .150-.312  | 312           | 17/32 (.5312)               | .595        | .022        | .805       | .530      | .425       | 1.390       | 1.025       | 1.120         |

| THREAD SIZE   | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+0,15<br>-0,00 | HD<br>±0,25 | HH<br>±0,05 | L<br>±0,38 | D<br>MAX | IL<br>MAX | LB<br>±0,38 | ILB<br>MAX | IATD*<br>MAX |
|---------------|-----------------|------------|---------------|-----------------------------|-------------|-------------|------------|----------|-----------|-------------|------------|--------------|
| M4 x 0,7 ISO  | 470             | 0,50-2,00  | 2.0           | 6,75                        | 7,87        | 0,48        | 10,67      | 6,73     | 7,75      | 18,80       | 16,26      | 15,49        |
| M4 x 0,7 ISO  | 470             | 2,00-3,30  | 3.3           | 6,75                        | 7,87        | 0,48        | 11,94      | 6,73     | 7,75      | 18,80       | 14,73      | 17,02        |
| M5 x 0,8 ISO  | 580             | 0,50-3,30  | 3.3           | 7,60                        | 8,64        | 0,48        | 12,07      | 7,52     | 8,00      | 25,15       | 21,46      | 18,54        |
| M5 x 0,8 ISO  | 580             | 3,30-5,70  | 5.7           | 7,60                        | 8,64        | 0,48        | 14,86      | 7,52     | 8,00      | 25,15       | 18,67      | 21,34        |
| M6 x 1,0 ISO  | 610             | 0,70-4,20  | 4.2           | 10,00                       | 11,56       | 0,55        | 14,73      | 9,91     | 9,65      | 30,23       | 25,53      | 22,73        |
| M6 x 1,0 ISO  | 610             | 4,20-6,60  | 6.6           | 10,00                       | 11,56       | 0,55        | 17,27      | 9,91     | 9,65      | 30,23       | 22,99      | 26,29        |
| M8 x 1,25 ISO | 8125            | 0,70-3,80  | 3.8           | 13,50                       | 15,11       | 0,55        | 17,53      | 13,46    | 11,94     | 35,31       | 29,85      | 25,27        |
| M8 x 1,25 ISO | 8125            | 3,80-7,90  | 7.9           | 13,50                       | 15,11       | 0,55        | 20,45      | 13,46    | 10,80     | 35,31       | 26,04      | 28,45        |
| M10 x 1,5 ISO | 1015            | 0,70-3,80  | 3.8           | 13,50                       | 15,11       | 0,55        | 17,53      | 13,46    | 11,94     | 35,31       | 29,85      | 25,27        |
| M10 x 1,5 ISO | 1015            | 3,80-7,90  | 7.9           | 13,50                       | 15,11       | 0,55        | 20,45      | 13,46    | 10,80     | 35,31       | 26,04      | 28,45        |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Additional UNF fine threads are available. Contact AVK for details. **NOTE 3:** Additional grip lengths are available in certain thread sizes. Contact AVK for details. \* Dimensions in minimum grip condition.

## PART NUMBERING SYSTEM

| AK                 |               | ( )                |                                                                                 | ( )                                                                                                  |                 | - ( ) - ( )   |          | ( )      |            |
|--------------------|---------------|--------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------|---------------|----------|----------|------------|
| MATERIAL           |               | FINISH             |                                                                                 | Thread Call Out                                                                                      |                 | Grip Call Out |          | TYPE     |            |
| Product Series     | CALL OUT      | Grade              | CALL OUT                                                                        | Specification                                                                                        | Thread Call Out | Grip Call Out | CALL OUT | Type     |            |
|                    | S             | Steel 1010/1008    | 4                                                                               | STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING |                 |               | BLANK    | Open End |            |
| * BY SPECIAL ORDER |               | * BY SPECIAL ORDER |                                                                                 | SAMPLE NUMBER:<br>AKS4-420-165                                                                       |                 |               |          | B        | Closed End |
| A                  | Aluminum 5056 | 9                  | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING |                                                                                                      |                 |               |          |          |            |
| B                  | Brass 270/260 | 1                  | ALUMINUM OR MONEL - CLEAR PROTECTIVE COATING                                    |                                                                                                      |                 |               |          |          |            |
| M                  | Monel 400     | 2                  | BRASS - CAD PER QQP 416 TYPE I, CLASS 3 WITH CLEAR PROTECTIVE COATING           |                                                                                                      |                 |               |          |          |            |

\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 28 and 30

## A-H SERIES INSERT PROFILE

The **A-H Series Insert** features a radius corner hex body. When installed into a corresponding hex hole, the radius corners of the A-H Series Insert expand and fill the hole corners providing exceptional resistance to spinning in the panel. The A-H Series is designed to be used with Grade 5 or Metric Class 8.8/9.8 mating screws.

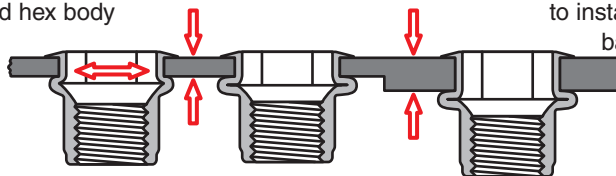
The A-H Series Insert can be installed using AVK's ARO brand pneumatic tools or AVK's SPP<sup>TM</sup> pneumatic/hydraulic tooling. These tools can be located at any position on your assembly line. The A-H Series Insert can be installed either before or after finish.



## SPINWALL TECHNOLOGY<sup>TM</sup>

### HOW HOLE FILL WORKS FOR YOU

As the A-H Series is installed, the radius cornered hex body expands **FILLING THE HOLE**. This feature provides exceptional torque strength and vibration resistance.



The installation tool then continues to install the insert forming a backside flange even in multiple or variable thickness materials **—WITHOUT ADJUSTMENT.**

## DESIGN BENEFITS

- ◆ **EXCEPTIONAL** resistance to spinning in the panel is achieved as the A-H Series' hex body expands **FILLING THE HOLE**.
- ◆ **AVOID STRESS FRACTURES** of your material and prolong punch and die life by specifying a radius corner in your hex hole. This is possible when using the A-H Series radius hex body insert.
- ◆ **QUALITY INSTALLATIONS** even in variable thickness materials are assured by AVK's spin/spin ARO pneumatic tools and our SPP<sup>TM</sup> pneumatic/hydraulic tools.
- ◆ **SUPERIOR THREAD STRENGTH** is provided due to our internal rolled thread manufacturing process.
- ◆ **THREADS GAUGE** before and after installation due to the increased cross-sectional thickness of the thread area. Thread dilation is prevented.
- ◆ **INVENTORY REDUCTION** is possible because of the A-H Series' wide grip range capacity. It is 2.5 times greater than conventional rivet nuts.
- ◆ **SUPERIOR CORROSION RESISTANCE** is provided by our standard zinc/yellow dichromate finish (96 hours. salt spray to white corrosion). For exceptional corrosion protection we offer a tin/zinc alloy finish.
- ◆ **AVAILABLE IN STEEL.**

## ADDITIONAL DESIGN TYPES

### CLOSED END

Thread area is enclosed eliminating leakage past the threads from either side of the application. See page 15.



### SEALED HEAD

A PVC foam seal is bonded to the underside of the head and when installed provides a weathertight seal. Also available in the closed end version. See page 19 for important grip information.

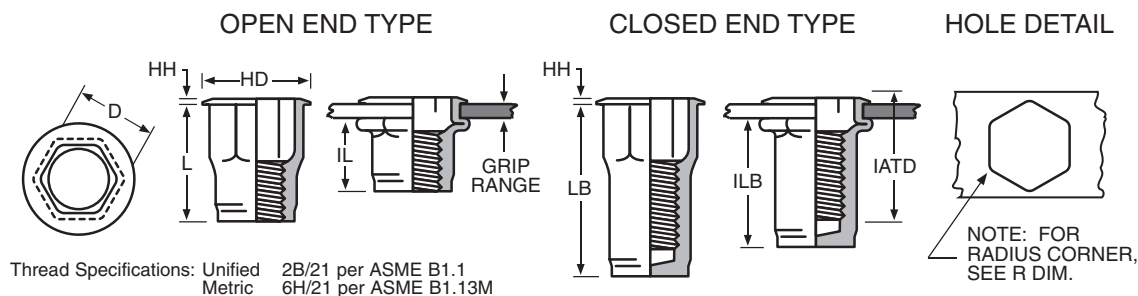


### STAINLESS STEEL

Stainless Steel Hex body insert available in specific thread ranges only. Please contact your AVK Sales Representative for more information.



# UNIFIED (INCH) AND METRIC THREAD SIZES



| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE (ACROSS FLATS)<br>+.004 -.000 | HD<br>±.010<br>±.025* | HH<br>±.003 | L<br>±.015 | D<br>MAX. | IL<br>MAX. | LB<br>±.015 | ILB<br>MAX. | IATD**<br>MAX. | R<br>MAX. |
|-------------|-----------------|------------|---------------|-----------------------------------------|-----------------------|-------------|------------|-----------|------------|-------------|-------------|----------------|-----------|
| 6-32 UNC    | 632             | .020-.080  | 80            | .250                                    | .375                  | .027        | .385       | .249      | .295       | .740        | .640        | .575           | .015      |
| 6-32 UNC    | 632             | .080-.130  | 130           | .250                                    | .375                  | .027        | .435       | .249      | .295       | .740        | .580        | .640           | .015      |
| 8-32 UNC    | 832             | .020-.080  | 80            | .250                                    | .375                  | .027        | .385       | .249      | .295       | .740        | .640        | .575           | .015      |
| 8-32 UNC    | 832             | .080-.130  | 130           | .250                                    | .375                  | .027        | .435       | .249      | .295       | .740        | .580        | .640           | .015      |
| 10-24 UNC   | 1024            | .020-.130  | 130           | .281                                    | .390                  | .027        | .435       | .280      | .275       | 1.030       | .845        | .695           | .020      |
| 10-24 UNC   | 1024            | .130-.225  | 225           | .281                                    | .390                  | .027        | .535       | .280      | .275       | 1.030       | .735        | .805           | .020      |
| 10-32 UNF   | 1032            | .020-.130  | 130           | .281                                    | .390                  | .027        | .435       | .280      | .275       | 1.030       | .845        | .695           | .020      |
| 10-32 UNF   | 1032            | .130-.225  | 225           | .281                                    | .390                  | .027        | .535       | .280      | .275       | 1.030       | .735        | .805           | .020      |
| 1/4-20 UNC  | 420             | .027-.165  | 165           | .375                                    | .510                  | .030        | .585       | .374      | .400       | 1.190       | 1.015       | .945           | .040      |
| 1/4-20 UNC  | 420             | .165-.260  | 260           | .375                                    | .510                  | .030        | .685       | .374      | .400       | 1.190       | .915        | 1.085          | .040      |
| 5/16-18 UNC | 518             | .027-.150  | 150           | .500                                    | .655*                 | .035        | .685       | .499      | .530       | 1.445       | 1.235       | 1.045          | .040      |
| 5/16-18 UNC | 518             | .150-.312  | 312           | .500                                    | .655*                 | .035        | .845       | .499      | .515       | 1.445       | 1.220       | 1.170          | .040      |
| 3/8-16 UNC  | 616             | .027-.150  | 150           | .500                                    | .655*                 | .035        | .685       | .499      | .530       | 1.445       | 1.235       | 1.045          | .040      |
| 3/8-16 UNC  | 616             | .150-.312  | 312           | .500                                    | .655*                 | .035        | .845       | .499      | .515       | 1.445       | 1.220       | 1.170          | .040      |

| THREAD SIZE   | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE (ACROSS FLATS)<br>+.010 -.000 | HD<br>±.025<br>±.064* | HH<br>±.008 | L<br>±.038 | D<br>MAX. | IL<br>MAX. | LB<br>±.038 | ILB<br>MAX. | IATD**<br>MAX. | R<br>MAX. |
|---------------|-----------------|------------|---------------|-----------------------------------------|-----------------------|-------------|------------|-----------|------------|-------------|-------------|----------------|-----------|
| M4 x 0.7 ISO  | 470             | 0.50-2.00  | 2.0           | 6.35                                    | 9.53                  | 0.68        | 9.78       | 6.35      | 7.49       | 18.80       | 16.26       | 14.61          | .38       |
| M4 x 0.7 ISO  | 470             | 2.00-3.30  | 3.3           | 6.35                                    | 9.53                  | 0.68        | 11.05      | 6.35      | 7.49       | 18.80       | 14.73       | 16.26          | .38       |
| M5 x 0.8 ISO  | 580             | 0.50-3.30  | 3.3           | 7.14                                    | 9.91                  | 0.68        | 11.05      | 7.10      | 6.99       | 26.16       | 21.46       | 17.65          | .50       |
| M5 x 0.8 ISO  | 580             | 3.30-5.70  | 5.7           | 7.14                                    | 9.91                  | 0.68        | 13.59      | 7.10      | 6.99       | 26.16       | 18.67       | 20.45          | .50       |
| M6 x 1.0 ISO  | 610             | 0.70-4.20  | 4.2           | 9.53                                    | 12.96                 | 0.76        | 14.86      | 9.50      | 10.16      | 30.23       | 25.78       | 24.00          | 1.0       |
| M6 x 1.0 ISO  | 610             | 4.20-6.60  | 6.6           | 9.53                                    | 12.96                 | 0.76        | 17.40      | 9.50      | 10.16      | 30.23       | 23.24       | 27.56          | 1.0       |
| M8 x 1.25 ISO | 8125            | 0.70-3.8   | 3.8           | 12.70                                   | 16.64*                | 0.89        | 17.40      | 12.70     | 13.46      | 36.70       | 31.37       | 26.54          | 1.0       |
| M8 x 1.25 ISO | 8125            | 3.8-7.90   | 7.9           | 12.70                                   | 16.64*                | 0.89        | 21.46      | 12.70     | 13.08      | 36.70       | 30.99       | 29.72          | 1.0       |
| M10 x 1.5 ISO | 1015            | 0.70-3.8   | 3.8           | 12.70                                   | 16.64*                | 0.89        | 17.40      | 12.70     | 13.46      | 36.70       | 31.37       | 26.54          | 1.0       |
| M10 x 1.5 ISO | 1015            | 3.8-7.90   | 7.9           | 12.70                                   | 16.64*                | 0.89        | 21.46      | 12.70     | 13.08      | 36.70       | 30.99       | 29.72          | 1.0       |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Available in additional materials and sizes. Contact AVK for details.  
\*\*Dimensions in minimum grip condition.

## PART NUMBERING SYSTEM

| AH             |                 | ( )      |                                                                                                      | ( )                |               | - ( ) - ( )                 |                    | ( )                |  |
|----------------|-----------------|----------|------------------------------------------------------------------------------------------------------|--------------------|---------------|-----------------------------|--------------------|--------------------|--|
| Product Series |                 | MATERIAL |                                                                                                      | FINISH             |               | Thread Call Out             |                    | TYPE               |  |
| CALL OUT       | Grade           | CALL OUT | Specification                                                                                        | Thread Call Out    | Grip Call Out | CALL OUT                    | Type               |                    |  |
| S              | Steel 1010/1008 | 4        | STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING |                    |               | BLANK                       | Open End           |                    |  |
|                |                 |          |                                                                                                      | * BY SPECIAL ORDER |               | SAMPLE NUMBER: AHS4-420-165 |                    | * BY SPECIAL ORDER |  |
| 9              |                 | 9        | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING                      |                    |               | B                           | Closed End         |                    |  |
|                |                 |          |                                                                                                      |                    |               | S                           | Sealed, See pg. 19 |                    |  |

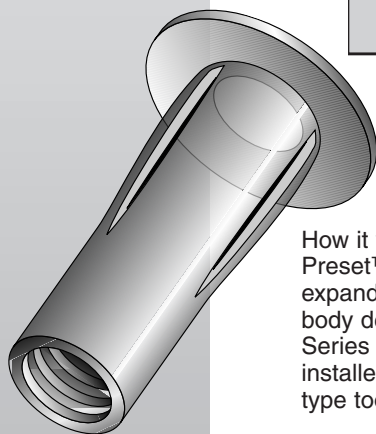
\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 28 and 30

## A-R SERIES<sup>TM</sup> INSERT PROFILE

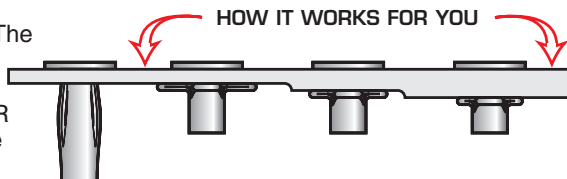
The **A-R Series**<sup>TM</sup> threaded insert has been designed for use in plastics and thin gauge sheet metal applications where increased pull-out resistance is required.

The A-R Series features a **PreSet**<sup>TM</sup> slotted body design that when installed folds into four segments gripping the backside of the parent material. This design feature allows the A-R Series to be installed into single, variable or multiple thickness materials using AVK's ARO torque-stall pneumatic tools or AVK's SPP<sup>TM</sup> pneumatic/hydraulic tools.



### A-R Series<sup>TM</sup> PreSet<sup>TM</sup> Design

How it works for you: The PreSet<sup>TM</sup> slightly expanded slotted body design of the A-R Series enables it to be installed using torque type tools.



HOW IT WORKS FOR YOU

Hand or pneumatic torque tools will install the A-R Series in single, variable or multiple thickness materials.

## DESIGN BENEFITS

- ◆ **INSTALLS USING TORQUE** stall type tooling due to the slightly expanded slotted body design. This is important when working with plastics that vary in thickness. No adjustment of the tool is necessary when installing the part into variable thickness materials.
- ◆ **INSTALLS USING HAND WRENCHED TORQUE** type tools. Ideal for use in kits and consumer installation applications
- ◆ **PROVIDES EXCEPTIONAL** pull out resistance in soft plastics or thin sheet metal applications even if holes are hand drilled and oversized.
- ◆ **SUPERIOR CORROSION RESISTANCE** because all surfaces of the slotted body are plated. Standard plating is zinc/yellow dichromate finish (96 hours to white corroding). For exceptional corrosion protection we offer a tin/zinc alloy finish.
- ◆ Superior thread strength due to AVK's internal roll threading process.
- ◆ **AVAILABLE IN STEEL.**

## ADDITIONAL DESIGN TYPES

**Square Headed A-R Series:** The head is square in shape and can be placed into an embossed or molded square recess in the parent material for exceptional spin out resistance.

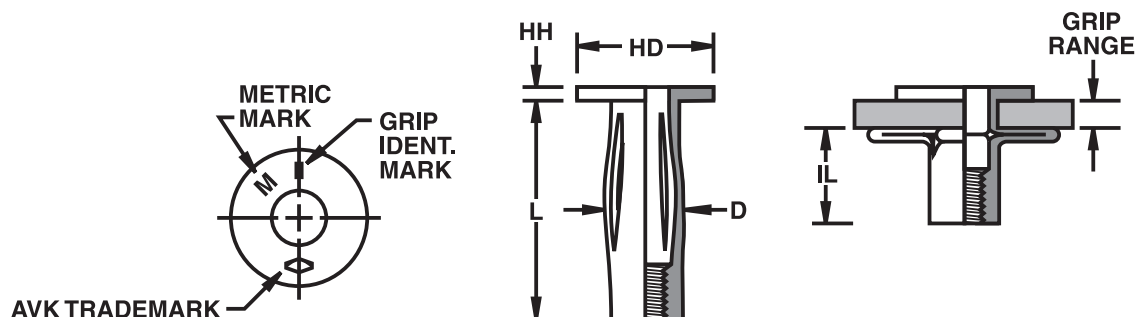
## AIR TOOL SELECTION SPP<sup>TM</sup> TOOL

The A-R has been designed to install with either the SPP Tool or the ARO type tool. The SPP Tool will install the A-R per the suggested grip ranges shown on page 17. See page 33 for SPP tool information.

## AIR TOOL SELECTION ARO<sup>TM</sup> TOOL

The ARO pneumatic tool shown on pages 28 and 29 will install the A-R Series threaded insert. It will affect the published grip range of the part based on the tools' RPM and the density of the parent material. See the chart on page 17 for grip range information. AVK suggests trial installations in the actual application before specifying the optimum ARO tool.

# UNIFIED (INCH) AND METRIC THREAD SIZES



**THREAD CLASS:** Unified 2B/21 per ASME B1.1  
Metric 6H/21 per ASME B1.13M

**MATERIAL:** S=Steel C1010/1008

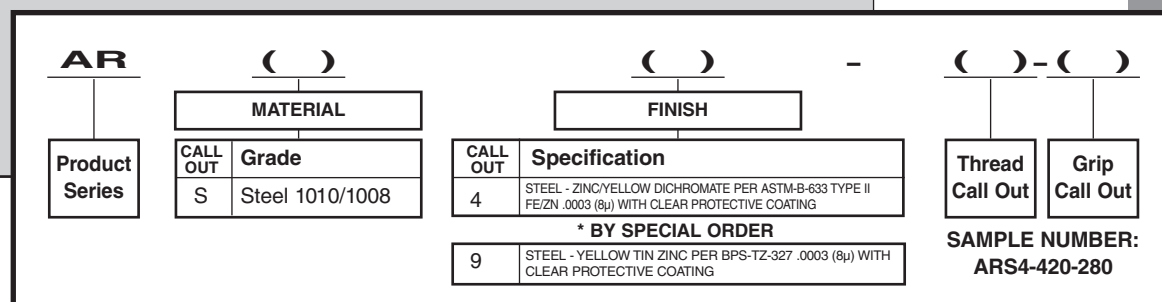
**PLATING:** 4=Yellow Zinc Plate per ASTM-B633TYII, FE/ZN 8, .0003 (8μ) with clear protective coating

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE    | HD           | HH*          | L              | D            | IL MAX | GRIP ID MARK |
|-------------|-----------------|------------|---------------|--------------|--------------|--------------|----------------|--------------|--------|--------------|
| 1/4-20 UNC  | 420             | .020-.280  | .280          | .390<br>.396 | .645<br>.610 | .063<br>.053 | 1.015<br>.985  | .382<br>.368 | .520   | Blank        |
| 1/4-20 UNC  | 420             | .280-.500  | .500          | .390<br>.396 | .645<br>.610 | .063<br>.053 | 1.249<br>1.219 | .382<br>.368 | .520   | 1 Rad        |
| 5/16-18 UNC | 518             | .020-.280  | .280          | .500<br>.506 | .770<br>.740 | .067<br>.057 | 1.156<br>1.126 | .495<br>.490 | .775   | Blank        |

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE      | HD             | HH*          | L              | D              | IL MAX | GRIP ID MARK |
|-------------|-----------------|------------|---------------|----------------|----------------|--------------|----------------|----------------|--------|--------------|
| M6x1,0 ISO  | 610             | 0,5-7,1    | 7.1           | 10,00<br>10,15 | 16,38<br>15,49 | 1,60<br>1,35 | 25,78<br>25,02 | 9,8<br>9,35    | 13,21  | Blank        |
| M6x1,0 ISO  | 610             | 7,1-12,7   | 12.7          | 10,00<br>10,15 | 16,38<br>15,49 | 1,60<br>1,35 | 31,72<br>30,96 | 9,8<br>9,35    | 13,21  | 1 Rad        |
| M8x1,25 ISO | 8125            | 0,5-7,1    | 7.1           | 12,70<br>12,85 | 19,56<br>18,80 | 1,70<br>1,45 | 29,63<br>28,60 | 12,57<br>12,47 | 19,69  | Blank        |

**NOTE 1:** Grip range stated in the dimensional chart above can be achieved using pull type installation tools and may be variable based on hole size and parent material density. AVK recommends trial installations to determine actual grip range in the application. **NOTE 2:** Grip ranges will be less than stated above when using torque type installation tools. Grip range will be affected by the tool RPM speed, stall torque, hole size and parent material density. AVK recommends trial installations to determine actual grip. See page 33 for torque tool selection guidelines.

## PART NUMBERING SYSTEM

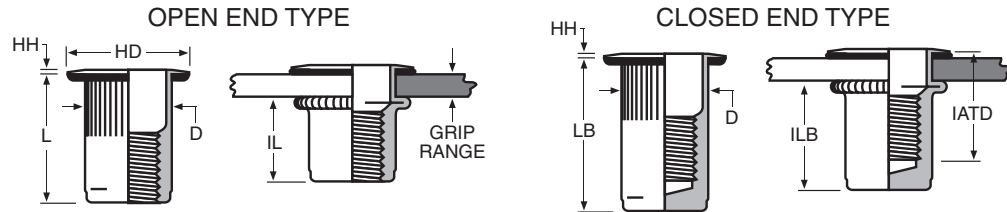


\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 33 and 35

# A-L SERIES SEALED HEAD INSERT PROFILE

The **A-L Series Sealed Head Insert** provides all the features of the standard A-L Series Insert plus the addition of a PVC foam seal that is bonded to the underside head of the insert. This feature provides a weather resistant seal that will withstand 50 PSI - 3.4 BARS of pressure.



## UNIFIED (INCH) AND METRIC THREAD

| THREAD SIZE  | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+.006 -.000<br>(+.015 -.000) | HD<br>±.010<br>(±.025)<br>(±.064) | HH<br>±.003<br>(±.008) | L<br>±.015<br>(±.038) | D<br>MAX. | IL<br>MAX. | LB<br>±.015<br>(±.038) | ILB<br>MAX. | IATD*<br>MAX. |
|--------------|-----------------|------------|---------------|-------------------------------------------|-----------------------------------|------------------------|-----------------------|-----------|------------|------------------------|-------------|---------------|
| 6-32 UNC     | 632             | .020-.060  | 80            | 17/64 (.2656)                             | .390                              | .030                   | .420                  | .265      | .305       | .740                   | .640        | .610          |
| 6-32 UNC     | 632             | .060-.100  | 130           | 17/64 (.2656)                             | .390                              | .030                   | .470                  | .265      | .305       | .740                   | .580        | .670          |
| 8-32 UNC     | 832             | .020-.060  | 80            | 17/64 (.2656)                             | .390                              | .030                   | .420                  | .265      | .305       | .740                   | .640        | .610          |
| 8-32 UNC     | 832             | .060-.100  | 130           | 17/64 (.2656)                             | .390                              | .030                   | .470                  | .265      | .305       | .740                   | .580        | .670          |
| 10-24 UNC    | 1024            | .020-.100  | 130           | 19/64 (.2969)                             | .415                              | .030                   | .475                  | .296      | .315       | .990                   | .845        | .730          |
| 10-24 UNC    | 1024            | .100-.175  | 225           | 19/64 (.2969)                             | .415                              | .030                   | .585                  | .296      | .315       | .990                   | .735        | .840          |
| 10-32 UNF    | 1032            | .020-.100  | 130           | 19/64 (.2969)                             | .415                              | .030                   | .475                  | .296      | .315       | .990                   | .845        | .730          |
| 10-32 UNF    | 1032            | .100-.175  | 225           | 19/64 (.2969)                             | .415                              | .030                   | .585                  | .296      | .315       | .990                   | .735        | .840          |
| 1/4-20 UNC   | 420             | .027-.125  | 165           | 25/64 (.3906)                             | .500                              | .030                   | .580                  | .390      | .380       | 1.190                  | 1.005       | .895          |
| 1/4-20 UNC   | 420             | .125-.195  | 260           | 25/64 (.3906)                             | .500                              | .030                   | .680                  | .390      | .380       | 1.190                  | .905        | 1.035         |
| 5/16-18 UNC  | 518             | .027-.115  | 150           | 17/32 (.5312)                             | .685*                             | .035                   | .690                  | .530      | .470       | 1.390                  | 1.175       | .995          |
| 5/16-18 UNC  | 518             | .115-.250  | 312           | 17/32 (.5312)                             | .685*                             | .035                   | .805                  | .530      | .425       | 1.390                  | 1.025       | 1.120         |
| 3/8-16 UNC   | 616             | .027-.115  | 150           | 17/32 (.5312)                             | .685*                             | .035                   | .690                  | .530      | .470       | 1.390                  | 1.175       | .995          |
| 3/8-16 UNC   | 616             | .115-.250  | 312           | 17/32 (.5312)                             | .685*                             | .035                   | .805                  | .530      | .425       | 1.390                  | 1.025       | 1.120         |
| 1/2-13 UNC   | 813             | .063-.150  | 200           | 11/16 (.6875)                             | .865*                             | .047                   | 1.150                 | .685      | .850       | 2.365                  | 2.070       | 1.505         |
| 1/2-13 UNC   | 813             | .150-.280  | 350           | 11/16 (.6875)                             | .865*                             | .047                   | 1.300                 | .685      | .850       | 2.365                  | 1.920       | 1.505         |
| 1/2-13 UNC   | 813             | .280-.400  | 500           | 11/16 (.6875)                             | .865*                             | .047                   | 1.450                 | .685      | .860       | 3.365                  | 1.770       | 1.505         |
| M4x0.7 ISO   | 470             | 0.50-1.52  | 2.0           | 6.75                                      | 9.91                              | 0.76                   | 10.67                 | 6.73      | 7.75       | 18.80                  | 16.26       | 15.49         |
| M4x0.7 ISO   | 470             | 1.52-2.54  | 3.3           | 6.75                                      | 9.91                              | 0.76                   | 11.94                 | 6.73      | 7.75       | 18.80                  | 14.73       | 17.02         |
| M5x0.8 ISO   | 580             | 0.50-2.54  | 3.3           | 7.60                                      | 10.54                             | 0.76                   | 12.07                 | 7.52      | 8.00       | 25.15                  | 21.46       | 18.54         |
| M5x0.8 ISO   | 580             | 2.54-4.45  | 5.7           | 7.60                                      | 10.54                             | 0.76                   | 14.86                 | 7.52      | 8.00       | 25.15                  | 18.67       | 21.34         |
| M6x1.0 ISO   | 610             | 0.70-3.17  | 4.2           | 10.00                                     | 12.70                             | 0.76                   | 14.73                 | 9.91      | 9.65       | 30.23                  | 25.53       | 22.73         |
| M6x1.0 ISO   | 610             | 3.17-4.95  | 6.6           | 10.00                                     | 12.70                             | 0.76                   | 17.27                 | 9.91      | 9.65       | 30.23                  | 22.99       | 26.29         |
| M8x1.25 ISO  | 8125            | 0.70-2.92  | 3.8           | 13.50                                     | 17.40*                            | 0.89                   | 17.53                 | 13.46     | 11.94      | 35.31                  | 29.85       | 25.27         |
| M8x1.25 ISO  | 8125            | 2.92-6.35  | 7.9           | 13.50                                     | 17.40*                            | 0.89                   | 20.45                 | 13.46     | 10.80      | 35.31                  | 26.04       | 28.45         |
| M10x1.5 ISO  | 1015            | 0.70-2.92  | 3.8           | 13.50                                     | 17.40*                            | 0.89                   | 17.53                 | 13.46     | 11.94      | 35.31                  | 29.85       | 25.27         |
| M10x1.5 ISO  | 1015            | 2.92-6.35  | 7.9           | 13.50                                     | 17.40*                            | 0.89                   | 20.45                 | 13.46     | 10.80      | 35.31                  | 26.04       | 28.45         |
| M12x1.75 ISO | 12175           | 1.60-3.81  | 5.1           | 17.45                                     | 21.97*                            | 1.19                   | 29.21                 | 17.4      | 21.59      | 60.07                  | 52.58       | 38.23         |
| M12x1.75 ISO | 12175           | 3.81-7.11  | 8.9           | 17.45                                     | 21.97*                            | 1.19                   | 33.02                 | 17.4      | 21.59      | 60.07                  | 48.77       | 38.23         |
| M12x1.75 ISO | 12175           | 7.11-10.16 | 12.7          | 17.45                                     | 21.97*                            | 1.19                   | 36.83                 | 17.4      | 21.84      | 60.07                  | 44.96       | 38.23         |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Additional UNF fine thread materials are available. Contact AVK for details. **NOTE 3:** The A-L Series shown on this page incorporates an underhead seal which reduces the standard grip range of the part based on the seal thickness. The grip call out remains based on the standard part for part numbering simplicity. **NOTE 4:** The PVC foam seal is not recommended for use with petroleum based liquids. \* Dimensions in minimum grip condition.

## PART NUMBERING SYSTEM

| AL                        |          | ( )                                                                             |          | ( )                                                                                                  |                       | - ( ) - ( )   |                           | ( )      |  |
|---------------------------|----------|---------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------|-----------------------|---------------|---------------------------|----------|--|
| MATERIAL                  |          | FINISH                                                                          |          | TYPE                                                                                                 |                       |               |                           |          |  |
| Product Series            | CALL OUT | Grade                                                                           | CALL OUT | Specification                                                                                        | Thread Call Out       | Grip Call Out | CALL OUT                  | Type     |  |
|                           | S        | Steel 1010/1008                                                                 | 4        | STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING |                       |               | BLANK                     | Open End |  |
| <b>* BY SPECIAL ORDER</b> |          |                                                                                 |          |                                                                                                      | <b>SAMPLE NUMBER:</b> |               | <b>* BY SPECIAL ORDER</b> |          |  |
|                           | 9        | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING |          |                                                                                                      | ALS4-420-165S         |               | S                         | Sealed   |  |

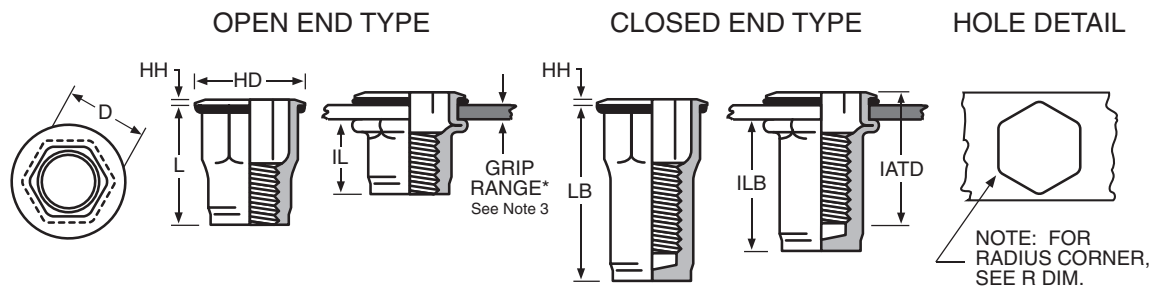
\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 28 and 30

Thread Specifications: Unified 2B/21 per ASME B1.1 Metric 6H/21 per ASME B1.13M

# A-H SERIES SEALED HEAD INSERT PROFILE

The **A-H Series Sealed Head Insert** provides all the features of the standard A-H Series Insert plus the addition of a PVC foam seal that is bonded to the underside head of the insert. This feature provides a weather resistant seal that will withstand 50 PSI–3.4 bars of pressure.



## UNIFIED (INCH) AND METRIC THREAD SIZES

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE (ACROSS FLATS)<br>+.004 -.000<br>(+.010 -.000) | HD<br>±.010<br>±.025*<br>(±.025)<br>(±.064) | HH<br>±.003<br>(±0.08) | L<br>±.015<br>(±0.38) | D<br>MAX. | IL<br>MAX. | LB<br>±.015<br>(±0.38) | ILB<br>MAX. | IATD**<br>MAX. | R<br>MAX. |
|-------------|-----------------|------------|---------------|----------------------------------------------------------|---------------------------------------------|------------------------|-----------------------|-----------|------------|------------------------|-------------|----------------|-----------|
| 6-32 UNC    | 632             | .020-.060  | 80            | .250                                                     | .375                                        | .027                   | .385                  | .249      | .295       | .740                   | .640        | .575           | .015      |
| 6-32 UNC    | 632             | .060-.100  | 130           | .250                                                     | .375                                        | .027                   | .435                  | .249      | .295       | .740                   | .580        | .640           | .015      |
| 8-32 UNC    | 832             | .020-.060  | 80            | .250                                                     | .375                                        | .027                   | .385                  | .249      | .295       | .740                   | .640        | .575           | .015      |
| 8-32 UNC    | 832             | .060-.100  | 130           | .250                                                     | .375                                        | .027                   | .435                  | .249      | .295       | .740                   | .580        | .640           | .015      |
| 10-24 UNC   | 1024            | .020-.100  | 130           | .281                                                     | .390                                        | .027                   | .435                  | .280      | .275       | 1.030                  | .845        | .695           | .020      |
| 10-24 UNC   | 1024            | .100-.175  | 225           | .281                                                     | .390                                        | .027                   | .535                  | .280      | .275       | 1.030                  | .735        | .805           | .020      |
| 10-32 UNF   | 1032            | .020-.100  | 130           | .281                                                     | .390                                        | .027                   | .435                  | .280      | .275       | 1.030                  | .845        | .695           | .020      |
| 10-32 UNF   | 1032            | .100-.175  | 225           | .281                                                     | .390                                        | .027                   | .535                  | .280      | .275       | 1.030                  | .735        | .805           | .020      |
| 1/4-20 UNC  | 420             | .027-.125  | 165           | .375                                                     | .510                                        | .030                   | .585                  | .374      | .400       | 1.190                  | 1.015       | .945           | .040      |
| 1/4-20 UNC  | 420             | .125-.195  | 260           | .375                                                     | .510                                        | .030                   | .685                  | .374      | .400       | 1.190                  | .915        | 1.085          | .040      |
| 5/16-18 UNC | 518             | .027-.115  | 150           | .500                                                     | .655*                                       | .035                   | .685                  | .499      | .530       | 1.445                  | 1.235       | 1.045          | .040      |
| 5/16-18 UNC | 518             | .115-.250  | 312           | .500                                                     | .655*                                       | .035                   | .845                  | .499      | .515       | 1.445                  | 1.220       | 1.170          | .040      |
| 3/8-16 UNC  | 616             | .027-.115  | 150           | .500                                                     | .655*                                       | .035                   | .685                  | .499      | .530       | 1.445                  | 1.235       | 1.045          | .040      |
| 3/8-16 UNC  | 616             | .115-.250  | 312           | .500                                                     | .655*                                       | .035                   | .845                  | .499      | .515       | 1.445                  | 1.220       | 1.170          | .040      |
| M4x0,7 ISO  | 470             | 0,50-1,52  | 2.0           | 6,35                                                     | 9,53                                        | 0,68                   | 9,78                  | 6,35      | 7,49       | 18,80                  | 16,26       | 14,61          | ,38       |
| M4x0,7 ISO  | 470             | 1,52-2,54  | 3.3           | 6,35                                                     | 9,53                                        | 0,68                   | 11,05                 | 6,35      | 7,49       | 18,80                  | 14,73       | 16,26          | ,38       |
| M5x0,8 ISO  | 580             | 0,50-2,54  | 3.3           | 7,14                                                     | 9,91                                        | 0,68                   | 11,05                 | 7,10      | 6,99       | 26,16                  | 21,46       | 17,65          | ,50       |
| M5x0,8 ISO  | 580             | 2,54-4,45  | 5.7           | 7,14                                                     | 9,91                                        | 0,68                   | 13,59                 | 7,10      | 6,99       | 26,16                  | 18,67       | 20,45          | ,50       |
| M6x1,0 ISO  | 610             | 0,70-3,17  | 4.2           | 9,53                                                     | 12,96                                       | 0,76                   | 14,86                 | 9,50      | 10,16      | 30,23                  | 25,78       | 24,00          | 1,0       |
| M6x1,0 ISO  | 610             | 3,17-4,95  | 6.6           | 9,53                                                     | 12,96                                       | 0,76                   | 17,40                 | 9,50      | 10,16      | 30,23                  | 23,24       | 27,56          | 1,0       |
| M8x1,25 ISO | 8125            | 0,70-2,92  | 3.8           | 12,70                                                    | 16,64*                                      | 0,89                   | 17,40                 | 12,70     | 13,46      | 36,70                  | 31,37       | 26,54          | 1,0       |
| M8x1,25 ISO | 8125            | 2,92-6,35  | 7.9           | 12,70                                                    | 16,64*                                      | 0,89                   | 21,46                 | 12,70     | 13,08      | 36,70                  | 30,99       | 29,72          | 1,0       |
| M10x1,5 ISO | 1015            | 0,70-2,92  | 3.8           | 12,70                                                    | 16,64*                                      | 0,89                   | 17,40                 | 12,70     | 13,46      | 36,70                  | 31,37       | 26,54          | 1,0       |
| M10x1,5 ISO | 1015            | 2,92-6,35  | 7.9           | 12,70                                                    | 16,64*                                      | 0,89                   | 21,46                 | 12,70     | 13,08      | 36,70                  | 30,99       | 29,72          | 1,0       |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** UNF fine threads are available. Contact AVK for details. **NOTE 3:** The A-H Series shown on this page incorporates an underhead seal which reduces the standard grip range of the part based on the seal thickness. The grip call out remains based on the standard part for part numbering simplicity. **NOTE 4:** The PVC foam seal is not recommended for use with petroleum based liquids. \*Dimensions in minimum grip condition.

## PART NUMBERING SYSTEM

| AH                                                                                |               | ( )                      |               | ( )                                                                                                                   |                                 | - ( ) - ( )   |                    | ( )              |  |
|-----------------------------------------------------------------------------------|---------------|--------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|--------------------|------------------|--|
| MATERIAL                                                                          |               | FINISH                   |               | Thread Call Out                                                                                                       |                                 | Grip Call Out |                    | TYPE             |  |
| Product Series                                                                    | CALL OUT<br>S | Grade<br>Steel 1010/1008 | CALL OUT<br>4 | Specification<br>STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING | Thread Call Out                 | Grip Call Out | CALL OUT<br>BLANK  | Type<br>Open End |  |
| * BY SPECIAL ORDER                                                                |               |                          |               |                                                                                                                       | SAMPLE NUMBER:<br>AHS4-420-165S |               | * BY SPECIAL ORDER |                  |  |
| 9 STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING |               |                          |               |                                                                                                                       |                                 |               | S Sealed           |                  |  |

\* Special order items are subject to minimum order requirements. Contact AVK for details.

**For air tool selection see pages 28 and 30**

Thread Specifications: Unified 2B/21 per ASME B1.1 Metric 6H/21 per ASME B1.13M

## A-S SERIES STUD PROFILE

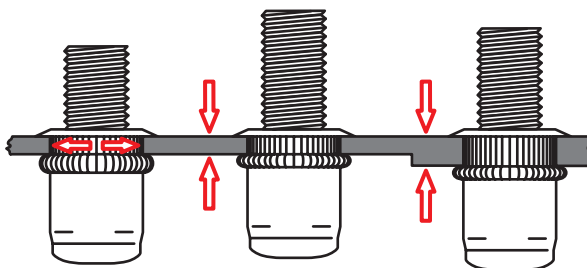
The **A-S Series Stud** offers a unique design advantage in that once installed, a threaded stud is left protruding from the workpiece. Component parts can be located on the stud until final assembly is accomplished with a mating nut. The A-S series is an ideal alternative to clinch or weld studs. The A-S Series is designed to be used with Grade 5 or Metric Class 8.8/9.8 non thread locking type nuts.

The A-S Series Stud is installed using AVK's ARO brand pneumatic tools or AVK's SPP<sup>TM</sup> pneumatic/hydraulic tools. These tools can be located at any position on your assembly line. The A-S Series Stud can be installed either before or after finish.

## SPINWALL TECHNOLOGY<sup>TM</sup>

### HOW HOLE FILL WORKS FOR YOU

As the A-S Series is installed, the knurled body expands 360° **FILLING THE HOLE**. This feature provides exceptional torque strength and vibration resistance.



The installation tool then continues to install the stud forming a backside flange even in variable thickness material—**WITHOUT ADJUSTMENT**.

## DESIGN BENEFITS

- ◆ **PROTRUDING STUD** allows component parts to be located on the stud until final assembly is accomplished with a mating nut.
- ◆ **EXCEPTIONAL TORQUE STRENGTH** is achieved as the stud's knurled body expands **FILLING THE HOLE**.
- ◆ **QUALITY INSTALLATIONS** even in variable thickness materials are assured by AVK's spin/spin ARO pneumatic or AVK's SPP pneumatic/hydraulic tools.
- ◆ **ELIMINATE PAINT MASKING** procedures as required with weld or clinch studs. The A-S Series Stud can be installed after painting.
- ◆ **SUPERIOR CORROSION RESISTANCE** is provided by our standard zinc/yellow dichromate finish (96 hours. salt spray to white corrosion). Alternative finishes are available.
- ◆ **AVAILABLE** in Steel 1010/1008 shell - Steel 1038 threaded stud.

## ADDITIONAL DESIGN TYPES

### SEALED HEAD

A PVC foam seal is bonded to the underside of the head and when installed provides a weathertight seal. Note that the addition of a seal reduces the parts grip range. Contact AVK for a sales drawing.

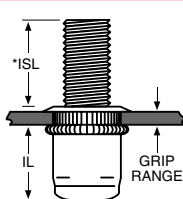
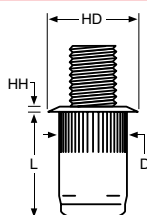


### WEDGE HEAD

The addition of wedges under the head provides even greater torque capability, especially in soft or thin materials, and is excellent for electrical grounding applications. Contact AVK for a sales drawing.



# UNIFIED (INCH) AND METRIC THREAD SIZES



\* **NOTE:** The ISL Dimension shown below is the height of the installed stud at max grip. The height of the stud will increase if it is installed into thinner material. To calculate actual ISL use this formula: **Max grip – actual grip + ISL = Actual ISL**

Thread Specifications: Unified 2A/21 per ASME B1.1  
Metric 6G/21 per ASME B1.13M

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | STUD LENGTHS    |                 |                 | HOLE SIZE<br>+.006<br>-.000 | HD<br>±.010<br>±.025* | HH<br>±.003 | L<br>±.020 | D<br>MAX. | IL<br>MAX. |
|-------------|-----------------|------------|---------------|-----------------|-----------------|-----------------|-----------------------------|-----------------------|-------------|------------|-----------|------------|
|             |                 |            |               | ISL<br>CALL OUT | ISL<br>CALL OUT | ISL<br>CALL OUT |                             |                       |             |            |           |            |
| 6-32 UNC    | 632             | .020-.080  | 80            | .500            | .625            | .750            | 17/64 (.2656)               | .390                  | .030        | .485       | .265      | .360       |
| 6-32 UNC    | 632             | .080-.130  | 130           | .450            | .575            | .700            | 17/64 (.2656)               | .390                  | .030        | .535       | .265      | .360       |
| 8-32 UNC    | 832             | .020-.080  | 80            | .500            | .625            | .750            | 17/64 (.2656)               | .390                  | .030        | .485       | .265      | .360       |
| 8-32 UNC    | 832             | .080-.130  | 130           | .450            | .575            | .700            | 17/64 (.2656)               | .390                  | .030        | .535       | .265      | .360       |
| 10-24 UNC   | 1024            | .020-.130  | 130           | .500            | .625            | .750            | 19/64 (.2969)               | .415                  | .030        | .545       | .296      | .380       |
| 10-24 UNC   | 1024            | .130-.225  | 225           | .405            | .530            | .655            | 19/64 (.2969)               | .415                  | .030        | .655       | .296      | .380       |
| 10-32 UNF   | 1032            | .020-.130  | 130           | .500            | .625            | .750            | 19/64 (.2969)               | .415                  | .030        | .545       | .296      | .380       |
| 10-32 UNF   | 1032            | .130-.225  | 225           | .405            | .530            | .655            | 19/64 (.2969)               | .415                  | .030        | .655       | .296      | .380       |
| 1/4-20 UNC  | 420             | .027-.165  | 165           | .625            | .8125           | 1.000           | 25/64 (.3906)               | .500                  | .030        | .670       | .390      | .465       |
| 1/4-20 UNC  | 420             | .165-.260  | 260           | .625            | .8125           | 1.000           | 25/64 (.3906)               | .500                  | .030        | .770       | .390      | .465       |
| 5/16-18 UNC | 518             | .027-.150  | 150           | .625            | .875            | 1.125           | 17/32 (.5312)               | .685*                 | .035        | .810       | .530      | .600       |
| 5/16-18 UNC | 518             | .150-.312  | 312           | .463            | .713            | .963            | 17/32 (.5312)               | .685*                 | .035        | .925       | .530      | .555       |
| 3/8-16 UNC  | 616             | .027-.150  | 150           | .750            | 1.000           | 1.250           | 17/32 (.5312)               | .685*                 | .035        | .810       | .530      | .600       |
| 3/8-16 UNC  | 616             | .150-.312  | 312           | .588            | .838            | 1.088           | 17/32 (.5312)               | .685*                 | .035        | .925       | .530      | .535       |

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | STUD LENGTHS    |                 |                 | HOLE SIZE<br>+.015<br>-.000 | HD<br>±0.25<br>±0.64* | HH<br>±0.08 | L<br>±0.50 | D<br>MAX. | IL<br>MAX. |
|-------------|-----------------|------------|---------------|-----------------|-----------------|-----------------|-----------------------------|-----------------------|-------------|------------|-----------|------------|
|             |                 |            |               | ISL<br>CALL OUT | ISL<br>CALL OUT | ISL<br>CALL OUT |                             |                       |             |            |           |            |
| M4x0,7 ISO  | 470             | 0,50-2,00  | 2.0           | 12,0            | 15,0            | 20,0            | 6,75                        | 9,91                  | 0,76        | 12,32      | 6,73      | 9,15       |
| M4x0,7 ISO  | 470             | 2,00-3,30  | 3.3           | 10,7            | 13,7            | 18,7            | 6,75                        | 9,91                  | 0,76        | 13,59      | 6,73      | 9,15       |
| M5x0,8 ISO  | 580             | 0,50-3,30  | 3.3           | 12,0            | 15,0            | 20,0            | 7,60                        | 10,54                 | 0,76        | 13,84      | 7,52      | 9,65       |
| M5x0,8 ISO  | 580             | 3,30-5,70  | 5.7           | 9,6             | 12,6            | 17,6            | 7,60                        | 10,54                 | 0,76        | 16,64      | 7,52      | 9,65       |
| M6x1,0 ISO  | 610             | 0,70-4,20  | 4.2           | 15,0            | 20,0            | 25,0            | 10,00                       | 12,70                 | 0,76        | 17,02      | 9,91      | 11,81      |
| M6x1,0 ISO  | 610             | 4,20-6,60  | 6.6           | 12,6            | 17,6            | 22,6            | 10,00                       | 12,70                 | 0,76        | 19,56      | 9,91      | 11,81      |
| M8x1,25 ISO | 8125            | 0,70-3,8   | 3.8           | 16,0            | 22,0            | 28,0            | 13,50                       | 17,40*                | 0,89        | 20,57      | 13,46     | 15,24      |
| M8x1,25 ISO | 8125            | 3,8-7,90   | 7.9           | 11,9            | 17,9            | 23,9            | 13,50                       | 17,40*                | 0,89        | 23,50      | 13,46     | 14,10      |
| M10x1,5 ISO | 1015            | 0,70-3,8   | 3.8           | 20,0            | 25,0            | 30,0            | 13,50                       | 17,40*                | 0,89        | 20,57      | 13,46     | 15,24      |
| M10x1,5 ISO | 1015            | 3,8-7,90   | 7.9           | 15,9            | 20,9            | 25,9            | 13,50                       | 17,40*                | 0,89        | 23,50      | 13,46     | 13,60      |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Additional UNF fine threads are available. Contact AVK for details. \* Dimensions in maximum grip condition.

## PART NUMBERING SYSTEM

|                |          |                       |  |                    |                                                                                                         |                                                                                 |               |                      |          |                               |  |
|----------------|----------|-----------------------|--|--------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------|----------------------|----------|-------------------------------|--|
| AS             |          | ( )                   |  | ( )                |                                                                                                         | -                                                                               |               | ( )-( )-( )          |          | ( )                           |  |
|                |          | MATERIAL              |  | FINISH             |                                                                                                         |                                                                                 |               |                      |          | TYPE                          |  |
| Product Series | CALL OUT | Grade                 |  | CALL OUT           | Specification                                                                                           | Thread Call Out                                                                 | Grip Call Out | Stud Length Call Out | CALL OUT | Type                          |  |
|                | S        | Steel Shell 1010/1008 |  | 4                  | STEEL - ZINC/YELLOW DICHROMATE PER ASTM-B-633 TYPE II<br>FE/ZN .0003 (8µ) WITH CLEAR PROTECTIVE COATING |                                                                                 |               |                      | BLANK    | Standard                      |  |
|                |          | Steel Stud 1038       |  |                    |                                                                                                         |                                                                                 |               |                      |          |                               |  |
|                |          |                       |  | * BY SPECIAL ORDER |                                                                                                         | SAMPLE NUMBER:                                                                  |               | * BY SPECIAL ORDER   |          |                               |  |
|                |          |                       |  | 9                  |                                                                                                         | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 .0003 (8µ) WITH CLEAR PROTECTIVE COATING |               | ASS4-610-4.2-15      |          | S Sealed Head<br>(See Pg. 18) |  |

\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see page 32

## A-T SERIES INSERT PROFILE

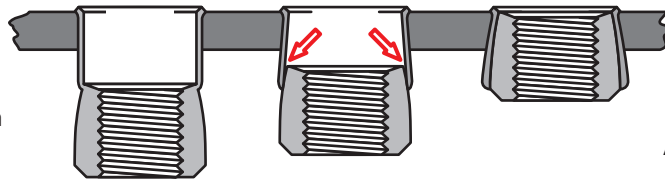
The **A-T Series Insert** is unique in that it can be installed into most any material above .030/.76 mm in thickness. As the A-T Series is installed, the threaded portion is completely swaged 360° into the sleeve portion and the hole. This permits the A-T Series to be used with Grade 8/Metric 12.9 mating screws.

The A-T Series Insert is installed using lightweight, handheld pneumatic ARO tools that can be located at any position in your product's assembly sequence. The A-T Series Insert can be installed either prior to or after finish.



### 360° SWAGING HOW IT WORKS FOR YOU

As the A-T Series Insert is installed, the threaded nut portion is drawn into the upper sleeve portion.



As this occurs, a 360° swaging action takes place anchoring the A-T Series in the parent material.

## DESIGN BENEFITS

- ◆ **REDUCED OVERALL LENGTH** of the installed A-T Series Insert allows it to be used in limited clearance applications.
- ◆ **QUALITY INSTALLATIONS** even in variable thickness materials are assured by our spin/spin torque stall tools (featured on page 29).
- ◆ **INVENTORY REDUCTION** is possible because one A-T Series Insert will work in any thickness material above .030/.76 mm.
- ◆ **INSTALLS INTO MOST ANY MATERIAL** with a thickness over .030/.76 mm.
- ◆ **CAN BE USED WITH GRADE 8/METRIC CLASS 12.9 SCREWS** due to the A-T Series high shear load capability.
- ◆ **AVAILABLE** in Steel from stock. Aluminum, Brass and Series 304 Stainless Steel are available by special order. Contact AVK for details.

## ADDITIONAL DESIGN TYPES

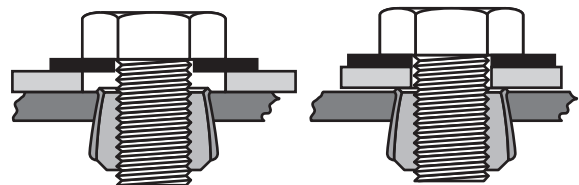
### CLOSED END

Thread area is enclosed eliminating leakage past the threads from either side of the application. See page 23.



## JOINT DESIGN PRACTICES

AVK recommends that the mating part comes in contact with the head of the A-T Series Insert. If a gap or clearance hole exists between the mating part and the A-T Series Insert, the threaded nut portion may rotate or pull through the parent material.



# UNIFIED (INCH) AND METRIC THREAD SIZES



| THREAD SIZE | THREAD CALL OUT | HD $\pm .005$ | L $\pm .015$ | D MAX. | IL MAX. | LB $\pm .015$ | ILB MAX. | IATD** MAX. | HOLE DEPTH MIN. |
|-------------|-----------------|---------------|--------------|--------|---------|---------------|----------|-------------|-----------------|
| 4-40 UNC    | 440             | .211          | .370         | .1875  | .205    | .660          | .495     | .395        | .400            |
| 6-32 UNC    | 632             | .240          | .370         | .2185  | .205    | .675          | .505     | .410        | .400            |
| 8-32 UNC    | 832             | .269          | .370         | .2495  | .205    | .675          | .505     | .410        | .400            |
| 10-24 UNC   | 1024            | .306          | .370         | .2805  | .205    | .685          | .520     | .385        | .400            |
| 10-32 UNF   | 1032            | .306          | .370         | .2805  | .205    | .685          | .520     | .385        | .400            |
| 1/4-20 UNC  | 420             | .400          | .515         | .3745  | .275    | 1.005         | .760     | .615        | .540            |
| 5/16-18 UNC | 518             | .528          | .615         | .4995  | .325    | 1.065         | .770     | .630        | .640            |
| 3/8-16 UNC  | 616             | .588          | .745         | .5615  | .390    | 1.450         | 1.095    | .890        | .770            |
| 1/2-13 UNC  | 813             | .800          | .935         | .7485  | .485    | NA            | NA       | NA          | .960            |

\*\* Dimensions in minimum grip condition. Additional UNF thread sizes available. Contact AVK for details.

| THREAD SIZE  | THREAD CALL OUT | HD $\pm 0.13$ | L $\pm 0.38$ | D MAX. | IL MAX. | LB $\pm 0.38$ | ILB MAX. | IATD** MAX. | HOLE DEPTH MIN. |
|--------------|-----------------|---------------|--------------|--------|---------|---------------|----------|-------------|-----------------|
| M3x0,5 ISO   | 350             | 5,36          | 9,40         | 4,76   | 5,21    | 16,77         | 12,57    | 10,03       | 10,16           |
| M4x0,7 ISO   | 470             | 6,83          | 9,40         | 6,34   | 5,21    | 17,15         | 12,83    | 10,41       | 10,16           |
| M5x0,8 ISO   | 580             | 7,77          | 9,40         | 7,12   | 5,21    | 17,40         | 13,21    | 9,78        | 10,16           |
| M6x1,0 ISO   | 610             | 10,16         | 13,08        | 9,51   | 6,99    | 25,53         | 19,30    | 15,62       | 13,72           |
| M8x1,25 ISO  | 8125            | 13,41         | 15,62        | 12,69  | 8,26    | 27,05         | 19,56    | 16,00       | 16,26           |
| M10x1,5 ISO  | 1015            | 14,94         | 18,92        | 14,26  | 9,91    | 36,83         | 27,81    | 22,61       | 19,56           |
| M12x1,75 ISO | 12175           | 20,32         | 23,75        | 19,01  | 12,32   | NA            | NA       | NA          | 24,38           |

\*\* Dimensions in minimum grip condition.

## HOLE SIZE / MATERIAL THICKNESS CHART

Installation hole size for the A-T Series Insert is determined by the parent material's thickness and density. The thicker the material the larger the hole required to allow full 360° installation swaging. The application should be tested before hole size is specified.

| THREAD SIZE | .030-.090 MAT. THICKNESS |         | .091-.124 MAT. THICKNESS |         | .125-.186 MAT. THICKNESS |         | .187-OVER MAT. THICKNESS |         |
|-------------|--------------------------|---------|--------------------------|---------|--------------------------|---------|--------------------------|---------|
|             | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL |
| 4-40 UNC    | 3/16                     | .1875   | #10                      | .1935   | #10                      | .1935   | #9                       | .1960   |
| 6-32 UNC    | 7/32                     | .2188   | #2                       | .2210   | #1                       | .2280   | #1                       | .2280   |
| 8-32 UNC    | 1/4                      | .2500   | "F"                      | .2570   | 17/64                    | .2656   | 17/64                    | .2656   |
| 10-24 UNC   | 9/32                     | .2812   | "L"                      | .2900   | "L"                      | .2900   | 19/64                    | .2969   |
| 10-32 UNF   | 9/32                     | .2812   | "L"                      | .2900   | "L"                      | .2900   | 19/64                    | .2969   |
| 1/4-20 UNC  | 3/8                      | .3750   | 3/8                      | .3750   | "W"                      | .3860   | 25/64                    | .3906   |
| 5/16-18 UNC | 1/2                      | .5000   | 1/2                      | .5000   | 33/64                    | .5156   | 33/64                    | .5156   |
| 3/8-16 UNC  | 9/16                     | .5625   | 9/16                     | .5625   | 37/64                    | .5781   | 37/64                    | .5781   |
| 1/2-13 UNC  | 3/4                      | .7500   | 49/64                    | .7656   | 25/32                    | .7810   | 51/64                    | .7970   |

| THREAD SIZE  | 0,76-2,29 MAT. THICKNESS |         | 2,31-3,15 MAT. THICKNESS |         | 3,17-4,72 MAT. THICKNESS |         | 4,72-OVER MAT. THICKNESS |         |
|--------------|--------------------------|---------|--------------------------|---------|--------------------------|---------|--------------------------|---------|
|              | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL | DRILL SIZE               | DECIMAL |
| M3x0,5 ISO   | 4,75                     | .1875   | 4,90                     | .1935   | 4,90                     | .1935   | 4,97                     | .1960   |
| M4x0,7 ISO   | 6,35                     | .2500   | 6,52                     | .2570   | 6,74                     | .2656   | 6,74                     | .2656   |
| M5x0,8 ISO   | 7,14                     | .2812   | 7,36                     | .2900   | 7,36                     | .2900   | 7,54                     | .2969   |
| M6x1,0 ISO   | 9,52                     | .3750   | 9,52                     | .3750   | 9,80                     | .3860   | 9,92                     | .3906   |
| M8x1,25 ISO  | 12,70                    | .5000   | 12,70                    | .5000   | 13,09                    | .5156   | 13,09                    | .5156   |
| M10x1,5 ISO  | 14,28                    | .5625   | 14,28                    | .5625   | 14,68                    | .5781   | 14,68                    | .5781   |
| M12x1,75 ISO | 19,05                    | .7500   | 19,44                    | .7656   | 19,83                    | .7810   | 20,24                    | .7970   |

**FINISH:** The standard specified finishes for the A-T Series Insert are cadmium and tin. Alteration to these finishes will reduce performance. **\*THREAD CLASS:** The A-T Series Insert's internal threads are manufactured oversized to compensate for resulting thread portion shrinkage during the installation swaging process. They are not gaugeable prior to or after installation but will be compatible with Class 2A/3A or 6g screws after installation.



All materials for the A-T Series are plated cadmium or tin and look similar. Radial grooves are machined into the part for material identification.

**MATERIAL TYPE IDENTIFICATION GROOVES**  
 NONE .....STEEL 2 .....BRASS  
 1 .....STAINLESS 3 .....ALUMINUM

## PART NUMBERING SYSTEM

| AT                      | ( )                                                                       | ( )                                                                                                                                        | ( )                                                                                                                                                                                               | ( )             |
|-------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Product Series          | TYPE                                                                      | MATERIAL                                                                                                                                   | FINISH                                                                                                                                                                                            | Thread Call Out |
|                         | CALL OUT Design<br>Blank Open End<br>* BY SPECIAL ORDER<br>502 Closed End | CALL OUT Grade<br>S STEEL C1010 OR C1110<br>* BY SPECIAL ORDER<br>A ALUMINUM 6061-T6<br>B BRASS 360 OR 464<br>C STAINLESS STEEL 304 SERIES | CALL OUT Specification<br>2 STEEL, ALUMINUM, STAINLESS AND BRASS - CADMIUM PER QQ-P-416 TYPE I CLASS 3<br>* BY SPECIAL ORDER<br>5 STEEL, ALUMINUM, STAINLESS AND BRASS PER MIL-T-10727B TIN PLATE |                 |
| SAMPLE NUMBER: ATS2-610 |                                                                           |                                                                                                                                            |                                                                                                                                                                                                   |                 |

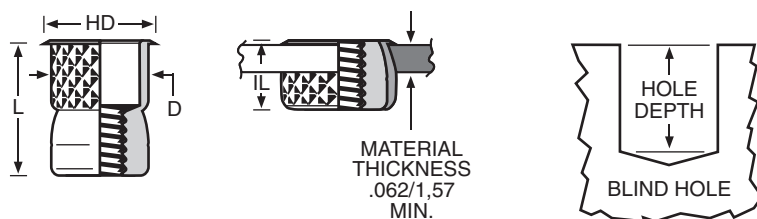
\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see page 31

## A-W SERIES INSERT PROFILE

The **A-W Series Insert** can be installed into most any material softer than itself that is thicker than .062/1,57. The A-W Series provides exceptional shear strength and pull out in fiberglass and plywoods. The brass A-W Series Insert is particularly useful for the fiberglass boat industry.

The A-W Series Insert is installed using lightweight, handheld pneumatic tools that can be located at any position in your product's assembly sequence. The A-W Series Insert can be installed either prior to or after finish.



## UNIFIED (INCH) AND METRIC THREAD

SEE THREAD  
CLASS NOTE \*

| THREAD SIZE | THREAD CALL OUT | HOLE SIZE<br>+.005 -.000<br>(+0,13 -0,00) | HD<br>±.005<br>(±0,13) | L<br>±.015<br>(±0,38) | D<br>MAX. | IL<br>MAX. | HOLE DEPTH<br>MIN. |
|-------------|-----------------|-------------------------------------------|------------------------|-----------------------|-----------|------------|--------------------|
| 6-32 UNC    | 632             | 15/64 (.234)                              | .255                   | .370                  | .233      | .205       | .400               |
| 8-32 UNC    | 832             | 17/64 (.266)                              | .285                   | .370                  | .264      | .205       | .400               |
| 10-24 UNC   | 1024            | 19/64 (.297)                              | .320                   | .370                  | .295      | .205       | .400               |
| 10-32 UNF   | 1032            | 19/64 (.297)                              | .320                   | .370                  | .295      | .205       | .400               |
| 1/4-20 UNC  | 420             | 25/64 (.391)                              | .415                   | .515                  | .389      | .275       | .540               |
| 5/16-18 UNC | 518             | 17/32 (.531)                              | .550                   | .615                  | .528      | .325       | .640               |
| 3/8-16 UNC  | 616             | 19/32 (.594)                              | .615                   | .740                  | .590      | .390       | .770               |
| M4x0,7 ISO  | 470             | 6,75                                      | 7,24                   | 9,40                  | 6,71      | 5,21       | 10,16              |
| M5x0,8 ISO  | 580             | 7,54                                      | 8,13                   | 9,40                  | 7,50      | 5,21       | 10,16              |
| M6x1,0 ISO  | 610             | 9,92                                      | 10,54                  | 13,08                 | 9,88      | 6,99       | 13,72              |
| M8x1,25 ISO | 8125            | 13,49                                     | 13,97                  | 15,62                 | 13,41     | 8,26       | 16,26              |
| M10x1,5 ISO | 1015            | 15,00                                     | 15,62                  | 18,80                 | 14,99     | 9,91       | 19,56              |

**NOTE 1:** Additional UNF thread sizes available

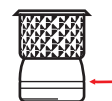
**NOTE 2: HOLE SIZE:** The A-W Series Insert hole size will be dependent on parent material density. Experimentation is required for optimum performance.

**NOTE 3: FINISH:** The standard specified finishes for the A-W Series Insert are cadmium and tin. Alteration to these finishes will reduce performance.

**\*THREAD CLASS:** The A-W Series Insert's internal threads are manufactured oversized to compensate for resulting thread portion shrinkage during the installation swaging process. They are not gaugeable prior to or after installation but will be compatible with Class 2A/3A or 6g screws after installation.

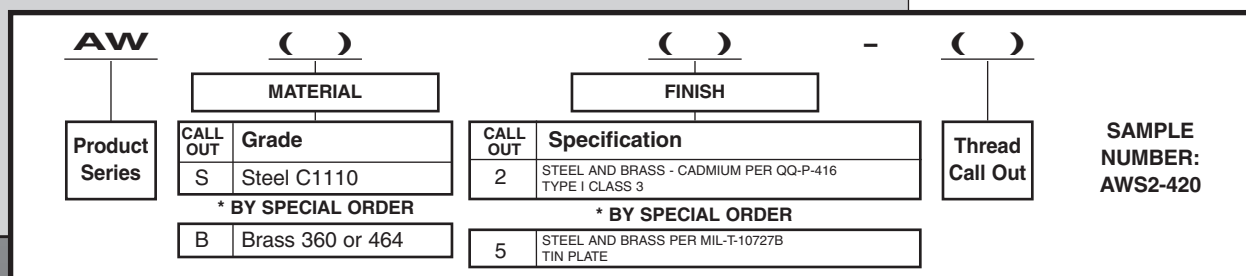
All materials for the A-W Series are plated cadmium and look similar. Radial grooves are machined into the part for material identification.

### MATERIAL TYPE IDENTIFICATION GROOVES



NONE — STEEL  
2 — BRASS

## PART NUMBERING SYSTEM



\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see page 31

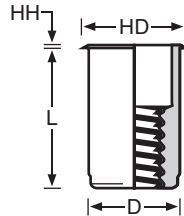
# A-O SERIES PROFILE

**A-O**<sup>TM</sup>  
**SERIES**

**INSERT**  
THREADED INSERT

The **A-O Series Insert** features a reduced profile head design which is similar to the superior A-K Series Insert. It also has a smaller body diameter than the A-K Series Insert. The A-O Series Insert can be specified when the application design parameters require a smaller hole or closer hole to edge tolerances.

The A-O Series can be installed with AVK's ARO brand pneumatic tools or AVK's SPP™ pneumatic/hydraulic tools. These tools can be located at any position on your assembly line. The A-O Series can be installed either prior to or after finish further enhancing its flexibility in your manufacturing environment.



Thread Specifications: Unified 2B/21 per ASME B1.1  
Metric 6H/21 per ASME B1.13M

## UNIFIED (INCH) AND METRIC THREAD SIZES

| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | HOLE SIZE<br>+.006 -.000<br>(+.015 -.000) | HD<br>±.010<br>(±0,25) | HH<br>±.003<br>(±0,08) | L<br>±.015<br>(±0,38) | D<br>MAX. | IL<br>MAX. |
|-------------|-----------------|------------|---------------|-------------------------------------------|------------------------|------------------------|-----------------------|-----------|------------|
| 6-32 UNC    | 632             | .020-.080  | 80            | 1/4 (.250)                                | .295                   | .018                   | .385                  | .249      | .315       |
| 8-32 UNC    | 832             | .020-.080  | 80            | 1/4 (.250)                                | .295                   | .018                   | .385                  | .249      | .315       |
| 10-24 UNC   | 1024            | .020-.130  | 130           | 9/32 (.2812)                              | .320                   | .020                   | .440                  | .280      | .330       |
| 10-32 UNF   | 1032            | .020-.130  | 130           | 9/32 (.2812)                              | .320                   | .020                   | .440                  | .280      | .330       |
| 1/4-20 UNC  | 420             | .030-.165  | 165           | 3/8 (.375)                                | .425                   | .022                   | .580                  | .374      | .440       |
| 5/16-18 UNC | 518             | .040-.200  | 200           | 1/2 (.500)                                | .560                   | .022                   | .690                  | .499      | .540       |
| 3/8-16 UNC  | 616             | .040-.200  | 200           | 1/2 (.500)                                | .560                   | .022                   | .690                  | .499      | .540       |
| M4x0,7 ISO  | 470             | 0,50-2,00  | 2.0           | 6,4                                       | 7,49                   | 0,46                   | 9,78                  | 6,32      | 8,00       |
| M5x0,8 ISO  | 580             | 0,50-3,30  | 3.3           | 7,2                                       | 8,13                   | 0,51                   | 11,18                 | 7,11      | 8,38       |
| M6x1,0 ISO  | 610             | 0,76-4,20  | 4.2           | 9,6                                       | 10,80                  | 0,56                   | 14,73                 | 9,50      | 11,18      |
| M8x1,25 ISO | 8125            | 1,02-5,1   | 5.1           | 12,7                                      | 14,22                  | 0,56                   | 17,53                 | 12,67     | 13,72      |
| M10x1,5 ISO | 1015            | 1,02-5,1   | 5.1           | 12,7                                      | 14,22                  | 0,56                   | 17,53                 | 12,67     | 13,72      |

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** UNF fine threads are available. Contact AVK for details.

## PART NUMBERING SYSTEM

|                    |             |                 |             |                                                                                                            |                    |                  |           |  |                                   |
|--------------------|-------------|-----------------|-------------|------------------------------------------------------------------------------------------------------------|--------------------|------------------|-----------|--|-----------------------------------|
| AO                 |             | ( )             |             | ( )                                                                                                        |                    | -                | ( ) - ( ) |  | SAMPLE<br>NUMBER:<br>AOS4-420-165 |
| Product<br>Series  | MATERIAL    |                 | FINISH      |                                                                                                            | Thread<br>Call Out | Grip<br>Call Out |           |  |                                   |
|                    | CALL<br>OUT | Grade           | CALL<br>OUT | Specification                                                                                              |                    |                  |           |  |                                   |
|                    | S           | Steel 1010/1008 | 3           | UNIFIED (INCH) PRODUCT ZINC PLATE PER ASTM-B633<br>TYPE III Fe/Zn (.0003/8μ) WITH CLEAR PROTECTIVE COATING |                    |                  |           |  |                                   |
|                    |             |                 | 4           | METRIC - ZINC/YELLOW DICHROMATE PER ASTM-B-633<br>TYPE II FE/ZN (.0003/8μ) WITH CLEAR PROTECTIVE COATING   |                    |                  |           |  |                                   |
| * BY SPECIAL ORDER |             |                 |             |                                                                                                            |                    |                  |           |  |                                   |
|                    |             |                 | 9           | STEEL - YELLOW TIN ZINC PER BPS-TZ-327 (.0003/8μ) WITH<br>CLEAR PROTECTIVE COATIN                          |                    |                  |           |  |                                   |

\* Special order items are subject to minimum order requirements. Contact AVK for details.

For air tool selection see pages 28 and 30

## R-N SERIES RIVET NUT PROFILE

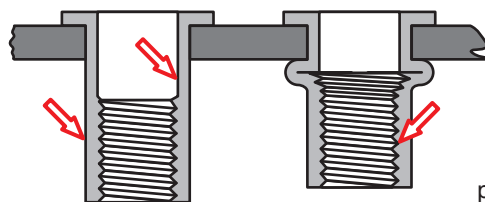
The **R-N Series Rivet Nut** features a heavy duty head profile and increased wall thickness in the collapse area. This makes the R-N Series ideal for leg leveling applications as shown on page 7.

The R-N Series Rivet Nut can be installed using AVK's SPP™ pneumatic/hydraulic tools or the specific rivet nut tools shown on page 37. The R-N Series Rivet Nut's heavier wall thickness and resulting upset load requires this type of tool be used for installation. The R-N Series can be installed either prior to or after finish.

## COLD FORMING TECHNOLOGY™

### HOW IT WORKS FOR YOU

The R-N Series Rivet Nuts are manufactured using state-of-the-art cold forming technology. This provides very precise tolerances. All surfaces of the R-N Series are **FORMED**, not machined. This provides excellent quality.



The internal thread of the R-N Series Rivet Nut is **roll FORMED** not machined. This provides excellent thread strength.

## DESIGN BENEFITS

- ◆ **INCREASED PUSH-OUT LOADS** are achievable in leg leveling applications when using the R-N Series due to its heavy duty head profile and thick wall construction.
- ◆ **SUPERIOR THREAD STRENGTH** is provided due to our internal rolled thread manufacturing process.
- ◆ **SUPERIOR CORROSION RESISTANCE** is provided by our standard cadmium finish (72 hours. salt spray).
- ◆ **UNIFORM INSTALLATION** is guaranteed because of the dimensional tolerances and concentricity tolerances built into our product made possible by our cold forming technology.
- ◆ **AVAILABLE** in steel and aluminum. For additional materials, contact AVK for availability.

## ADDITIONAL DESIGN TYPES

### CLOSED END

Thread area is enclosed eliminating leakage past the threads from either side of the application.

Contact AVK for availability.



### KEYED HEAD

An underside of the head "key" projection when placed into a matching "keyed" hole design provides additional torque resistance.

Contact AVK for availability.



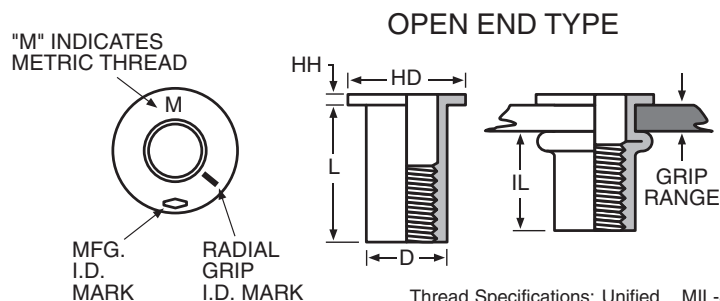
### 100° COUNTERSUNK HEAD

A 100° countersunk head profile when installed into a matching countersunk hole provides a flush installation.

Contact AVK for availability.



# FLATHEAD UNIFIED (INCH) AND METRIC THREAD SIZES



| THREAD SIZE | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | I.D. MARK | HOLE SIZE<br>+ .003<br>- .000 | HD<br>±.015 | HH<br>NOM. | L<br>±.015 | D**<br>+.000<br>-.004 | IL<br>REF. |
|-------------|-----------------|------------|---------------|-----------|-------------------------------|-------------|------------|------------|-----------------------|------------|
| 4-40 UNC    | 440             | .010-.060  | 60            | BLANK     | 5/32 (.155)                   | .270        | .025       | .345       | .155                  | .230       |
| 4-40 UNC    | 440             | .060-.085  | 85            | 1-RAD.    | 5/32 (.155)                   | .270        | .025       | .370       | .155                  | .230       |
| 4-40 UNC    | 440             | .085-.110  | 110           | 2-RAD.    | 5/32 (.155)                   | .270        | .025       | .400       | .155                  | .230       |
| 6-32 UNC    | 632             | .010-.075  | 75            | 1-RAD.    | #12 (.189)                    | .325        | .032       | .438       | .189                  | .300       |
| 6-32 UNC    | 632             | .075-.120  | 120           | 3-RAD.    | #12 (.189)                    | .325        | .032       | .500       | .189                  | .315       |
| 6-32 UNC    | 632             | .120-.160  | 160           | 5-RAD.    | #12 (.189)                    | .325        | .032       | .500       | .189                  | .270       |
| 8-32 UNC    | 832             | .010-.075  | 75            | 1-RAD.    | #2 (.221)                     | .357        | .032       | .438       | .221                  | .300       |
| 8-32 UNC    | 832             | .075-.120  | 120           | 3-RAD.    | #2 (.221)                     | .357        | .032       | .500       | .221                  | .315       |
| 8-32 UNC    | 832             | .120-.160  | 160           | 5-RAD.    | #2 (.221)                     | .357        | .032       | .500       | .221                  | .270       |
| 10-32 UNF   | 1032            | .010-.080  | 80            | BLANK     | 1/4 (.250)                    | .406        | .038       | .531       | .250                  | .380       |
| 10-32 UNF   | 1032            | .080-.130  | 130           | 1-RAD.    | 1/4 (.250)                    | .406        | .038       | .594       | .250                  | .390       |
| 10-32 UNF   | 1032            | .130-.180  | 180           | 2-RAD.    | 1/4 (.250)                    | .406        | .038       | .641       | .250                  | .390       |
| 1/4-20 UNC  | 420             | .020-.080  | 80            | BLANK     | Q (.332)                      | .475        | .058       | .625       | .332                  | .450       |
| 1/4-20 UNC  | 420             | .080-.140  | 140           | 1-RAD.    | Q (.332)                      | .475        | .058       | .687       | .332                  | .450       |
| 1/4-20 UNC  | 420             | .140-.200  | 200           | 2-RAD.    | Q (.332)                      | .475        | .058       | .750       | .332                  | .450       |
| 5/16-18 UNC | 518             | .030-.125  | 125           | BLANK     | Z (.413)                      | .665        | .062       | .750       | .413                  | .505       |
| 5/16-18 UNC | 518             | .125-.200  | 200           | 1-RAD.    | Z (.413)                      | .665        | .062       | .875       | .413                  | .555       |
| 5/16-18 UNC | 518             | .200-.275  | 275           | 2-RAD.    | Z (.413)                      | .665        | .062       | .937       | .413                  | .540       |
| 3/8-16 UNC  | 616             | .030-.115  | 115           | BLANK     | 12.5 mm (.490)                | .781        | .088       | .844       | .490**                | .585       |
| 3/8-16 UNC  | 616             | .115-.200  | 200           | 1-RAD.    | 12.5 mm (.490)                | .781        | .088       | .938       | .490**                | .595       |
| 3/8-16 UNC  | 616             | .200-.285  | 285           | 2-RAD.    | 12.5 mm (.490)                | .781        | .088       | 1.031      | .490**                | .605       |
| 1/2-13 UNC  | 813             | .050-.150  | 150           | BLANK     | 5/8 (.625)                    | .906        | .085       | .906       | .625                  | .605       |
| 1/2-13 UNC  | 813             | .150-.250  | 250           | 1-RAD.    | 5/8 (.625)                    | .906        | .085       | 1.031      | .625                  | .630       |
| 1/2-13 UNC  | 813             | .250-.350  | 350           | 2-RAD.    | 5/8 (.625)                    | .906        | .085       | 1.141      | .625                  | .640       |

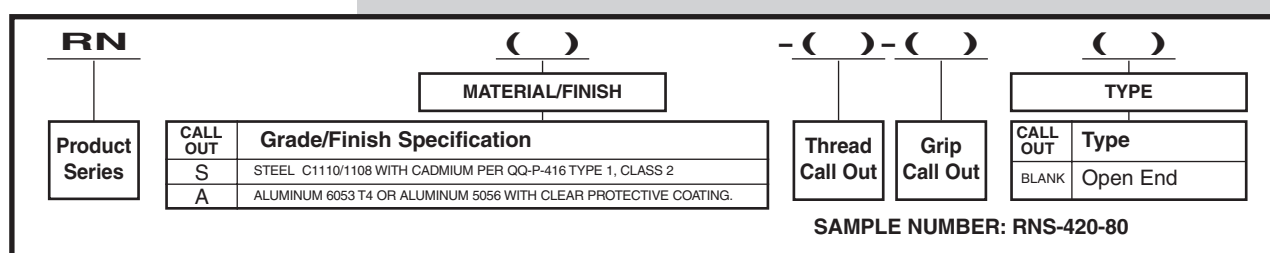
Note: \*\* +0.00 -.007

| THREAD SIZE  | THREAD CALL OUT | GRIP RANGE | GRIP CALL OUT | I.D. MARK | HOLE SIZE<br>+0.08<br>-0.00 | HD<br>±0.38 | HH<br>NOM. | L<br>±0.38 | D**<br>+0.00<br>-0.10 | IL<br>REF. |
|--------------|-----------------|------------|---------------|-----------|-----------------------------|-------------|------------|------------|-----------------------|------------|
| M3x0.5 ISO   | 350             | 0.25-1.00  | 1.0           | BLANK     | 3.94                        | 6.68        | 0.63       | 8.00       | 3.93                  | 5.61       |
| M3x0.5 ISO   | 350             | 1.00-1.75  | 1.75          | 1-RAD.    | 3.94                        | 6.68        | 0.63       | 8.75       | 3.93                  | 5.61       |
| M3x0.5 ISO   | 350             | 1.75-2.50  | 2.5           | 2-RAD.    | 3.94                        | 6.68        | 0.63       | 9.50       | 3.93                  | 5.61       |
| M4x0.7 ISO   | 470             | 0.25-2.00  | 2.0           | BLANK     | 5.60                        | 9.01        | 0.81       | 11.00      | 5.61                  | 7.08       |
| M4x0.7 ISO   | 470             | 2.00-3.00  | 3.0           | 1-RAD.    | 5.60                        | 9.01        | 0.81       | 12.00      | 5.61                  | 7.08       |
| M4x0.7 ISO   | 470             | 3.00-4.00  | 4.0           | 2-RAD.    | 5.60                        | 9.01        | 0.81       | 13.00      | 5.61                  | 7.08       |
| M5x0.8 ISO   | 580             | 0.25-2.00  | 2.0           | BLANK     | 7.20                        | 11.17       | 1.22       | 14.50      | 7.13                  | 10.09      |
| M5x0.8 ISO   | 580             | 2.00-3.50  | 3.5           | 1-RAD.    | 7.20                        | 11.17       | 1.22       | 16.00      | 7.13                  | 10.09      |
| M5x0.8 ISO   | 580             | 3.50-5.00  | 5.0           | 2-RAD.    | 7.20                        | 11.17       | 1.22       | 17.50      | 7.13                  | 10.09      |
| M6x1.0 ISO   | 610             | 0.75-2.00  | 2.0           | BLANK     | 8.50                        | 13.43       | 1.47       | 15.50      | 8.43                  | 10.58      |
| M6x1.0 ISO   | 610             | 2.00-3.50  | 3.5           | 1-RAD.    | 8.50                        | 13.43       | 1.47       | 17.00      | 8.43                  | 10.58      |
| M6x1.0 ISO   | 610             | 3.50-5.00  | 5.0           | 2-RAD.    | 8.50                        | 13.43       | 1.47       | 18.50      | 8.43                  | 10.58      |
| M8x1.25 ISO  | 8125            | 1.00-3.00  | 3.0           | BLANK     | 10.50                       | 16.65       | 1.57       | 18.00      | 10.48                 | 11.83      |
| M8x1.25 ISO  | 8125            | 3.00-5.00  | 5.0           | 1-RAD.    | 10.50                       | 16.65       | 1.57       | 20.00      | 10.48                 | 11.83      |
| M8x1.25 ISO  | 8125            | 5.00-7.00  | 7.0           | 2-RAD.    | 10.50                       | 16.65       | 1.57       | 22.00      | 10.48                 | 11.83      |
| M10x1.5 ISO  | 1015            | 1.00-3.00  | 3.0           | BLANK     | 12.50                       | 19.50       | 2.23       | 20.00      | 12.44**               | 13.20      |
| M10x1.5 ISO  | 1015            | 3.00-5.50  | 5.5           | 1-RAD.    | 12.50                       | 19.50       | 2.23       | 22.50      | 12.44**               | 13.20      |
| M10x1.5 ISO  | 1015            | 5.50-8.00  | 8.0           | 2-RAD.    | 12.50                       | 19.50       | 2.23       | 25.00      | 12.44**               | 13.20      |
| M12x1.75 ISO | 12175           | 1.00-3.00  | 3.0           | BLANK     | 15.50                       | 22.79       | 2.23       | 24.00      | 15.46                 | 16.45      |
| M12x1.75 ISO | 12175           | 3.00-5.50  | 5.5           | 1-RAD.    | 15.50                       | 22.79       | 2.23       | 26.50      | 15.46                 | 16.45      |
| M12x1.75 ISO | 12175           | 5.50-8.00  | 8.0           | 2-RAD.    | 15.50                       | 22.79       | 2.23       | 29.00      | 15.46                 | 16.45      |

Note: \*\* +0.00 -.007

**NOTE 1:** Grip range can be affected by parent material density and actual hole size. AVK suggests trial installations to determine optimum grip. **NOTE 2:** Additional UNF and UNC threads are available. Contact AVK for details. **NOTE 3:** RN Series threads are not gaugeable after installation. **NOTE 4:** Additional grip sizes, materials, head styles and closed end versions are available by special order. Contact AVK for details.

## PART NUMBERING SYSTEM

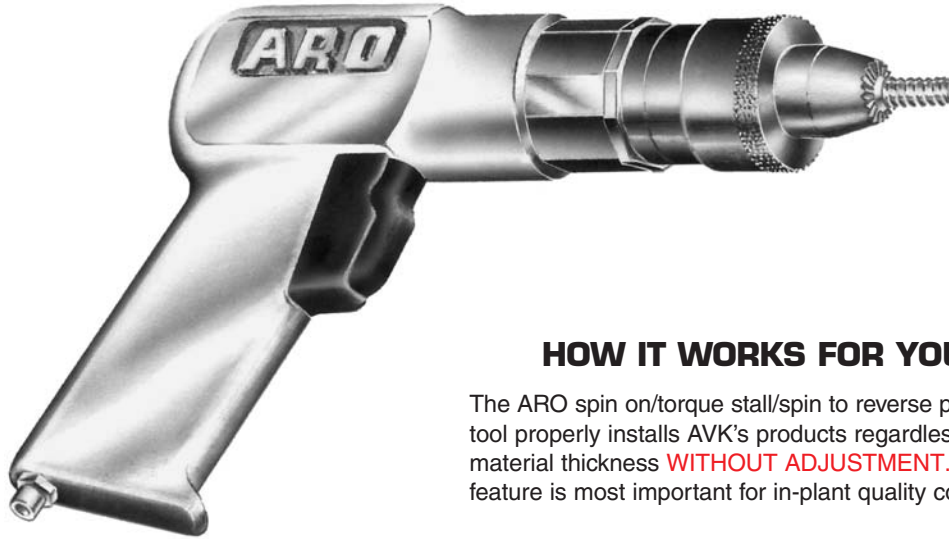


For air tool selection see pages 34 and 37

\*\* Products with \*\* (two asteriks) do not meet or comply with MS/NAS Specifications

## AVK PNEUMATIC TOOLS PROFILE

AVK has selected the **ARO Brand Pneumatic Tool** for its ergonomic design and outstanding dependability. The rocker style forward and reverse trigger is easy and comfortable to use. AVK's Quick-Change thread adaption kit assembly allows for easy removal of the tool's nose assembly without the need for wrenches.



### HOW IT WORKS FOR YOU

The ARO spin on/torque stall/spin to reverse pneumatic tool properly installs AVK's products regardless of material thickness **WITHOUT ADJUSTMENT**. This feature is most important for in-plant quality control.

## DESIGN BENEFITS

- ◆ The AVK ARO tool uses torque to install the AVK product. It needs no adjustment to install the product in variable thickness material.
- ◆ The ergonomic design of the AVK ARO tool makes it feel comfortable to the operator and weighs in at 3 lbs. (1.36 kg).
- ◆ The rocker style trigger is easy to use and minimizes operator fatigue.
- ◆ The knurled nose assembly eliminates any torque "kick" during installation.
- ◆ Preventative maintenance is quick and easy with AVK's patented\* quick-change thread adaption kit. No tools are required to access these parts.
- ◆ The rugged design of the tool casing, its components and the AVK thread adaption kit provides you with reliability.

## ADDITIONAL TOOL DESIGNS

### INLINE DESIGN

The Inline Style Tool is designed for vertical installations. Contact AVK for information.



### RIGHT ANGLE DESIGN

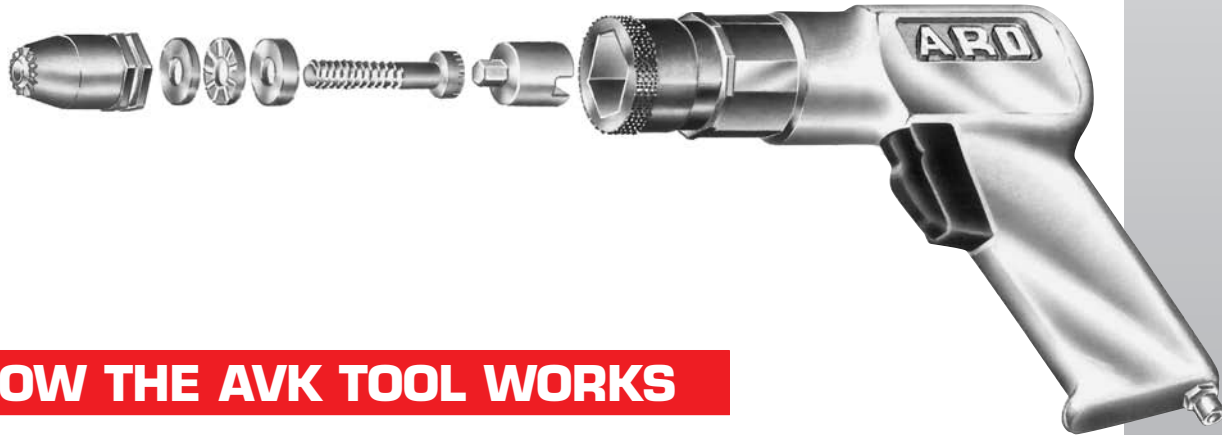
The Right Angle Inline Style Tool is designed for limited access applications.



Contact AVK for information.

## AVK PNEUMATIC TOOLS PROFILE

The AVK pneumatic tool features a Quick-Change thread adaption kit. This patented\* feature allows for easy access to the thread size component parts. AVK's Pneumatic tools provide the highest RPM for the thread size selected resulting in optimum installation speed.



## HOW THE AVK TOOL WORKS

- ◆ The operator quarter turns an AVK threaded insert onto the tool mandrel and places it into the hole in the parent material.
- ◆ The top trigger is depressed and the tool mandrel spins into the insert.
- ◆ The head of the threaded insert is gripped by the knurling at the nose piece preventing it from spinning as the threaded area of the insert "walks" up the mandrel.
- ◆ As this occurs, the AVK insert expands within the hole wall providing hole fill and then forms a secondary flange against the backside of the parent material.
- ◆ The tool continues to torque the insert's threads causing the knurling to bite into the backside edge of the parent material.
- ◆ The tool then stalls when it reaches its predetermined stall torque. This occurs even if the parent material varies in thickness. The tool needs no adjustment to install AVK's products in variable thickness materials.
- ◆ The operator then presses the lower reverse trigger and the mandrel spins out of the installed part.

## ADDITIONAL TOOL DESIGNS

### HEAD FORMING THREAD ADAPTION KIT

A Head Forming T.A.K. is available to contour the AVK insert's head to the radius of a tube to increase torque capability.



Contact AVK for information.

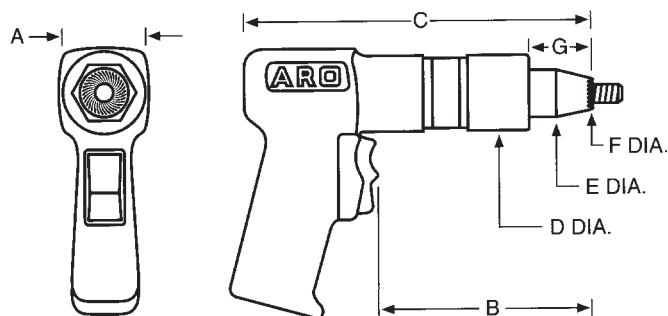
### SPECIAL DESIGNS

AVK can design a thread adaption kit nose assembly to fit your particular application needs.



Contact AVK for information.

## AVK PNEUMATIC TOOL SPECIFICATIONS



The tool shown on this page has been specifically designed to install the A-L, A-K, A-H, A-P and A-O Series Inserts.

Once you have selected the type of insert and thread size required for your application, select the appropriate RPM tool from the chart below.

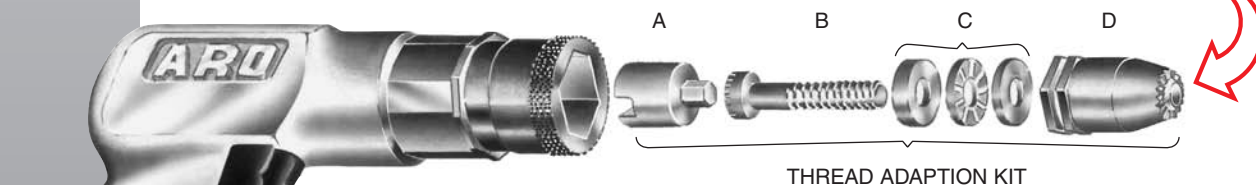
## DIMENSIONAL DATA/TOOL SET-UP REQUIREMENTS

| RPM   | WEIGHT<br>LBS.-Kg | A             | B             | C             | D<br>DIA.     | E<br>DIA.     | F<br>DIA. MAX. | G           |
|-------|-------------------|---------------|---------------|---------------|---------------|---------------|----------------|-------------|
| 3,000 | 2.55<br>1.15      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,8 | 1.57<br>39,87 | 1.00<br>25,4  | .400<br>10,16  | 1.3<br>33,0 |
| 1,500 | 2.58<br>1.17      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,8 | 1.57<br>39,87 | 1.00<br>25,4  | .400<br>10,16  | 1.3<br>33,0 |
| 900   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 9.00<br>228,6 | 1.57<br>39,87 | 1.00<br>25,4  | .500<br>12,7   | 1.3<br>33,0 |
| 600   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 9.00<br>228,6 | 1.57<br>39,87 | 1.00<br>25,4  | .640<br>16,25  | 1.3<br>33,0 |
| 350   | 3.25<br>1.46      | 1.86<br>47,24 | 5.37<br>136,3 | 8.37<br>212,5 | 1.50<br>38,1  | 1.42<br>36,06 | .900<br>22,86  | NA<br>NA    |

### PROPER AIR SUPPLY SET-UP REQUIRES:

- ◆ 90-110 PSI (6.2-7.5 BARS) dynamic (tool running) air pressure at 25 S.C.F.M.
- ◆ Inline oiler/separator
- ◆ Air pressure gauge and regulator
- ◆ 5/16 or 7,92 mm minimum hose ID
- ◆ 5/16 or 7,92 mm minimum fittings ID

**SPECIAL FEATURE**—The AKPT nose cone design incorporates a special pilot/serrated tip that is essential to proper insert installation. The “A-K” prefix in the tool part number designates this feature.



THREAD ADAPTION KIT

## AIR TOOL SELECTION/SPARE PARTS

| THREAD SIZE    | TOOL R.P.M. | COMPLETE TOOL PART NUMBER | THREAD ADAPTION KIT | A HEX DRIVE | B MANDREL 10 PER BAG | C BEARING SET | D NOSE CONE | DYNAMIC AIR PRESSURE SETTINGS PSI - BARS |                                                                                                                                                                                                                                                                |
|----------------|-------------|---------------------------|---------------------|-------------|----------------------|---------------|-------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6-32 UNC       | 3000        | AKPT30P632                | AKPT632TAK          | 29NPT22     | B3SH632-1500         | 32PT 1        | 77AKPT6     | 70-80                                    | This chart designates the tool, spare parts and dynamic (tool running) air pressure requirements for our most popular steel product. Consult the AVK tool catalog or contact AVK for tool RPM and air pressure settings for aluminum, brass and monel product. |
| 8-32 UNC       | 3000        | AKPT30P832                | AKPT832TAK          | 29NPT23     | B3SH832-1500         | 32PT 2        | 77AKPT8     | 75-90                                    |                                                                                                                                                                                                                                                                |
| 10-24 UNC      | 1500        | AKPT15P1024               | AKPT1024TAK         | 29NPT4      | B3SH1024-1750        | 32PT 4        | 77AKPT10    | 60-80                                    |                                                                                                                                                                                                                                                                |
| 10-32 UNF      | 1500        | AKPT15P1032               | AKPT1032TAK         | 29NPT4      | B3SH1032-1750        | 32PT 4        | 77AKPT10    | 60-80                                    |                                                                                                                                                                                                                                                                |
| 1/4-20 UNC     | 900         | AKPT9P420                 | AKPT420TAK          | 29NPT5      | B3SH420-1500         | 32PT 5        | 77AKPT250   | 70-90                                    |                                                                                                                                                                                                                                                                |
| 5/16-18 UNC    | 600         | AKPT6P518                 | AKPT518TAK          | 29NPT6      | B3SH518-2000         | 32PT 7        | 77AKPT3125  | 80-110                                   |                                                                                                                                                                                                                                                                |
| 3/8-16 UNC     | 600         | AKPT6P616                 | AKPT616TAK          | 29NPT7      | B3SH616-2000         | 32PT 8        | 77AKPT375   | 80-110                                   |                                                                                                                                                                                                                                                                |
| 1/2-13 UNC     | 350         | AKPT3P813                 | AKPT813CTA          | 29NPT26     | B3SH813-2500         | 30NPT500      | 77AKPT500   | 80-110                                   |                                                                                                                                                                                                                                                                |
| M4 x 0,7 ISO   | 3000        | AKPT30P470                | AKPT470TAK          | 29NPT24     | B3SH470-40           | 32PT 3        | 77AKPTM4    | 4.8-5.5                                  |                                                                                                                                                                                                                                                                |
| M5 x 0,8 ISO   | 1500        | AKPT15P580                | AKPT580TAK          | 29NPT10     | B3SH580-45           | 32PT 4        | 77AKPTM5    | 4.1-5.5                                  |                                                                                                                                                                                                                                                                |
| M6 x 1,0 ISO   | 900         | AKPT9P610                 | AKPT610TAK          | 29NPT11     | B3SH610-40           | 32PT 6        | 77AKPTM6    | 4.8-6.2                                  |                                                                                                                                                                                                                                                                |
| M8 x 1,25 ISO  | 600         | AKPT6P8125                | AKPT8125TAK         | 29NPT12     | B3SH8125-50          | 32PT 7        | 77AKPTM8    | 5.5-7.5                                  |                                                                                                                                                                                                                                                                |
| M10 x 1,5 ISO  | 600         | AKPT6P1015                | AKPT1015TAK         | 29NPT25     | B3SH1015-50          | 32PT 10       | 77AKPTM10   | 5.5-7.5                                  |                                                                                                                                                                                                                                                                |
| M12 x 1,75 ISO | 350         | AKPT3P12175               | AKPT12175CTA        | 29NPT27     | B3SH12175-60         | 30NPT500      | 77AKPTM12   | 5.5-7.5                                  |                                                                                                                                                                                                                                                                |

NOTE: UNF FINE THREAD COMPONENTS ARE AVAILABLE.

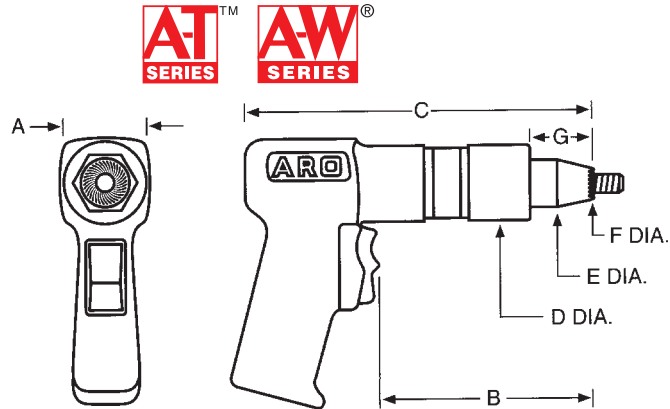
### PREVENTATIVE MAINTENANCE REQUIREMENTS:

- ◆ The bearing set must be kept in a WET lubricated condition to assure proper tool operation. AVK suggests the use of high temperature grease such as LUBRIPLATE BRAND 930 AA.
- ◆ The tool mandrel should be inspected for thread wear or damage and replaced. To test the condition of the mandrel, thread an AVK insert onto the mandrel backwards until it touches the pilot. If any drag is still felt, replace the mandrel with Unbrako socket head cap screws.

# AVK PNEUMATIC TOOL SPECIFICATIONS

The tool shown on this page has been specifically designed to install the A-T and A-W Series Inserts.

Once you have selected the type of insert and thread size required for your application, select the appropriate RPM tool from the chart below.



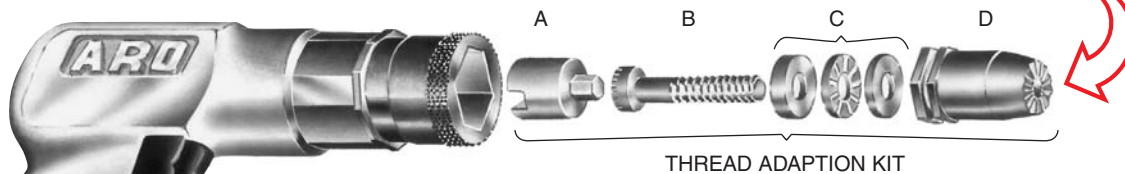
## DIMENSIONAL DATA/TOOL SET-UP REQUIREMENTS

PROPER AIR SUPPLY SET-UP REQUIRES:

- ◆ 90-110 PSI (6.2-7.5 BARS) dynamic (tool running) air pressure at 25 S.C.F.M.
- ◆ Inline oiler/separator
- ◆ Air pressure gauge and regulator
- ◆ 5/16 or 7,92 mm minimum hose ID
- ◆ 5/16 or 7,92 mm minimum fittings ID

| RPM   | WEIGHT<br>LBS.-Kg | A             | B             | C             | D<br>DIA.     | E<br>DIA.     | F<br>DIA. MAX. | G           |
|-------|-------------------|---------------|---------------|---------------|---------------|---------------|----------------|-------------|
| 3,000 | 2.55<br>1.15      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,8 | 1.57<br>39,87 | 1.00<br>25,4  | .350<br>8,89   | 1.3<br>33.0 |
| 1,500 | 2.58<br>1.17      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,8 | 1.57<br>39,87 | 1.00<br>25,4  | .600<br>15,24  | 1.3<br>33.0 |
| 600   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 9.0<br>228,6  | 1.57<br>39,87 | 1.00<br>25,4  | .625<br>15,87  | 1.3<br>33.0 |
| 350   | 3.25<br>1.46      | 1.86<br>47,24 | 6.00<br>152,4 | 8.37<br>212,5 | 1.50<br>38,1  | 1.43<br>36,32 | .900<br>22,86  | NA          |

SPECIAL FEATURE—This NPT nose cone design incorporates a special serrated tip that is essential to proper insert installation. The “N” prefix in the tool part number designates this feature.



## AIR TOOL SELECTION/SPARE PARTS

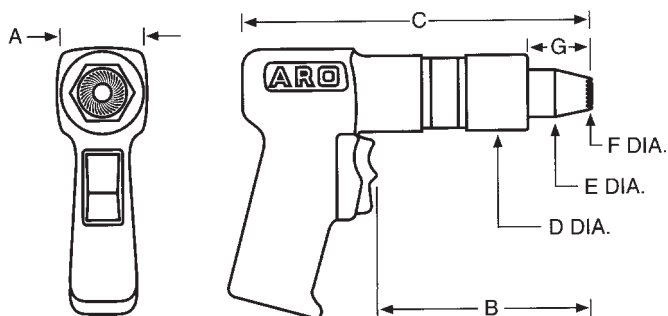
| THREAD SIZE  | TOOL R.P.M. | COMPLETE TOOL PART NUMBER | THREAD ADAPTION KIT | A HEX DRIVE | B MANDREL 10 PER BAG | C BEARING SET | D NOSE CONE | DYNAMIC AIR PRESSURE SETTINGS PSI - BARS | This chart designates the tool, spare parts and dynamic (tool running) air pressure requirements for our most popular steel product. Consult the AVK tool catalog or contact AVK for tool RPM and air pressure settings for aluminum, brass and stainless product. |
|--------------|-------------|---------------------------|---------------------|-------------|----------------------|---------------|-------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-40 UNC     | 3000        | NPT30P440                 | NPT440TAK           | 29NPT1      | B3SH440-750          | 30NPT 4       | 77NPT 4     | 36-40                                    |                                                                                                                                                                                                                                                                    |
| 6-32 UNC     | 3000        | NPT30P632                 | NPT632TAK           | 29NPT2      | B3SH632-875          | 30NPT 6       | 77NPT 6     | 70-80                                    |                                                                                                                                                                                                                                                                    |
| 8-32 UNC     | 3000        | NPT30P832                 | NPT832TAK           | 29NPT3      | B3SH832-875          | 30NPT 8       | 77NPT 8     | 60-85                                    |                                                                                                                                                                                                                                                                    |
| 10-24 UNC    | 1500        | NPT15P1024                | NPT1024TAK          | 29NPT4      | B3SH1024-1500        | 30NPT 10      | 77NPT 10    | 60-85                                    |                                                                                                                                                                                                                                                                    |
| 10-32 UNF    | 1500        | NPT15P1032                | NPT1032TAK          | 29NPT4      | B3SH1032-1500        | 30NPT 10      | 77NPT 10    | 60-85                                    |                                                                                                                                                                                                                                                                    |
| 1/4-20 UNC   | 1500        | NPT15P420                 | NPT420TAK           | 29NPT5      | B3SH420-1250         | 30NPT 250     | 77NPT 250   | 70-95                                    |                                                                                                                                                                                                                                                                    |
| 5/16-18 UNC  | 600         | NPT6P518                  | NPT518TAK           | 29NPT6      | B3SH518-1750         | 30NPT 3125    | 77NPT 3125  | 80-100                                   |                                                                                                                                                                                                                                                                    |
| 3/8-16 UNC   | 600         | NPT6P616                  | NPT616TAK           | 29NPT7      | B3SH616-1750         | 30NPT 375     | 77NPT 375   | 90-110                                   |                                                                                                                                                                                                                                                                    |
| 1/2-13 UNC   | 350         | NPT3P813                  | NPT813CTA           | 29NPT26     | B3SH813-2000         | 30NPT 500     | 77NPT 500   | 95-110                                   |                                                                                                                                                                                                                                                                    |
| M3x0,5 ISO   | 3000        | NPT30P350                 | NPT350TAK           | 29NPT8      | B3SH350-20           | 30NPT M3      | 77NPT M3    | 2.4-2.7                                  |                                                                                                                                                                                                                                                                    |
| M4x0,7 ISO   | 3000        | NPT30P470                 | NPT470TAK           | 29NPT9      | B3SH470-20           | 30NPT M4      | 77NPT M4    | 4.1-5.5                                  |                                                                                                                                                                                                                                                                    |
| M5x0,8 ISO   | 1500        | NPT15P580                 | NPT580TAK           | 29NPT10     | B3SH580-40           | 30NPT M5      | 77NPT M5    | 4.1-5.5                                  |                                                                                                                                                                                                                                                                    |
| M6x1,0 ISO   | 1500        | NPT15P610                 | NPT610TAK           | 29NPT11     | B3SH610-35           | 30NPT M6      | 77NPT M6    | 4.8-6.5                                  |                                                                                                                                                                                                                                                                    |
| M8x1,25 ISO  | 600         | NPT6P8125                 | NPT8125TAK          | 29NPT12     | B3SH8125-45          | 30NPT M8      | 77NPT M8    | 5.5-6.8                                  |                                                                                                                                                                                                                                                                    |
| M10x1,5 ISO  | 600         | NPT6P1015                 | NPT1015TAK          | 29NPT25     | B3SH1015-45          | 32PT 10       | 77NPT M10   | 6.2-7.5                                  |                                                                                                                                                                                                                                                                    |
| M12x1,75 ISO | 350         | NPT3P12175                | NPT12175CTA         | 29NPT27     | B3SH12175-50         | 30NPT 500     | 77NPT M12   | 6.2-7.5                                  |                                                                                                                                                                                                                                                                    |

NOTE: UNF FINE THREADS COMPONENTS ARE AVAILABLE

PREVENTATIVE MAINTENANCE REQUIREMENTS:

- ◆ The bearing set must be kept in a WET lubricated condition to assure proper tool operation. AVK suggests the use of high temperature grease such as LUBRIPLATE BRAND 930 AA.
- ◆ The tool mandrel should be inspected for thread wear or damage and replaced. To test the condition of the mandrel, thread an AVK insert onto the mandrel backwards until it touches the knurled nose cone. If any drag is still felt, replace the mandrel with Unbrako socket head cap screws.

## AVK PNEUMATIC TOOL SPECIFICATIONS



The tool shown on this page has been specifically designed to install the A-S Series Stud.

Once you have selected the type of stud and thread size required for your application refer to the chart below for air tool selection.

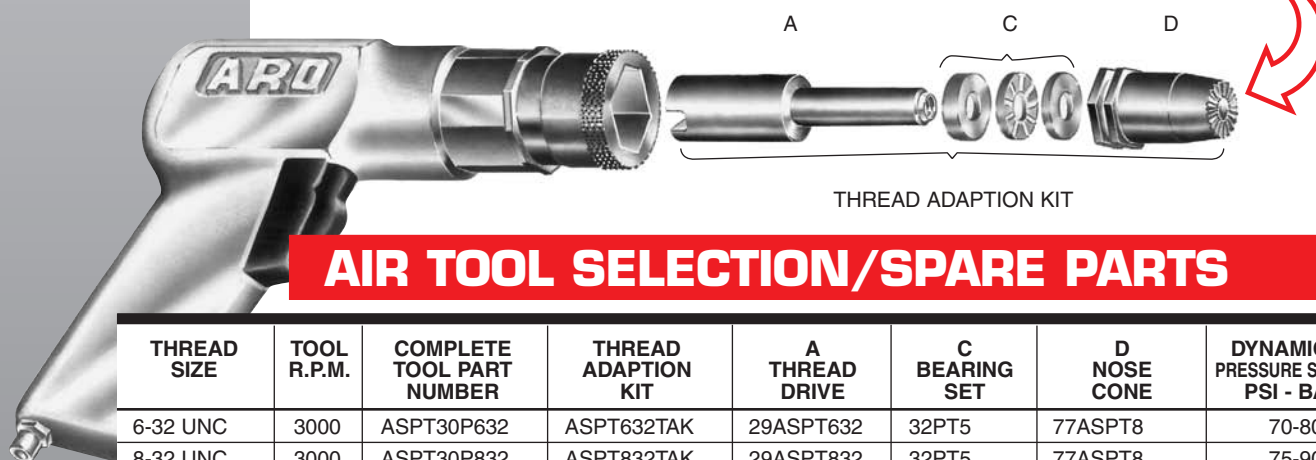
## DIMENSIONAL DATA/TOOL SET-UP REQUIREMENTS

| RPM   | WEIGHT<br>LBS.-Kg | A             | B             | C              | D<br>DIA.     | E<br>DIA.    | F<br>DIA. MAX. | G           |
|-------|-------------------|---------------|---------------|----------------|---------------|--------------|----------------|-------------|
| 3,000 | 2.55<br>1.15      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,85 | 1.57<br>39,87 | 1.00<br>25,4 | .400<br>10,16  | 1.3<br>33,0 |
| 1,500 | 2.58<br>1.17      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,85 | 1.57<br>39,87 | 1.00<br>25,4 | .400<br>10,16  | 1.3<br>33,0 |
| 900   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 9.00<br>228,6  | 1.57<br>39,87 | 1.00<br>25,4 | .500<br>12,7   | 1.3<br>33,0 |
| 600   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 8.37<br>212,6  | 1.57<br>39,87 | 1.00<br>25,4 | .640<br>16,25  | 1.3<br>33,0 |

### PROPER AIR SUPPLY SET-UP REQUIRES:

- ◆ 90-110 PSI (6.2-7.5 BARS) dynamic (tool running) air pressure at 25 S.C.F.M.
- ◆ Inline oiler/separator
- ◆ Air pressure gauge and regulator
- ◆ 5/16 or 7,92 mm minimum hose ID
- ◆ 5/16 or 7,92 mm minimum fittings ID

**SPECIAL FEATURE**—This tool nose cone design incorporates a special serration tip that is essential to proper insert installation.



THREAD ADAPTION KIT

## AIR TOOL SELECTION/SPARE PARTS

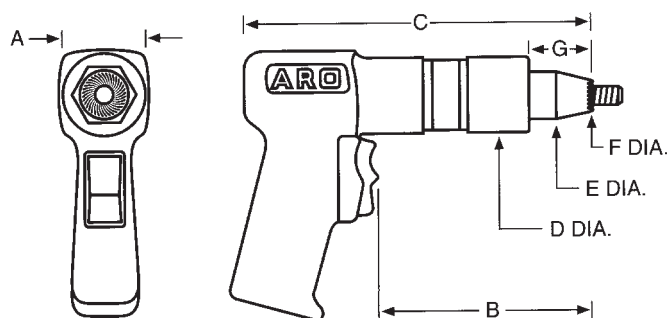
| THREAD SIZE   | TOOL R.P.M. | COMPLETE TOOL PART NUMBER | THREAD ADAPTION KIT | A THREAD DRIVE | C BEARING SET | D NOSE CONE | DYNAMIC AIR PRESSURE SETTINGS PSI - BARS |
|---------------|-------------|---------------------------|---------------------|----------------|---------------|-------------|------------------------------------------|
| 6-32 UNC      | 3000        | ASPT30P632                | ASPT632TAK          | 29ASPT632      | 32PT5         | 77ASPT8     | 70-80                                    |
| 8-32 UNC      | 3000        | ASPT30P832                | ASPT832TAK          | 29ASPT832      | 32PT5         | 77ASPT8     | 75-90                                    |
| 10-24 UNC     | 1500        | ASPT15P1024               | ASPT1024TAK         | 29ASPT1024     | 32PT12        | 77ASPT10    | 60-80                                    |
| 10-32 UNF     | 1500        | ASPT15P1032               | ASPT1032TAK         | 29ASPT1032     | 32PT12        | 77ASPT10    | 60-80                                    |
| 1/4-20 UNC    | 900         | ASPT9P420                 | ASPT420TAK          | 29ASPT420      | 32PT8         | 77ASPT4     | 70-90                                    |
| 5/16-18 UNC   | 600         | ASPT6P518                 | ASPT518CTA          | 29ASPT518      | 30NPT500      | 77ASPT8125  | 80-110                                   |
| 3/8-16 UNC    | 600         | ASPT6P616                 | ASPT616CTA          | 29ASPT616      | 30NPT500      | 77ASPT8125  | 80-110                                   |
| M4 x 0,7 ISO  | 3000        | ASPT30P470                | ASPT470TAK          | 29ASPT470      | 32PT5         | 77ASPT8     | 4.8-5.5                                  |
| M5 x 0,8 ISO  | 1500        | ASPT15P580                | ASPT580TAK          | 29ASPT580      | 32PT12        | 77ASPT10    | 4.1-5.5                                  |
| M6 x 1,0 ISO  | 900         | ASPT9P610                 | ASPT610TAK          | 29ASPT610      | 32PT8         | 77ASPT4     | 5.5-6.2                                  |
| M8 x 1,25 ISO | 600         | ASPT6P8125                | ASPT8125CTA         | 29ASPT8125     | 30NPT500      | 77ASPT8125  | 5.5-7.5                                  |
| M10 x 1,5 ISO | 600         | ASPT6P1015                | ASPT1015CTA         | 29ASPT1015     | 30NPT500      | 77ASPT8125  | 5.5-7.5                                  |

NOTE: Air pressure settings are specified dynamic (tool running).

### PREVENTATIVE MAINTENANCE REQUIREMENTS:

- ◆ The bearing set must be kept in a WET lubricated condition to assure proper tool operation. AVK suggests the use of high temperature grease such as LUBRIPLATE BRAND 930 AA.

# AVK PNEUMATIC TOOL SPECIFICATIONS



The tool shown on this page has been specifically designed to install the A-R Series Inserts.

AVK recommends that trial installations be performed to determine the optimum tool for the fastener selected using actual application materials and hole sizes. Tool RPM and parent material density will affect the grip range of the fastener. See the chart below for guidelines.

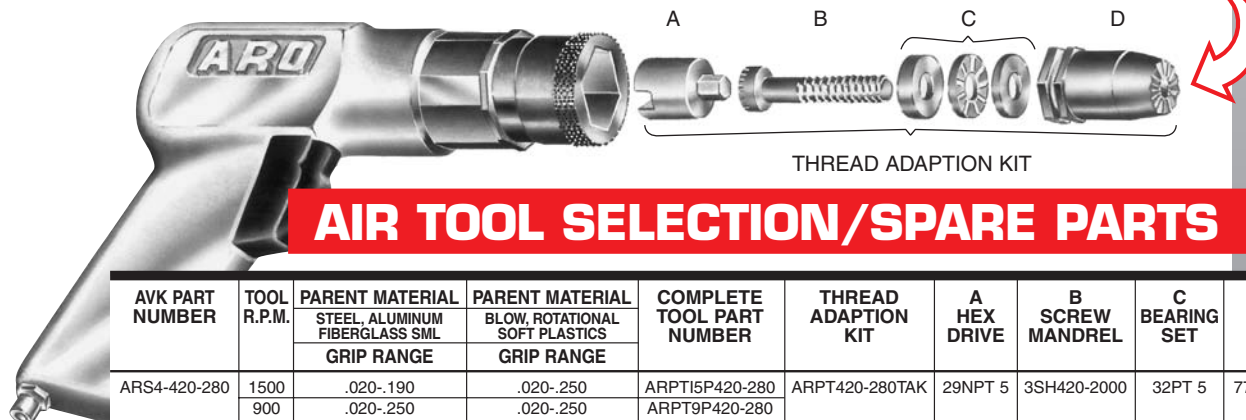
## DIMENSIONAL DATA/TOOL SET-UP REQUIREMENTS

PROPER AIR SUPPLY SET-UP REQUIRES:

- ◆ 90-110 PSI (6.2-7.5 BARS) dynamic (tool running) air pressure at 25 S.C.F.M.
- ◆ Inline oiler/separator
- ◆ Air pressure gauge and regulator
- ◆ 5/16 or 7,92 mm minimum hose ID
- ◆ 5/16 or 7,92 mm minimum fittings ID

| RPM   | WEIGHT<br>LBS.-Kg | A             | B             | C             | D<br>DIA.     | E<br>DIA.    | F<br>DIA. MAX. | G           |
|-------|-------------------|---------------|---------------|---------------|---------------|--------------|----------------|-------------|
| 1,500 | 2.58<br>1.17      | 1.86<br>47,24 | 4.75<br>120,6 | 7.75<br>196,8 | 1.57<br>39,87 | 1.00<br>25,4 | .400<br>10,16  | 1.3<br>33,0 |
| 900   | 3.18<br>1.44      | 1.86<br>47,24 | 6.00<br>152,4 | 9.00<br>228,6 | 1.57<br>39,87 | 1.00<br>25,4 | .500<br>12,7   | 1.3<br>33,0 |

**SPECIAL FEATURE**—This NPT nose cone design incorporates a special serrated tip that is essential to proper insert installation. The “N” prefix in the tool part number designates this feature.



## AIR TOOL SELECTION/SPARE PARTS

| AVK PART NUMBER | TOOL R.P.M. | PARENT MATERIAL<br>STEEL, ALUMINUM<br>FIBERGLASS SML | PARENT MATERIAL<br>BLOW, ROTATIONAL<br>SOFT PLASTICS | COMPLETE<br>TOOL PART<br>NUMBER | THREAD<br>ADAPTION<br>KIT | A<br>HEX<br>DRIVE | B<br>SCREW<br>MANDREL | C<br>BEARING<br>SET | D<br>NOSE<br>CONE | DYNAMIC AIR<br>PRESSURE<br>SETTINGS<br>PSI - BARS |
|-----------------|-------------|------------------------------------------------------|------------------------------------------------------|---------------------------------|---------------------------|-------------------|-----------------------|---------------------|-------------------|---------------------------------------------------|
|                 |             | GRIP RANGE                                           | GRIP RANGE                                           |                                 |                           |                   |                       |                     |                   |                                                   |
| ARS4-420-280    | 1500        | .020-.190                                            | .020-.250                                            | ARPTI5P420-280                  | ARPT420-280TAK            | 29NPT 5           | 3SH420-2000           | 32PT 5              | 77NPT 250         | 60-70                                             |
|                 | 900         | .020-.250                                            | .020-.250                                            | ARPT9P420-280                   |                           |                   |                       |                     |                   | 40-60                                             |
| ARS4-420-500    | 1500        | .250-.430                                            | .250-.470                                            | ARPTI5P420-500                  | ARPT420-500TAK            | 29NPT 5           | 3SH420-2250           | 32PT 5              | 77NPT 250         | 60-70                                             |
|                 | 900         | .250-.470                                            | .250-.470                                            | ARPT9P420-500                   |                           |                   |                       |                     |                   | 40-60                                             |
| ARS4-518-280    | 900         | .020-.260                                            | —                                                    | ARPT9P-518-280                  | ARPT518-280TAK            | 29NPT 6           | 3SH518-2500           | 32PT 7              | 77NPT 3125        | 70-90                                             |
|                 | —           | —                                                    | —                                                    | —                               |                           |                   |                       |                     |                   | —                                                 |
| ARS4-610-7.1    | 1500        | .5-4.8                                               | .5-6.3                                               | ARPT15P-610-7.1                 | ARPT610-7.1TAK            | 29NPT 11          | 3SH610-50             | 32PT 6              | 77NPT M6          | 4.1-4.8                                           |
|                 | 900         | .5-6.3                                               | .5-6.3                                               | ARPT9P-610-7.1                  |                           |                   |                       |                     |                   | 2.8-4.1                                           |
| ARS4-610-12.7   | 1500        | 6.3-10.9                                             | 6.3-11.9                                             | ARPT15P-610-12.7                | ARPT610-12.7TAK           | 29NPT 11          | 3SH610-55             | 32PT 6              | 77NPT M6          | 4.1-4.8                                           |
|                 | 900         | 6.3-11.9                                             | 6.3-11.9                                             | ARPT9P-610-12.7                 |                           |                   |                       |                     |                   | 2.8-4.1                                           |
| ARS4-8125-7.1   | 900         | .5-6.6                                               | —                                                    | ARPT9P-8125-7.1                 | ARPT8125-7.1TAK           | 29NPT 12          | 3SH8125-60            | 32PT 7              | 77NPT M8          | 70-90                                             |
|                 | —           | —                                                    | —                                                    | —                               |                           |                   |                       |                     |                   | —                                                 |

This chart designates the tool, spare parts and dynamic (tool running) air pressure requirements for our most popular steel product.

NOTE: UNF FINE THREAD COMPONENTS ARE AVAILABLE.

PREVENTATIVE MAINTENANCE REQUIREMENTS:

- ◆ The bearing set must be kept in a WET lubricated condition to assure proper tool operation. AVK suggests the use of high temperature grease such as LUBRIPLATE BRAND 930 AA.
- ◆ The tool mandrel should be inspected for thread wear or damage and replaced. To test the condition of the mandrel, thread an AVK insert onto the mandrel backwards until it touches the pilot. If any drag is still felt, replace the mandrel with Unbrako socket head cap screws.

The SPP<sup>TM</sup> Installation Tool has been designed to install AVK's full range of threaded inserts from 6-32 to 1/2-13 and M4 to M12 thread sizes in three seconds or less. The lightweight handheld tool features a 3000 rpm ARO pneumatic motor providing exceptional speed and up to 7,500 lbs. of pull force capacity .

The pneumatic hydraulic power control system exerts a pull load on the AVK fastener that is controlled by force. This feature assures that the AVK fastener is installed properly within its grip range even in variable or multiple thickness materials.

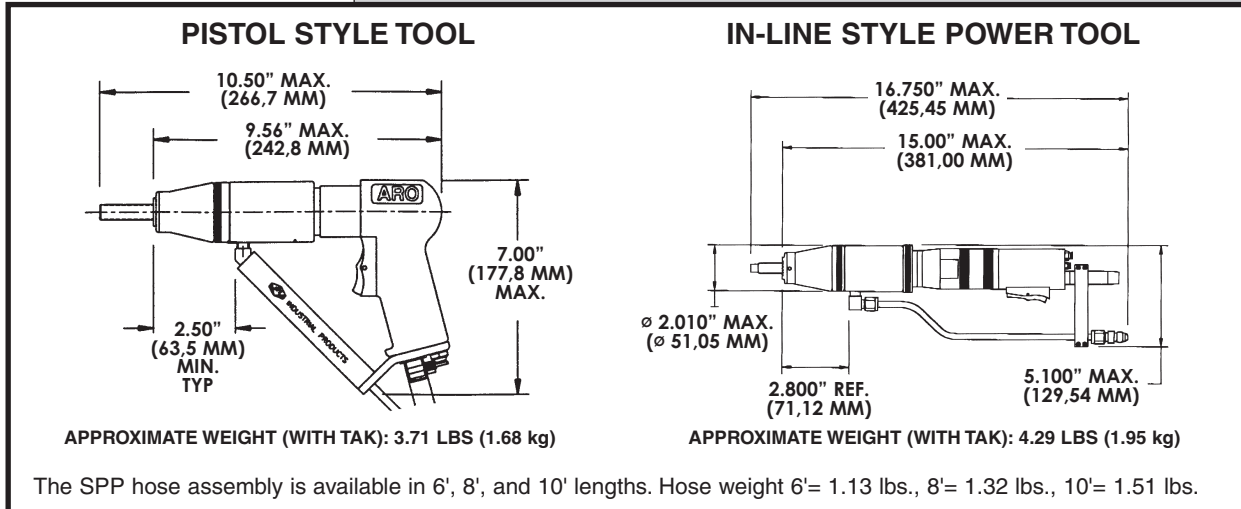
The SPP<sup>TM</sup> Tool hydraulic power control system is available in a suspended, bench top or caster model. The tool itself is available in the pistol or inline tool version.



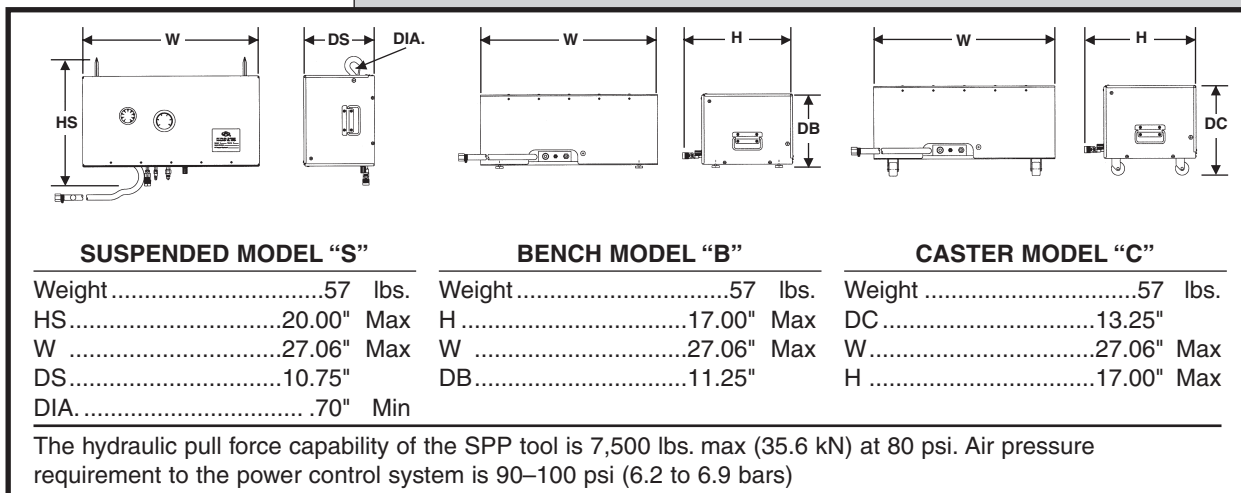
## **HERE'S HOW THE SPP<sup>TM</sup> TOOL WORKS**

- ◆ Select the appropriate quick release thread adaption kit and 1/4 turn lock it into the tool.
- ◆ Adjust the power control system's force control valve per the tool manual specifications for the AVK product and thread size you've selected.
- ◆ 1/4 turn the AVK product onto the tool mandrel. Place the AVK fastener into the hole in the parent material.
- ◆ Squeeze the top trigger of the SPP<sup>TM</sup> Tool and the tool mandrel will spin into the AVK fastener at 3000 rpm.
- ◆ When full thread engagement occurs the power control system automatically exerts its axial pull force installing the fastener properly regardless of parent material thickness.
- ◆ Squeeze the bottom trigger of the tool and the mandrel reverses at 3000 RPM from the installed fastener.
- ◆ The entire installation sequence takes approximately three seconds or less depending on fastener thread size and length.
- ◆ Size change over and power control system adjustment takes approximately three minutes.

## TOOL SPECIFICATIONS



## POWER CONTROL SYSTEM DIMENSIONS



## PRODUCT/TOOL SELECTION

Product tool section added note: SPP Tools per the part numbers below will be supplied with mandrels to install 1<sup>st</sup> and 2<sup>nd</sup> grip fasteners. If longer grip fasteners are being used contact AVK for appropriate mandrel part numbers.

| PRODUCT SERIES | THREAD SIZE          | STEEL | BRASS | ALUMINUM | MONEL | STAINLESS STEEL |
|----------------|----------------------|-------|-------|----------|-------|-----------------|
| A-K, A-L       | 6-32—1/2-13 M4-M10   | ✓     | ✓     | ✓        | ✓     | N/A             |
| A-H            | 6-32—3/8-16 M4-M10   | ✓     | ✓     | ✓        | ✓     | N/A             |
| A-P            | #10—3/8-16 M5-M10    | ✓     | ✓     | ✓        | ✓     | N/A             |
| A-R            | 1/4—20-5/16-18 M6-M8 | ✓     | N/A   | N/A      | N/A   | N/A             |
| A-T            | 6-32—1/2-13 M4-M12   | ✓     | ✓     | ✓        | N/A   | ✓               |
| A-W            | 6-32—3/8-16 M4-M10   | ✓     | ✓     | N/A      | N/A   | N/A             |
| A-O            | 6-32—3/8-16 M4-M10   | ✓     | N/A   | N/A      | N/A   | N/A             |
| R-N            | 6-32—1/2-13 M4-M10   | ✓     | N/A   | ✓        | N/A   | N/A             |

| SPP         | ( )        | ( )         | ( )                | ( )                        | ( )                        |
|-------------|------------|-------------|--------------------|----------------------------|----------------------------|
| System Type | TOOL TYPE  | HOSE LENGTH | POWER CONTROL TYPE | CALL OUT                   | AVK THREAD SIZE CALL OUT   |
|             | PISTOL P   | 6' 6        | SUSPENDED S        | AL A-L, A-K, A-H, A-P, A-O |                            |
|             | IN LINE IL | 8' 8        | BENCH B            | AR A-R                     |                            |
|             |            | 10' 10      | CASTER C           | RN R-N, A-T, A-W           |                            |
|             |            |             |                    |                            | SAMPLE NUMBER: SPPP8SAL420 |

**NOTE:** See SPP™ tool manual for set up, spare parts, preventative maintenance, and appropriate pressure settings.

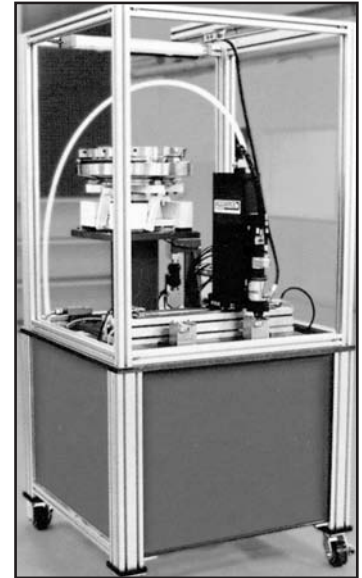


## AVK AUTOMATED INSTALLATION EQUIPMENT

AVK, in collaboration with Meikle Automation, has recognized the ever-increasing market demand for semi and fully automated custom insert installation equipment. Meikle's Dyna-System™ automated insert system is designed to meet high volume insert production assembly requirements maximizing facility output and reducing labor-intensive activities. Team Meikle/AVK can provide automated insert work stations and robotic arms utilizing micro-processors, electro-pneumatic circuitry, self diagnostic systems, light bar activated switches, light curtains and just about whatever your specification(s) may call for. For further information and to arrange a joint sales call and quotation inquiry, please contact your AVK Sales Representative today!



### **DYNA-SET™ AUTOMATED BLIND THREADED INSERT ASSEMBLY**



## OVERVIEW

Since 1983 AVK Industrial Products have been the vanguard of blind threaded structural insert and blind threaded stud technology. AVK is the design originator of "Spinwall Technology™" maximizing grip-range, blind side upset and ease of installation capability. AVK offers 12 high performance insert and stud product lines, as well as, customization of standard insert and stud products for your assembly system solution. Together, AVK and Meikle are expanding our fastening system abilities by offering semi and fully automated options for a multitude of sheet metal, composite, extrusion and hydro-formed structure assemblies; replacing among other things labor intensive weld nuts, sheet metal rivets, and self tapping screws. The Meikle Automation Dyna-Set™ System, combined with AVK inserts, provides the ultimate high-speed labor saving assembly solution.

### **Dyna-Set™ Technology**

The patent-pending design of the Dyna-Set™ automated insert system and Material Handling Module utilizes spin pull tool technology. The Dyna-Set™ will replace antiquated canister pneumatic hydraulic spin pull to stroke tools and provide greater reliability and maximize assembly capability.

Some of the Dyna-Set™ benefits are as follows:

- ◆ Labor savings
- ◆ Single or multi-simultaneous insert installation
- ◆ Maximum up-time providing optimal production output
- ◆ Multiple work station configurations are available
- ◆ Robotic arm with hole locating vision system is available
- ◆ Insert collapse load can be verified
- ◆ Dyna-Set™ systems are self diagnostic

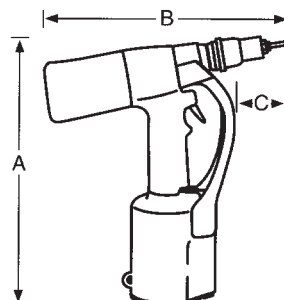
In addition to the above benefits, Dyna-Set's™ unique quick change chuck features the capability of replacing a mandrel without the use of tools. The Dyna-Set™ system can be mounted in any orientation, 360°, installing inserts from above, below or the side of a work piece. Its slim profile allows the Dyna-Set™ automated insert system and Material Handling Module to be mounted on three-inch centers, x,y,z slide configuration or a robotic arm providing maximum installation flexibility and labor reduction. The Dyna-Set™ System will handle AVK's A-L Series™ inserts, A-K Series™ inserts, A-H Series™ inserts, A-S Series™ and other customized components. AVK inserts are available in unified and Metric threads ranging in size from #440 through 5/16 unified and M3 through M8.

## AVK PNEUMATIC-HYDRAULIC TOOL SPECIFICATIONS

The tool shown on this page has been specifically designed to install the R-N Series Rivet Nuts.



Once you have selected the type of insert and thread size required for your application refer to the chart below for air tool selection.



RNPT 751

## DIMENSIONAL DATA/TOOL SET-UP REQUIREMENTS

PROPER AIR SUPPLY SET-UP REQUIRES:

- ◆ 85-100 PSI (5.8-6.8 BARS) dynamic air pressure at 17 S.C.F.M.
- ◆ Inline oiler/separator
- ◆ Air pressure gauge and regulator
- ◆ 5/16 or 7,92 mm minimum hose ID
- ◆ 5/16 or 7,92 mm minimum fittings ID

| TOOL PART NUMBER | WEIGHT<br>LBS.-kN. | PULL POWER<br>LBS.-kN. | A              | B              | C            |
|------------------|--------------------|------------------------|----------------|----------------|--------------|
| RNPT 751         | 6 LBS.<br>2.7 kN.  | 3,500 LBS.<br>15.6 kN. | 11.50<br>292,1 | 11.00<br>279,4 | 2.00<br>50,8 |

### TOOL SET-UP

The RNPT 751 tool installs the rivet nuts by applying a controlled length stroke. The length of stroke is determined by the fastener size and parent material thickness. Please contact AVK for stroke specifications.



RNPT 751 THREAD ADAPTION KIT

## AIR TOOL SELECTION/SPARE PARTS

| RNPT 751                                                                                                                                                                                           |                     |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|
| THREAD SIZE                                                                                                                                                                                        | THREAD ADAPTION KIT | A MANDREL      |
| 6-32 UNC                                                                                                                                                                                           | RNPT750-632TAK      | 18RNPT750-632  |
| 8-32 UNC                                                                                                                                                                                           | RNPT750-832TAK      | 18RNPT750-832  |
| 10-24 UNC                                                                                                                                                                                          | RNPT750-1024TAK     | 18RNPT750-1024 |
| 10-32 UNF                                                                                                                                                                                          | RNPT750-1032TAK     | 18RNPT750-1032 |
| 1/4-20 UNC                                                                                                                                                                                         | RNPT750-420TAK      | 18RNPT750-420  |
| 5/16-18 UNC                                                                                                                                                                                        | RNPT750-518TAK      | 18RNPT750-518  |
| 3/8-16 UNC                                                                                                                                                                                         | RNPT750-616TAK      | 18RNPT750-616  |
| M4x0,7 ISO                                                                                                                                                                                         | RNPT750-470TAK      | 18RNPT750-470  |
| M5x0,8 ISO                                                                                                                                                                                         | RNPT750-580TAK      | 18RNPT750-580  |
| M6x1,0 ISO                                                                                                                                                                                         | RNPT750-610TAK      | 18RNPT750-610  |
| M8x1,25 ISO                                                                                                                                                                                        | RNPT750-8125TAK     | 18RNPT750-8125 |
| M10x1,50 ISO                                                                                                                                                                                       | RNPT750-1015TAK     | 18RNPT750-1015 |
| The RNPT 751 is used to install Rivet-Nuts of the following materials.<br>Steel: 6-32 thru 5/16-18 and M4 thru M8<br>Aluminum: 6-32 thru 3/8-16 and M4 thru M10<br>Stainless Steel: SEE NOTE BELOW |                     |                |

**NOTE:** The RNPT751 is not recommended for installing stainless steel R-N Series. Please contact AVK for assistance.

# AVK HAND TOOL SPECIFICATIONS

The tools shown on this page are engineered to install AVK's...



## EXPENDABLE TOOLS

The following expendable tools are ideal for field repairs or consumer installation of AVK's products. Thread the AVK fastener onto the tool mandrel all the way up. Hold the tool with a box wrench and turn the mandrel with another box wrench or ratchet wrench until the AVK fastener is installed.

### A-K, A-L, A-P, AND A-O SERIES

#### 1st AND 2nd GRIP PRODUCT

##### MODEL AA184

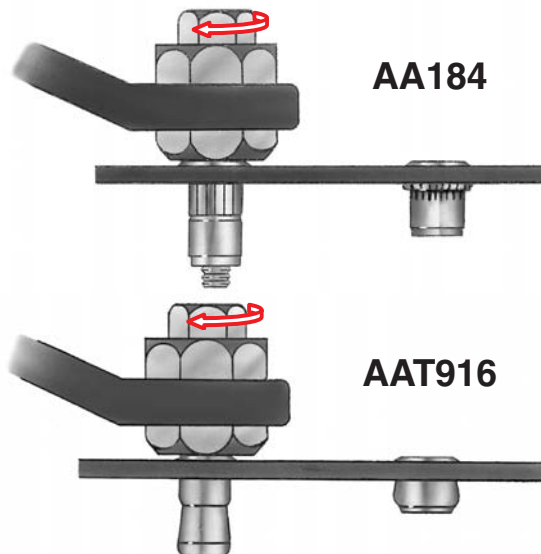
| THREAD SIZE    | TOOL PART NO. |
|----------------|---------------|
| 4-40 UNC       | N/A           |
| 6-32 UNC       | AA184-632     |
| 8-32 UNC       | AA184-832     |
| 10-24 UNC      | AA184-1024    |
| 10-32 UNF      | AA184-1032    |
| 1/4-20 UNC     | AA184-420     |
| 5/16-18 UNC    | AA184-518     |
| 3/8-16 UNC     | AA184-616     |
| 1/2-13 UNC     | AA184-813     |
| M3 x 0,5 ISO   | N/A           |
| M4 x 0,7 ISO   | AA184-470     |
| M5 x 0,8 ISO   | AA184-580     |
| M6 x 1,0 ISO   | AA184-610     |
| M8 x 1,25 ISO  | AA184-8125    |
| M10 x 1,5 ISO  | AA184-1015    |
| M12 x 1,75 ISO | AA184-12175   |

### A-T AND A-W SERIES

#### STANDARD PRODUCT

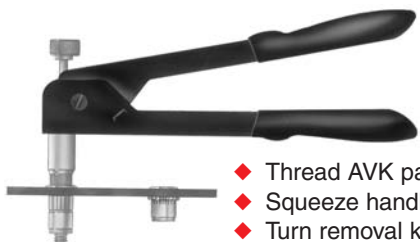
##### MODEL AAT916

| THREAD SIZE    | TOOL PART NO. |
|----------------|---------------|
| 4-40 UNC       | AAT916-440    |
| 6-32 UNC       | AAT916-632    |
| 8-32 UNC       | AAT916-832    |
| 10-24 UNC      | AAT916-1024   |
| 10-32 UNF      | AAT916-1032   |
| 1/4-20 UNC     | AAT916-420    |
| 5/16-18 UNC    | AAT916-518    |
| 3/8-16 UNC     | AAT916-616    |
| 1/2-13 UNC     | AAT916-813    |
| M3 x 0,5 ISO   | AAT916-350    |
| M4 x 0,7 ISO   | AAT916-470    |
| M5 x 0,8 ISO   | AAT916-580    |
| M6 x 1,0 ISO   | AAT916-610    |
| M8 x 1,25 ISO  | AAT916-8125   |
| M10 x 1,5 ISO  | AAT916-1015   |
| M12 x 1,75 ISO | AAT916-12175  |



## AA170 PLIER TOOL

The AA170 tool is ideal for prototyping or making repairs with the A-K, A-L, A-H, A-P and A-O Series threaded inserts.



- ◆ Thread AVK part onto tool
- ◆ Squeeze handles
- ◆ Turn removal knob

| THREAD SIZE   | CONVERSION KIT |
|---------------|----------------|
| 6-32 UNC      | AA271-632      |
| 8-32 UNC      | AA271-832      |
| 10-24 UNC     | AA271-1024     |
| 10-32 UNF     | AA271-1032     |
| 1/4-20 UNC    | AA271-420      |
| 5/16-18 UNC   | AA271-518*     |
| 3/8-16 UNC    | AA271-616*     |
| M4 x 0,7 ISO  | AA271-470      |
| M5 x 0,8 ISO  | AA271-580      |
| M6 x 1,0 ISO  | AA271-610      |
| M8 x 1,25 ISO | AA271-8125*    |
| M10 x 1,5 ISO | AA271-1015*    |

\*These sizes cannot be used with 2nd grip AVK fasteners.

## AA112 HI-TORQUER®

The AA112 is ideal for prototyping or making repairs with the A-T and A-W Series threaded inserts. It features a patented Quick Change thread size nose assembly.



### INSTALLATION:

- ◆ Thread insert fully onto mandrel.
- ◆ Place into hole.
- ◆ Hold grip bar while turning the "T" handle clockwise until insert is installed.
- ◆ Turn "T" handle counter clockwise to remove from insert.

| THREAD SIZE   | CONVERSION KIT |
|---------------|----------------|
| 4-40 UNC      | AAT202-440     |
| 6-32 UNC      | AAT202-632     |
| 8-32 UNC      | AAT202-832     |
| 10-24 UNC     | AAT202-1024    |
| 10-32 UNF     | AAT202-1032    |
| 1/4-20 UNC    | AAT202-420     |
| 5/16-18 UNC   | AAT202-518     |
| 3/8-16 UNC    | AAT202-616     |
| M3 x 0,5 ISO  | AAT202-350     |
| M4 x 0,7 ISO  | AAT202-470     |
| M5 x 0,8 ISO  | AAT202-580     |
| M6 x 1,0 ISO  | AAT202-610     |
| M8 x 1,25 ISO | AAT202-8125    |
| M10 x 1,5 ISO | AAT202-1015    |

## AVK HAND TOOL SPECIFICATIONS

The hand tools shown on this page are used to install the A-L, A-K, A-H, A-P, A-O, R-N and A-S Series Products.



### AA480 DOUBLE ACTION LEVER TOOL

The AA480 tool features a visual stroke indicator and a convenient spin-off removal knob. This tool installs the full range of AVK's spinwall technology product in steel and aluminum and limited sizes of steel and aluminum rivet nuts. See chart below.

#### INSTALLATION:

- ◆ Set the appropriate stroke per the AVK tool instruction sheet.
- ◆ Thread product fully onto tool mandrel.
- ◆ Place into the hole in your parent material.
- ◆ Squeeze tool handles together until product is fully collapsed.
- ◆ Spin center knob counter-clockwise unthreading tool mandrel from installed product.



### AA510 PLUNGER-DOUBLE ACTION LEVER TOOL

The AA510 tool features a visual stroke indicator and a quick acting pull to remove plunger. Due to its larger size, this tool installs the full range of AVK's Spinwall Technology™ Threaded Insert Product in all materials and various steel and aluminum rivet nuts. See chart below.

#### INSTALLATION:

- ◆ Set the appropriate stroke per the AVK tool instruction sheet.
- ◆ Withdraw plunger from tool.
- ◆ Hold product over tool mandrel while pushing plunger into tool fully threading product onto the tool mandrel.
- ◆ Place into hole in your parent material.
- ◆ Squeeze tool handles together until product is fully collapsed.
- ◆ Pull plunger from tool unthreading mandrel from installed product.



The chart below indicates the capability of the AA480 and AA510 to install a wide variety of AVK steel and aluminum products. Once you've selected the AVK product and thread size, refer to the chart and select the appropriate tool and conversion kit.

• Denotes the product can be installed with the AA480 tool. + Denotes the product can be installed with the AA510 tool.

| THREAD SIZE    | INSERT MATERIAL |      | A-L, A-K, A-H, A-P, A-O SERIES INSERTS |                       | INSERT MATERIAL |      | R-N SERIES RIVET NUTS |                       | STUD MATERIAL |                  | A-S SERIES STUDS      |  |
|----------------|-----------------|------|----------------------------------------|-----------------------|-----------------|------|-----------------------|-----------------------|---------------|------------------|-----------------------|--|
|                | STEEL           | ALUM | CONVERSION KIT #                       | REPLACEMENT MANDREL # | STEEL           | ALUM | CONVERSION KIT #      | REPLACEMENT MANDREL # | STEEL         | CONVERSION KIT # | REPLACEMENT MANDREL # |  |
| 4-40 UNC       |                 |      | NA                                     | NA                    | • +             | • +  | AA483-440             | 18AA481-440           |               | NA               | NA                    |  |
| 6-32 UNC       | • +             | • +  | AA481-632                              | 18AA481-632           | • +             | • +  | AA483-632             | 18AA481-632           | •             | AA485-632        | 18AA485-632           |  |
| 8-32 UNC       | • +             | • +  | AA481-832                              | 18AA481-832           | • +             | • +  | AA483-832             | 18AA481-832           | •             | AA485-832        | 18AA485-832           |  |
| 10-24 UNC      | • +             | • +  | AA481-1024                             | 18AA481-1024          | • +             | • +  | AA483-1024            | 18AA481-1024          | •             | AA485-1024       | 18AA485-1024          |  |
| 10-32 UNF      | • +             | • +  | AA481-1032                             | 18AA481-1032          | • +             | • +  | AA483-1032            | 18AA481-1032          | •             | AA485-1032       | 18AA485-1032          |  |
| 1/4-20 UNC     | • +             | • +  | AA481-420                              | 18AA481-420           | •               | • +  | AA483-420             | 18AA481-420           | •             | AA485-420        | 18AA485-420           |  |
| 5/16-18 UNC    | • +             | • +  | AA481-518                              | 18AA481-518           | •               | • +  | AA483-518             | 18AA481-518           | •             | AA485-518        | 18AA485-518           |  |
| 3/8-16 UNC     | • +             | • +  | AA481-616                              | 18AA481-616           |                 | •    | AA483-616             | 18AA481-616           | •             | AA485-616        | 18AA485-616           |  |
| 1/2-13 UNC     |                 | • +  | AA481-813                              | 18AA481-813           |                 |      | NA                    | NA                    |               | NA               | NA                    |  |
| M3 x 0,5 ISO   |                 |      | NA                                     | NA                    | • +             | • +  | AA483-350             | 18AA481-350           |               | NA               | NA                    |  |
| M4 x 0,7 ISO   | • +             | • +  | AA481-470                              | 18AA481-470           | • +             | • +  | AA483-470             | 18AA481-470           | •             | AA485-470        | 18AA485-470           |  |
| M5 x 0,8 ISO   | • +             | • +  | AA481-580                              | 18AA481-580           | • +             | • +  | AA483-580             | 18AA481-580           | •             | AA485-580        | 18AA485-580           |  |
| M6 x 1,0 ISO   | • +             | • +  | AA481-610                              | 18AA481-610           | •               | • +  | AA483-610             | 18AA481-610           | •             | AA485-610        | 18AA485-610           |  |
| M8 x 1,25 ISO  | • +             | • +  | AA481-8125                             | 18AA481-8125          | •               | • +  | AA483-8125            | 18AA481-8125          | •             | AA485-8125       | 18AA485-8125          |  |
| M10 x 1,5 ISO  | • +             | • +  | AA481-1015                             | 18AA481-1015          |                 | •    | AA483-1015            | 18AA481-1015          | •             | AA485-1015       | 18AA485-1015          |  |
| M12 x 1,75 ISO |                 | • +  | AA481-12175                            | 18AA481-12175         |                 |      | NA                    | NA                    |               | NA               | NA                    |  |

**NOTE:** Additional UNF fine thread sizes are available. Contact AVK for details.

**WARNING:** Failure to adjust the appropriate stroke setting per the AVK tool instruction sheet may damage the tool.

## AVK KIT SPECIFICATIONS

The Master Assortment Kits shown on this page have been designed for prototype, maintenance and repair applications.

### A-L SERIES INSERT MASTER ASSORTMENT KITS

The A-L Series Kit is ideal for prototype and maintenance repair applications. The rugged hard shell plastic kit contains quantities of the A-L Series threaded inserts and the AA170 plier tool shown on page 38. It also contains a full compliment of thread size conversion kits and instruction label.



INCH SIZE KIT PART NUMBER AVK 2292

| THREAD SIZE | INSERT QUANTITY | REFILL PAK PART NUMBER | CONVERSION KIT PART NUMBER |
|-------------|-----------------|------------------------|----------------------------|
| 6-32 UNC    | 50              | AALS4-632-80           | AA271-632                  |
| 8-32 UNC    | 50              | AALS4-832-80           | AA271-832                  |
| 10-32 UNF   | 50              | AALS4-1032-130         | AA271-1032                 |
| 1/4-20 UNC  | 50              | AALS4-420-165          | AA271-420                  |
| 5/16-18 UNC | 25              | AALS4-518-150          | AA271-518                  |
| 3/8-16 UNC  | 25              | AALS4-616-150          | AA271-616                  |

METRIC SIZE KIT PART NUMBER AVK 2293

| THREAD SIZE   | INSERT QUANTITY | REFILL PAK PART NUMBER | CONVERSION KIT PART NUMBER |
|---------------|-----------------|------------------------|----------------------------|
| M4 x 0,7 ISO  | 50              | AALS4-470-2.0          | AA271-470                  |
| M5 x 0,8 ISO  | 50              | AALS4-580-3.3          | AA271-580                  |
| M6 x 1,0 ISO  | 50              | AALS4-610-4.2          | AA271-610                  |
| M8 x 1,25 ISO | 25              | AALS4-8125-3.8         | AA271-8125                 |
| M10 x 1,5 ISO | 25              | AALS4-1015-3.8         | AA271-1015                 |

### A-T SERIES INSERT MASTER ASSORTMENT KITS

The A-T Series master assortment kit is ideal for prototype and maintenance repair applications. The rugged hard shell plastic kit contains quantities of the A-T Series Inserts and the AA 112 Hi-torquer tool shown on page 38. It also contains a full compliment of thread size conversion kits and instruction label.



INCH SIZE KIT PART NUMBER AAT312A

| THREAD SIZE | INSERT QUANTITY | REFILL PAK PART NUMBER | CONVERSION KIT PART NUMBER |
|-------------|-----------------|------------------------|----------------------------|
| 4-40 UNC    | 50              | AAT400-440             | AAT202-440                 |
| 6-32 UNC    | 50              | AAT400-632             | AAT202-632                 |
| 8-32 UNC    | 50              | AAT400-832             | AAT202-832                 |
| 10-32 UNF   | 50              | AAT400-1032            | AAT202-1032                |
| 1/4-20 UNC  | 50              | AAT400-420             | AAT202-420                 |
| 5/16-18 UNC | 25              | AAT400-518             | AAT202-518                 |

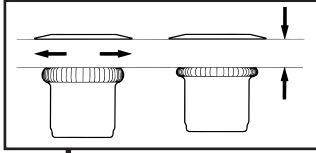
METRIC SIZE KIT PART NUMBER AAT312B

| THREAD SIZE   | INSERT QUANTITY | REFILL PAK PART NUMBER | CONVERSION KIT PART NUMBER |
|---------------|-----------------|------------------------|----------------------------|
| M3 x 0,5 ISO  | 50              | AAT400-350             | AAT202-350                 |
| M4 x 0,7 ISO  | 50              | AAT400-470             | AAT202-470                 |
| M5 x 0,8 ISO  | 50              | AAT400-580             | AAT202-580                 |
| M6 x 1,0 ISO  | 50              | AAT400-610             | AAT202-610                 |
| M8 x 1,25 ISO | 25              | AAT400-8125            | AAT202-8125                |

## DESIGN CONSIDERATIONS/APPLICATION CHECKLIST

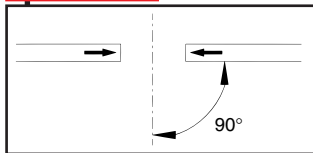
The following BEST PRACTICE information should be considered to insure proper application design when using AVK fasteners. If you have any application questions, please contact AVK.

### PARENT MATERIAL



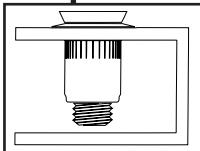
The parent material should be dense enough to support the hole fill and clamp load applied by the AVK fastener during installation. Its thickness should be within the grip range of the AVK fastener. Experimentation is suggested to determine optimum fastener selection for plastics.

### HOLE SIZE



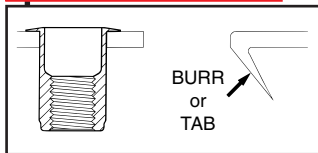
The hole produced in the parent material should be per the AVK catalog specifications. Tolerance for paint or coating buildup should be included to avoid an undersized hole condition. The hole should be square to the parent material.

### AVK PART ACCESS



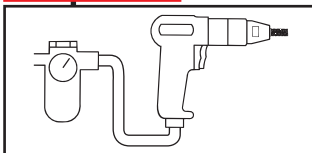
The AVK fastener head should sit flat on the parent material. An obstruction to the tail of the AVK product or to the access of the power tool can be corrected by contacting AVK for alternative fastener designs and tooling configurations. The AVK tool should be held perpendicular to the application to avoid excess mandrel wear.

### BACKSIDE SURFACE



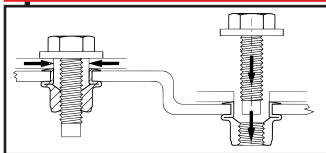
The backside surface of the parent material should not contain a punch slug or excessive burr that exceeds the grip range of the AVK fastener. Such large obstructions may prohibit the AVK fastener from installing properly.

### AIR SUPPLY



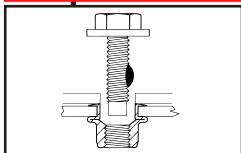
AVK's Spinwall Technology™ ARO brand installation tools require a pressure range of 60-110 PSI (4,1-7,5 BARS) at 25 S.C.F.M. of volume. Hose and fitting inside diameters need to be a minimum of 5/16 (7,92). An inline oiler and pressure regulator is required. Pressure should be measured as dynamic with the tool running.

### MATING PART HOLE SIZE/ALIGNMENT



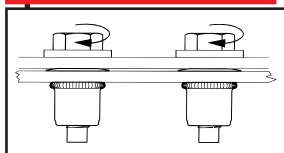
The mating part being attached should be non-rotational and contact the head of the AVK fastener. Its hole size should be .040 (1,0) smaller than the head diameter of the AVK fastener. The alignment of the mating part must provide perpendicular entry of the mating fastener into the AVK fastener.

### MATING FASTENER



For the A-K, A-L, A-T, A-W and A-O Series™ the mating fastener should be of a "free-spinning" design and of the grade or class as indicated in this catalog. If a mechanical, chemical locking or prevailing torque element is required, AVK recommends the design selection of the A-H Series™ in a hex punched hole. Specification of a dog point screw will minimize cross threading and speed the assembly process in any AVK application. Mating screws should be hand started and then power driven to minimize cross threading.

### ASSEMBLY TORQUE



For appropriate assembly torques, see the suggested assembly torque data contained on page 42. AVK's products, when used per the data provided in this catalog, have been designed to be compatible with the torque requirements of Grade 5 or Metric Class 8.8/9.8 screws. AVK suggests customer testing to determine the optimum torque due to mating component fit and mating fastener lubrication/finish variations.

## AVK TEST DATA

The test data on this page is intended to provide the designer with approximate strength values in various materials and thicknesses. The figures shown are averages of multiple tests. AVK recommends that this data be used only as a guide since various application factors may affect AVK product performance. We recommend testing your application when an exact strength figure is required or the load to be applied comes close to the published data.

Unified (Inch) thread size data is provided in pounds (lbs.) for force and inch pounds (in-lbs.) for torque. Metric data is provided in kilonewtons (kN) for force and newton meters (Nm) for torque.



**PULL-OUT** is the force required to pull the AVK product from the parent material. The parent material is restrained by a hold down ring 3x the AVK part "D" dim.

**PUSH-OUT** is the force required to push the AVK product through the parent material. The parent material is supported by a hold down ring 3x the AVK part "D" dim.

**SPIN-OUT** is the torque required to turn the AVK product in the parent material after installation without influencing clamp load on the AVK product.

**ASSEMBLY TORQUE** is the amount of torque suggested for Grade 5 or Metric Class 8.8/9.8 mating hardware based on industry standards.

A-L, A-K, A-H, A-S™ product was tested with the -4 finish. R-N Series® was Cadmium plated.

|                  |             | PULL-OUT       |      |      |      |                   |      |      |      | PUSH-OUT |            | SPIN-OUT* |     | ASSEMBLY TORQUE |
|------------------|-------------|----------------|------|------|------|-------------------|------|------|------|----------|------------|-----------|-----|-----------------|
|                  |             | IN STEEL SHEET |      |      |      | IN ALUMINUM SHEET |      |      |      | IN STEEL | STEEL ALUM |           |     |                 |
|                  |             | .030           | .062 | .090 | .125 | .030              | .062 | .090 | .125 | .125     | .062       | .062      |     |                 |
| A-L / A-K SERIES | THREAD SIZE | .76            | 1.57 | 2.28 | 3.17 | .76               | 1.57 | 2.28 | 3.17 | 3.17     | 1.57       | 1.57      |     |                 |
|                  | 6-32        | 316            | 817  | 1347 | 1952 | 276               | 898  | 1464 | 1604 |          |            |           | 12  |                 |
|                  | 8-32        | 316            | 773  | 1428 | 2000 | 275               | 828  | 1341 | 1571 |          |            |           | 22  |                 |
|                  | 10-24       | 473            | 971  | 1694 | 2127 | 325               | 964  | 1626 | 1967 |          |            |           | 32  |                 |
|                  | 10-32       | 473            | 971  | 1694 | 2127 | 325               | 964  | 1626 | 1967 |          |            |           | 36  |                 |
|                  | 1/4-20      | 570            | 1196 | 2244 | 2983 | 411               | 1157 | 2269 | 2892 |          |            |           | 75  |                 |
|                  | 5/16-18     | 703            | 2034 | 3066 | 4501 | 463               | 1435 | 2591 | 3896 |          |            |           | 156 |                 |
|                  | 3/8-16      | 703            | 1509 | 2551 | 4501 | 601               | 1554 | 2709 | 3850 |          |            |           | 276 |                 |
|                  | 1/2-13      | NA             | 1693 | 2855 | 4480 | NA                | 1866 | 3055 | 4184 |          |            |           | 660 |                 |
|                  | M4          | 1.4            | 3.4  | 6.3  | 8.8  | 1.2               | 3.6  | 5.9  | 6.9  |          |            |           | 2.5 |                 |
|                  | M5          | 2.1            | 4.3  | 7.5  | 9.4  | 1.4               | 4.2  | 7.2  | 8.7  |          |            |           | 5.0 |                 |
|                  | M6          | 2.5            | 5.3  | 9.9  | 13.2 | 1.8               | 5.1  | 10.0 | 12.8 |          |            |           | 8.6 |                 |
| M8               | 3.1         | 9.0            | 13.6 | 20.0 | 2.0  | 6.3               | 11.5 | 17.3 |      |          |            | 21        |     |                 |
| M10              | 3.1         | 6.7            | 11.3 | 20.0 | 2.6  | 6.9               | 12.0 | 17.1 |      |          |            | 42        |     |                 |
| M12              | NA          | 7.5            | 12.7 | 19.9 | NA   | 8.3               | 13.5 | 18.6 |      |          |            | 72        |     |                 |
| A-H SERIES       | 6-32        | 284            | 735  | NA   |      | 248               | 898  | NA   |      |          | 37         | 40        | 12  |                 |
|                  | 8-32        | 284            | 696  | NA   |      | 247               | 745  | NA   |      |          | 36         | 39        | 22  |                 |
|                  | 10-24       | 426            | 874  | 1525 |      | 292               | 867  | 1463 |      |          | 69         | 71        | 32  |                 |
|                  | 10-32       | 426            | 874  | 1525 |      | 292               | 867  | 1463 |      |          | 69         | 71        | 36  |                 |
|                  | 1/4-20      | 513            | 1176 | 2020 |      | 369               | 1041 | 2042 |      |          | 130        | 143       | 75  |                 |
|                  | 5/16-18     | 633            | 1831 | 2759 |      | 416               | 1292 | 2332 |      |          | 190        | 186       | 156 |                 |
|                  | 3/8-16      | 633            | 1358 | 2296 |      | 541               | 1399 | 2438 |      |          | 190        | 188       | 276 |                 |
|                  | M4          | 1.3            | 3.1  | 5.7  |      | 1.0               | 3.2  | 5.3  |      |          | 4.2        | 4.5       | 2.5 |                 |
|                  | M5          | 1.9            | 3.9  | 6.7  |      | 1.2               | 3.8  | 6.4  |      |          | 7.8        | 8.0       | 5.0 |                 |
|                  | M6          | 2.2            | 4.8  | 8.9  |      | 1.6               | 4.6  | 9.0  |      |          | 14.7       | 16.2      | 8.6 |                 |
|                  | M8          | 2.8            | 8.1  | 12.2 |      | 1.8               | 5.6  | 10.3 |      |          | 21.5       | 21.0      | 21  |                 |
|                  | M10         | 2.8            | 6.0  | 10.2 |      | 2.3               | 6.2  | 10.8 |      |          | 21.6       | 21.0      | 42  |                 |
| A-S SERIES       | 6-32        | 316            | 817  | 1437 | 1952 | 276               | 898  | 1464 | 1604 | 628      |            |           | 12  |                 |
|                  | 8-32        | 316            | 773  | 1428 | 2000 | 275               | 828  | 1341 | 1571 | 623      |            |           | 22  |                 |
|                  | 10-24       | 473            | 971  | 1694 | 2127 | 325               | 964  | 1626 | 1967 | 578      |            |           | 32  |                 |
|                  | 10-32       | 473            | 971  | 1694 | 2127 | 325               | 964  | 1626 | 1967 | 578      |            |           | 36  |                 |
|                  | 1/4-20      | 570            | 1196 | 2244 | 2983 | 411               | 1157 | 2269 | 2892 | 874      |            |           | 75  |                 |
|                  | 5/16-18     | 703            | 2034 | 3066 | 4501 | 463               | 1435 | 2591 | 3896 | 1134     |            |           | 156 |                 |
|                  | 3/8-16      | 703            | 1509 | 2551 | 4501 | 601               | 1554 | 2709 | 3850 | 1155     |            |           | 276 |                 |
|                  | M4          | 1.4            | 3.4  | 6.3  | 8.8  | 1.2               | 3.6  | 5.9  | 6.9  | 2.7      |            |           | 2.5 |                 |
|                  | M5          | 2.1            | 4.3  | 7.5  | 9.4  | 1.4               | 4.2  | 7.2  | 8.7  | 2.5      |            |           | 5.0 |                 |
|                  | M6          | 2.5            | 5.3  | 9.9  | 13.2 | 1.8               | 5.1  | 10.0 | 12.8 | 3.8      |            |           | 8.6 |                 |
|                  | M8          | 3.1            | 9.0  | 13.6 | 20.0 | 2.0               | 6.3  | 11.5 | 17.3 | 5.0      |            |           | 21  |                 |
|                  | M10         | 3.1            | 6.7  | 11.3 | 20.0 | 2.6               | 6.9  | 20.0 | 17.1 | 5.1      |            |           | 42  |                 |
| R-N SERIES*      | 1/4-20      | 347            | 854  | 1768 |      | 398               | 845  | 2100 |      | 1991     | 93         | 55        | 75  |                 |
|                  | 5/16-18     | 574            | 1201 | 2008 |      | 533               | 1106 | 2244 |      | 2688     | 176        | 101       | 156 |                 |
|                  | 3/8-16      | 623            | 1229 | 2041 |      | 585               | 1174 | 2110 |      | 3693     | 316        | 116       | 276 |                 |
|                  | 1/2-13      | 611            | 1314 | 2236 |      | 774               | 1385 | 2643 |      | 3877     | 450        | 216       | 660 |                 |
|                  | M6          | 1.5            | 3.7  | 7.8  |      | 1.7               | 3.7  | 9.3  |      | 8.8      | 16.8       | 10.6      | 8.6 |                 |
|                  | M8          | 2.5            | 5.3  | 8.9  |      | 2.6               | 2.9  | 9.9  |      | 11.9     | 30.3       | 12.7      | 21  |                 |
|                  | M10         | 2.7            | 5.4  | 9.0  |      | 2.6               | 5.2  | 9.3  |      | 16.4     | 40.2       | 13.5      | 42  |                 |
|                  | M12         | 2.7            | 5.8  | 9.9  |      | 3.4               | 6.1  | 11.7 |      | 17.2     | 53.3       | 48.8      | 72  |                 |

**NOTE 1:** Ultimate torque testing should be done using actual customer components and mating hardware due to plating/lubrication variables. AVK's fasteners have been designed to exceed the ultimate torque strength of the appropriate grade/class of mating hardware.

**NOTE 2:** For test data on other AVK products, contact AVK.

**NOTE 3:** R-N Series® Spin-out. Test drop values are keyed steel rivet nuts tested in steel plates and keyed aluminum rivet nut tested in 2024-T4 aluminum plates.

## DECIMAL EQUIVALENTS & DRILL SIZE CHART

| DRILL SIZE | INCH (Dec.) | METRIC (mm) | DRILL SIZE | INCH (Dec.) | METRIC (mm) | DRILL SIZE | INCH (Dec.) | METRIC (mm) | DRILL SIZE | INCH (Dec.) | METRIC (mm) | DRILL SIZE | INCH (Dec.) | METRIC (mm) |
|------------|-------------|-------------|------------|-------------|-------------|------------|-------------|-------------|------------|-------------|-------------|------------|-------------|-------------|
| 80         | .0135       | .343        | 50         | .0700       | 1,778       | 22         | .1570       | 3,988       | G          | .2610       | 6,630       | 31/64      | .4844       | 12,304      |
| 79         | .0145       | .368        | 49         | .0730       | 1,854       | 21         | .1590       | 4,039       | 17/64      | .2656       | 6,746       | 1/2        | .5000       | 12,700      |
| 1/64       | .0156       | .396        | 48         | .0760       | 1,930       | 20         | .1610       | 4,089       | H          | .2660       | 6,756       | 33/64      | .5156       | 13,096      |
| 78         | .0160       | .406        | 5/64       | .0781       | 1,984       | 19         | .1660       | 4,216       | I          | .2720       | 6,909       | 17/32      | .5312       | 13,492      |
| 77         | .0180       | .457        | 47         | .0785       | 1,994       | 18         | .1695       | 4,305       | J          | .2770       | 7,036       | 35/64      | .5469       | 13,891      |
| 76         | .0200       | .508        | 46         | .0810       | 2,057       | 11/64      | .1719       | 4,366       | K          | .2810       | 7,137       | 9/16       | .5625       | 14,288      |
| 75         | .0210       | .533        | 45         | .0820       | 2,083       | 17         | .1730       | 4,394       | 9/32       | .2812       | 7,142       | 37/64      | .5781       | 14,684      |
| 74         | .0225       | .572        | 44         | .0860       | 2,184       | 16         | .1770       | 4,496       | L          | .2900       | 7,366       | 19/32      | .5938       | 15,083      |
| 73         | .0240       | .609        | 43         | .0890       | 2,261       | 15         | .1800       | 4,572       | M          | .2950       | 7,493       | 39/64      | .6094       | 15,479      |
| 72         | .0250       | .635        | 42         | .0935       | 2,375       | 14         | .1820       | 4,623       | 19/64      | .2969       | 7,541       | 5/8        | .6250       | 15,875      |
| 71         | .0260       | .660        | 3/32       | .0938       | 2,383       | 13         | .1850       | 4,700       | N          | .3020       | 7,671       | 41/64      | .6406       | 16,271      |
| 70         | .0280       | .711        | 41         | .0960       | 2,438       | 3/16       | .1875       | 4,763       | 5/16       | .3125       | 7,938       | 21/32      | .6562       | 16,667      |
| 69         | .0292       | .742        | 40         | .0980       | 2,489       | 12         | .1890       | 4,801       | O          | .3160       | 8,026       | 43/64      | .6719       | 17,066      |
| 68         | .0310       | .787        | 39         | .0995       | 2,527       | 11         | .1910       | 4,851       | P          | .3230       | 8,204       | 11/16      | .6875       | 17,463      |
| 1/32       | .0312       | .792        | 38         | .1015       | 2,578       | 10         | .1935       | 4,915       | 21/64      | .3281       | 8,334       | 45/64      | .7031       | 17,859      |
| 67         | .0320       | .813        | 37         | .104        | 2,642       | 9          | .1960       | 4,978       | Q          | .3320       | 8,433       | 23/32      | .7188       | 18,258      |
| 66         | .0330       | .838        | 36         | .1065       | 2,705       | 8          | .1990       | 5,055       | R          | .3390       | 8,611       | 47/64      | .7344       | 18,654      |
| 65         | .0350       | .889        | 7/64       | .1094       | 2,779       | 7          | .2010       | 5,105       | 11/32      | .3438       | 8,733       | 3/4        | .7500       | 19,050      |
| 64         | .0360       | .914        | 35         | .1100       | 2,794       | 13/64      | .2031       | 5,159       | S          | .3480       | 8,839       | 49/64      | .7656       | 19,446      |
| 63         | .0370       | .940        | 34         | .1110       | 2,819       | 6          | .2040       | 5,182       | T          | .3580       | 9,093       | 25/32      | .7812       | 19,842      |
| 62         | .0380       | .965        | 33         | .1130       | 2,870       | 5          | .2055       | 5,220       | 23/64      | .3594       | 9,129       | 51/64      | .7969       | 20,241      |
| 61         | .0390       | .991        | 32         | .1160       | 2,946       | 4          | .2090       | 5,309       | U          | .3680       | 9,347       | 13/16      | .8125       | 20,638      |
| 60         | .0400       | 1,016       | 31         | .1200       | 3,048       | 3          | .2130       | 5,410       | 3/8        | .3750       | 9,525       | 53/64      | .8281       | 21,034      |
| 59         | .0410       | 1,041       | 1/8        | .1250       | 3,175       | 7/32       | .2188       | 5,558       | V          | .3770       | 9,576       | 27/32      | .8438       | 21,433      |
| 58         | .0420       | 1,067       | 30         | .1285       | 3,264       | 2          | .2210       | 5,613       | W          | .3860       | 9,804       | 55/64      | .8594       | 21,829      |
| 57         | .0430       | 1,092       | 29         | .1360       | 3,454       | 1          | .2280       | 5,791       | 25/64      | .3906       | 9,921       | 7/8        | .8750       | 22,225      |
| 56         | .0465       | 1,181       | 28         | .1405       | 3,569       | A          | .2340       | 5,944       | X          | .3970       | 10,084      | 57/64      | .8906       | 22,621      |
| 3/64       | .0469       | 1,191       | 9/63       | .1406       | 3,571       | 15/64      | .2344       | 5,954       | Y          | .4040       | 10,262      | 29/32      | .9062       | 23,017      |
| 55         | .0520       | 1,321       | 27         | .1440       | 3,658       | B          | .2380       | 6,045       | 13/32      | .4062       | 10,317      | 59/64      | .9219       | 23,416      |
| 54         | .0550       | 1,397       | 26         | .1470       | 3,734       | C          | .2420       | 6,147       | Z          | .4130       | 10,490      | 15/16      | .9375       | 23,813      |
| 53         | .0595       | 1,511       | 25         | .1495       | 3,797       | D          | .2460       | 6,248       | 27/64      | .4219       | 10,716      | 61/64      | .9531       | 24,209      |
| 1/16       | .0625       | 1,588       | 24         | .1520       | 3,861       | 1/4        | .2500       | 6,350       | 7/16       | .4375       | 11,113      | 31/32      | .9688       | 24,608      |
| 52         | .0635       | 1,613       | 23         | .1540       | 3,912       | E          | .2500       | 6,350       | 29/64      | .4531       | 11,509      | 63/64      | .9844       | 25,004      |
| 51         | .0670       | 1,702       | 5/32       | .1562       | 3,967       | F          | .2570       | 6,528       | 15/32      | .4688       | 11,908      | 1          | 1.000       | 25,400      |

## DECIMAL EQUIVALENT OF STANDARD GAUGE SHEET ALUMINUM & SHEET METAL

| NO. OF GAUGE | GAUGE        |                   | NO. OF GAUGE | GAUGE        |                   | NO. OF GAUGE | GAUGE        |                   |
|--------------|--------------|-------------------|--------------|--------------|-------------------|--------------|--------------|-------------------|
|              | ALUM (B & S) | STEEL (U.S. Std.) |              | ALUM (B & S) | STEEL (U.S. Std.) |              | ALUM (B & S) | STEEL (U.S. Std.) |
| 10           | .1019        | .1345             | 17           | .0453        | .0538             | 24           | .0201        | .0239             |
| 11           | .0907        | .1196             | 18           | .0403        | .0478             | 25           | .0179        | .0209             |
| 12           | .0808        | .1046             | 19           | .0359        | .0418             | 26           | .0159        | .0179             |
| 13           | .0720        | .0897             | 20           | .0320        | .0359             | 27           | .0142        | .0164             |
| 14           | .0641        | .0747             | 21           | .0285        | .0329             | 28           | .0126        | .0149             |
| 15           | .0571        | .0673             | 22           | .0253        | .0299             | 29           | .0113        | .0135             |
| 16           | .0508        | .0598             | 23           | .0226        | .0269             | 30           | .0100        | .0120             |

## SUGGESTED ASSEMBLY TORQUE VALUES TO PRODUCE CORRESPONDING BOLT LOADS

| THREAD SIZE | SAE GRADE 5 BOLTS |                           |        |
|-------------|-------------------|---------------------------|--------|
|             | CLAMP LOAD (lbs.) | ASSEMBLY TORQUE (in-lbs.) |        |
|             |                   | DRY                       | PLATED |
| #4-40       | 380               | 8                         | 6      |
| #6-32       | 580               | 16                        | 12     |
| #8-32       | 900               | 30                        | 22     |
| #10-24      | 1120              | 43                        | 32     |
| #10-32      | 1285              | 49                        | 36     |
| 1/4-20      | 2000              | 96                        | 75     |
| 1/4-28      | 2300              | 120                       | 86     |
| 5/16-18     | 3350              | 204                       | 156    |
| 5/16-24     | 3700              | 228                       | 168    |
| 3/8-16      | 4950              | 360                       | 276    |
| 3/8-24      | 5600              | 420                       | 300    |

## METRIC CONVERSIONS

|                 |                      |            |                           |                             |           |                    |
|-----------------|----------------------|------------|---------------------------|-----------------------------|-----------|--------------------|
| <b>LINEAR</b>   | Multiply INCHES      | by 25.4    | to get MILLIMETERS (mm)   | Multiply MILLIMETERS (mm)   | by .03937 | to get INCHES      |
|                 | Multiply FEET        | by 0.3048  | to get METERS (m)         | Multiply METERS (m)         | by 3.281  | to get FEET        |
|                 | Multiply INCHES      | by 2.54    | to get CENTIMETERS (cm)   | Multiply CENTIMETERS (cm)   | by .3937  | to get INCHES      |
| <b>TORQUE</b>   | Multiply INCH-POUNDS | by 0.11298 | to get NEWTON-METERS (Nm) | Multiply NEWTON-METERS (Nm) | by 8.851  | to get INCH-POUNDS |
|                 | Multiply FOOT-POUNDS | by 1.3558  | to get NEWTON-METERS (Nm) | Multiply NEWTON-METERS (Nm) | by 0.7376 | to get FOOT-POUNDS |
| <b>FORCE</b>    | Multiply POUNDS      | by .00445  | to get KILONEWTONS (kN)   | Multiply KILONEWTONS (kN)   | by 224.72 | to get POUNDS      |
| <b>PRESSURE</b> | Multiply PSI         | by .069    | to get BARS               | Multiply BARS               | by 14.5   | to get PSI         |

## TROUBLE SHOOTING BEFORE PROCEEDING CHECK TO BE SURE THAT THE AVK PRODUCT BEING USED HAS THE PROPER GRIP RANGE FOR THE MATERIAL IT'S BEING INSTALLED INTO.

| SYMPTOM                                             | CHECK LIST                                                                                                                                                  |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Tool does not operate                            | Air lines clear? Tool inlet clear? Tool exhaust clear? Motor Lubrication?                                                                                   |
| 2. Tool runs but stalls before product is installed | Proper tool RPM for thread size being used? Adequate air pressure with tool running? TAK bearing in place and lubricated? Damaged mandrel? Wrong grip part? |
| 3. Tool strips product threads                      | Proper tool RPM for thread size being used? Too much air pressure? Has the mandrel worn too small?                                                          |
| 4. Mandrels wear/break                              | Is there too much air pressure? Is the tool being held perpendicular during installation?                                                                   |



A-L Series, A-K Series, A-H Series, A-S Series, A-T Series, A-P Series, A-O Series, A-R Series, PreSet, SPP Tool, "Improving The Way We Assemble The World," and Spinwall Technology are trademarks of Avibank Mfg., Inc. A-W Series, R-N Series, Hi Torquer, AVK and  are registered trademarks of Avibank Mfg., Inc.

## AVK INDUSTRIAL PRODUCTS

AVK Industrial Products, a Precision Castparts Company, produces all of its blind threaded captive fasteners at its 90,000 square foot factory in Valencia, California which is located just 35 miles northwest of downtown Los Angeles. We have been manufacturing high quality blind threaded captive fasteners for over 25 years.

**AVK FASTENERS ARE MADE IN THE U.S.A.**

ISO 9001-1994    ISO 14001-1996    QS-9000-1998

AVK sells its products through Authorized Distributors which are supported by AVK's factory-trained field sales staff. For additional information contact your local AVK distributor or contact AVK.



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# AVK INDUSTRIAL PRODUCTS

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